

MATH 1040 – Statistical Inference

Course Notes

Summer 2026

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Chapter 1: Introduction to Statistics

Topic 1

Data and Variables

Lecture

What is statistics? Statistics is the science of _____.

- **Data** is _____. This information can come from observations, counts, measurements, or responses. Typically data takes the form of observed measurements (e.g. height, temperature) or descriptions (e.g. dog, cat, blue, female).
- Any task involving collecting, processing, displaying, and interpreting data, especially for the purpose of making decisions, is part of the discipline of statistics.

Big picture: statistics is how we use data to say things about _____.

Sometimes we do this by displaying data, so that we can get a good visual idea of what is going on.

Variables: A **variable** is a characteristic of a subject that can be assigned a number or category. Data are the values taken on by some variable. There are two basic types of variables:

1. **Quantitative** (Measurement) – takes numerical values for which arithmetic comparisons make sense. (ex: height, weight, concentration, etc.)
 - If the variable can be counted (only whole numbers make sense), it is **discrete**.
 - If the variable must be measured (decimals make sense), it is **continuous**.
2. **Categorical** (qualitative) – records into which of several categories a subject falls. (ex: sex, political affiliation, etc.)
 - A categorical variable with exactly two categories (such as whether you are in class today) is said to be a **binary** variable.
 - The distinction between quantitative and categorical variables is important because determining how to analyze the data depends on the variable type.

Note that something is only a variable if it can be measured on each observational unit. For example, height is a variable because we can collect height on every student. Average height of students in the class is not a variable. Number of students over 6' is not a variable.

Usually, if the data collected are numbers, then the variable is quantitative. This is not always the case, however.

Example: the group to which you were assigned is a number, but it is a categorical variable. Think about whether taking an average makes sense: if taking an average makes sense, then

Observational units or **cases** are the actual subjects assigned the variable values.

Example: If we collect the heights of the students in this class, the students are the observational units. The notion that variables take on different values or categories from different observational units is known as **variability**.

A **research question** often looks for patterns in a variable or compares a variable across different group or looks for a relationship between variables.

Examples:

Do most students in your class get at least 7 hours of sleep on a typical night? Do females tend to study more than males? Is there an association between how much students study per week and how much sleep they get?

Activities

Activity 1-1: Text Messages

- a. For each student in the class, record the number of text messages (including using messaging apps) they have sent so far today.

The numbers that you have recorded are data. Not all numbers are data, however. Data are numbers collected in a particular context. For example, the numbers 10, 3, and 7 do not constitute data in and of themselves. They are data, however, if they refer to the number of texts sent by the first three students to walk into the classroom today.

- b. Did every student in your class send the same number of messages?

Although it might be obvious that different students will send a different number of messages, the fact that the same question (e.g., “number of texts sent”) can produce different responses illustrates one of the most fundamental ideas in statistics: variability.

Definitions

A **variable** is any characteristic of a person or thing that can be assigned a number or a category. The person or thing to which the number or category is assigned, such as a student in your class, is called the **observational unit**. **Data** consist of the numbers or categories recorded for the observational units in a study. **Variability** refers to the phenomenon of a variable taking on different values or categories from observational unit to observational unit.

A **quantitative** variable measures a numerical characteristic such as height, whereas a **categorical** variable records a group designation such as gender.

Binary variables are categorical variables with only two possible categories, for example, yes and no.

- c. Is *number of messages sent* a quantitative or a categorical variable?
- d. What if we recorded only *whether or not you have sent a message today*? Would that be a quantitative or a categorical variable?
- e. Suggest another categorical variable that we could record about each student in your class with regard to cell phone use today.
- f. Still considering the students in your class as the observational units, suppose each was asked the following questions. Classify each of the following variables as categorical or quantitative. If it is categorical, also indicate whether it is binary.
 - Have you sent more messages or received more messages, or the same number of each?
 - What is the average duration of the calls you have made today?
 - Is your phone an iPhone or Android?
 - At what time did you receive your first message today?
 - What was the area code to which you made your first message today?

We will continue to focus your attention on observational units and variables in virtually all of the studies and data that you encounter throughout this class. Along the way, keep the following points in mind:

- The distinction between categorical and quantitative variables is quite important because determining which statistical tools to use for analyzing a given set of data often depends on the type of variable involved.
 - Notice that some variables can often be measured and recorded in different ways. For example, the time of your first message: did you specify a time (e.g., 8:00), a measurement (10 minutes after I awoke), or a time period (e.g., before breakfast)? This is true of many variables: the classification of the variable often depends on how the quantity is measured more than on any intrinsic property. (Also try to write your variables so that how they are measured is clear!)
 - Also notice that a variable can take on numerical values that are really just category labels, like area code. If it does not make sense to sum or average these numbers, we will consider these to be categorical variables.
- g. Lambert and Pinheiro (2006) describe a study in which researchers try to identify characteristics of cell phone calls that suggest the phone is being used fraudulently. Suppose we want to know the average duration of all the calls you (and only you) have made in the past month as a way to create a profile of your cell phone usage. Identify the observational units and variable in this measurement that will help us find that average, and classify the variable as quantitative or categorical.

Observational units:

Variable:

Type:

Keep in Mind

- It is very important that you carefully think about the observational units and how to phrase the variable as a characteristic that varies from observational unit to observational unit. It might be helpful to force yourself to always fill in the blanks of the following sentence:

We are recording _____ from _____ to _____.
variable observational units

For example, we are recording duration from call to call. Or it might be helpful to think of the variable as a question you could pose to each observational unit, for example, “How long was the call?”

- Whether you have properly defined a variable depends on how the observational unit is defined. Although *average call duration* is a legitimate variable when the observational units are different students, if you want to focus on the calls of one individual, then the calls are the observational units, and *call duration* is

the variable. In this case, the summary (average duration across all calls) is not considered a variable from call to call.

- To determine whether something is actually a variable, ask yourself whether or not it represents a question that can be asked of each observational unit (e.g., how long was the call) and whether the values can potentially vary from observational unit to observational unit. Notice that asking the students in your class *whether or not you are enrolled in this class* is a variable, but it would not be a very interesting one!
- h. Suggest two more categorical variables and two more quantitative variables that could be measured about the cell phone calls you made in the past month to help describe how you use your phone. Make sure you state these as variables and not as summaries.

Activity 1-2: Student Data

- a. Again consider the students in your class as observational units. Classify each of the following as categorical or quantitative. If it is categorical, also indicate whether or not it is binary.
- How many hours you have slept in the past week.
 - Whether or not you have slept for at least 8 hours in the past 24 hours.
 - Number of *Lord of the Rings* books that you have read.
 - How many states you have visited.
 - Handedness (which hand you write with).
 - Political viewpoint (classified as liberal, conservative, libertarian, etc.).
 - Day of the week on which you were born.
 - Average study time per week.
 - How many birthday gifts you received on your last birthday.

Definitions
A research question often looks for patterns in a variable or compares a variable across different groups or looks for a relationship between variables.

Some research questions that you could investigate with data on the above variables include:

- Do most students in your class get at least 8 hours of sleep in a typical night?
- Do females tend to study more than males?
- Is there an association between how much students study per week and how much sleep they get?

Notice that though these are also phrased as questions, they summarize the variable(s) across the observational units rather than being posed to the individual observational unit.

- b. Suggest two other research questions that you could investigate using the variables in part a.

Research question 1:

Research question 2:

- c. Suggest three additional variables that you could record about yourself and your classmates, and then propose two research questions that you could address using those variables. [Hint: Be sure to distinguish the variables from the research questions; remember a variable is some characteristic that can be recorded for each student and can vary from student to student.]

Variable 1:

Variable 2:

Variable 3:

Research question 1:

Research question 2:

Activity 1-3: Research Scenarios

For each of the following studies, identify the observational units and variables. Also, classify each variable as quantitative or categorical.

- a. A research suspects that people with PhDs tend to come from a higher socioeconomic status than those without. She takes a random sample of 100 people with PhDs and records the socioeconomic status of their parents.

Observational units:

Variable:

Type:

- b. A data scientist wants to know whether people from Utah tend to be taller than those from other states. He records the heights of 212 Utahns and then takes their average height.

Observational units:

Variable:

Type:

- c. A psychologist gives an exam to 300 African-American college students. Half of the students are randomly assigned to use a version of the exam that asks them to indicate their race, and the other half are randomly assigned to use a version of the exam that does not ask them to indicate their race. The psychologist suspects that those students who are not asked to indicate their race will score significantly higher on the exam than those who are asked to indicate their race.

Observational units:

Variable 1:

Type:

Variable 2:

Type:

Extra Space for Notes

Topic 2

Data and Distributions

Lecture

One goal of statistics is to describe the variability in data (or how the data varies).

The pattern of variation of a variable is known as its **distribution**. There are two basic ways to describe distributions:

1. Visual – bar graph, dotplot, histogram, stemplot, etc.
2. Numerical – proportion, mean, median, variance, etc.

A **proportion** simply measures a part in comparative relation to the whole.

Example: if I had 139 students in the fall, 191 students in the spring, and then 44 students in the summer, what proportion of all my students took a class with me in the summer?

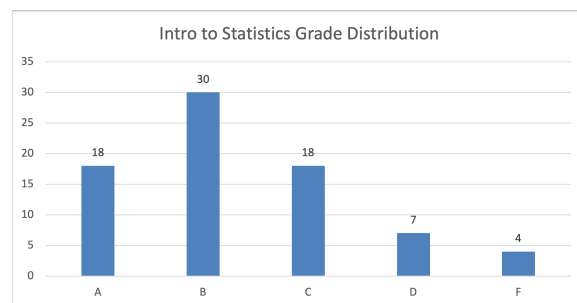
So, a proportion can be thought of as the number of “successes”, whatever it is we are interested in, divided by the total number of all.

Data Graphics

We will look at a couple charts and graphs now and then will focus on more later. A simple visual display for the distribution of a **categorical variable** is a **bar graph** or **bar chart**.

Bar charts

- Can be used to summarize one qualitative variable.
- Each bar in a bar chart shows you the frequency (or count) for the group (or category) it is associated with.
- The categories go on the x -axis.



Note: instructions on how to create all graphs we will be working with, among other things, can be found in the *Excel Guide* on Canvas.

A simple visual display for the distribution of a **quantitative variable** is a **dotplot**.

- We place a dot to represent each observation.
- If there are multiple observations with the same value, stack the dots.
- Dotplots are simple to construct and are most useful for small datasets.

- Can see unusually low and high values, typical values, and the variability (or spread) in the data.

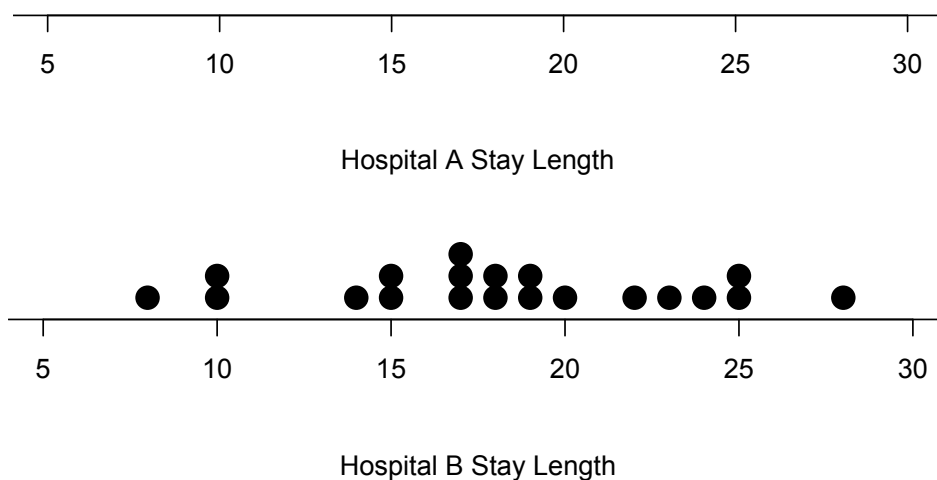
Example: Consider the following data on the length of stay for patients at two different hospitals.

Hospital A	18	26	22	22	20	24	20	17	21	24	22	24	20	21	25	23	15	23	22	27
Hospital B	22	19	20	17	14	10	18	25	17	17	25	28	23	19	24	18	8	15	10	15

Some questions:

- Which hospital tended to have longer stays?
- Which hospital was more consistent in terms of the stay lengths?
- How would you describe this distribution of hospital stay lengths?

We can more easily answer these questions by making some plots.



Although not every stay length for hospital B was shorter than every stay length for hospital A, there was a tendency for the stays to be shorter for hospital B.

Using the word “**tendency**” indicates that typically or on average, we expect hospital B to have shorter stay lengths based on these data. We will consider this notion of **statistical tendency** throughout this course.

Although hospital B had shorter stay lengths, they tended to be less consistent in their lengths. “**Less consistent**” means that there is more variability or spread in the stay lengths for hospital B.

Activities

Activity 2-1: Split up California?

- a. Guess what percentage of American adults favor splitting California into multiple states.
- b. Collect data on the class about whether they are in favor of splitting California. Write the raw responses here.
- c. Is this a categorical (also binary?) or quantitative variable?
- d. Count how many students responded to the California question and how many of them voted to split it up. Calculate the proportion who voted to split up California by dividing that count by the total count. (Note that a proportion is a number between 0 and 1. Report your answer to three decimal places.)
- e. How many and what proportion of the respondents voted to split up California?
- f. Create a visual display of these responses by drawing rectangles with heights corresponding to the proportions voting for each option. Be sure to give labels to the horizontal and vertical axes.

Definitions

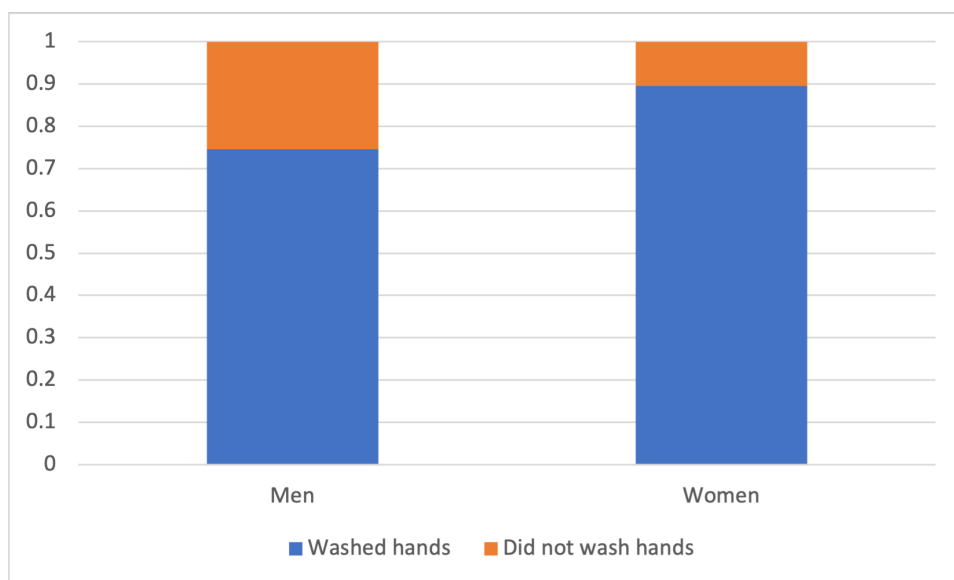
The visual display that you have constructed is called a **bar graph**. Bar graphs display the distribution of a categorical variable. The **distribution** of a variable refers to its pattern of variation. With a categorical variable, distribution means the variable's possible categories and the proportion of responses in each.

- g. Write a sentence or two describing what your analysis reveals about the attitudes of students in this class toward splitting California.

Activity 2-2: Hand Washing

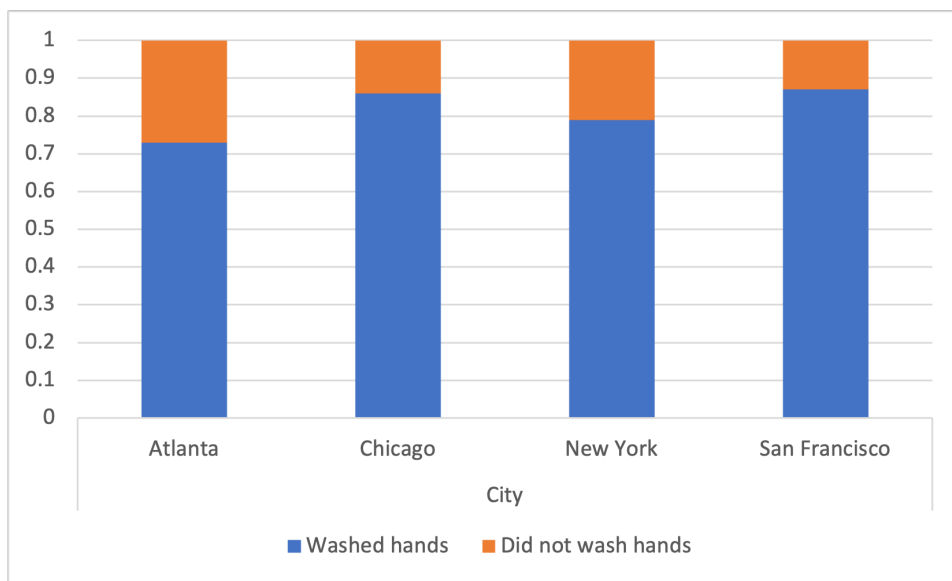
In August 2005, researchers for the American Society for Microbiology and the Soap and Detergent Association monitored the behavior of more than 6300 users of public restrooms. They observed people in public venues such as Turner Field in Atlanta and Grand Central Station in New York City. They found that 2393 of 3206 men washed their hands, compared to 2802 of 3130 women.

- What proportion of the men washed their hands? What proportion of the women washed their hands?
- Are these proportions consistent with the following **stacked or segmented bar graphs**?



- Comment on what your calculations and the bar graph reveal about whether or not one gender is more likely to wash their hands after using a public restroom.

- d. The following bar graphs reveal the distribution of the washed hands variable based on the city in which the person was observed:



For each city, estimate the proportion of people who washed their hands as accurately as you can from the graph.

Atlanta:

Chicago:

New York:

San Francisco:

- e. Comment on what the bar graphs reveal about how these cities compare with regard to hand washing.

Activity 2-3: Student Travels

- a. Guess the number of different states that a typical student at your school has visited. Also, guess what the largest and smallest numbers of states visited by the students in your class will be.

Smallest:

Typical:

Largest:

- b. Count the number of states you have visited and write that number on the white board.
- c. What type of variable is the *how many states have you visited* question (quantitative or categorical)?

- d. Create a visual display of the distribution of the number of states visited by you and your classmates. A horizontal scale has been drawn; place a dot for each student above the appropriate number of states visited. For repeated values, simply stack the dots on top of each other.



Definitions
The visual display that you constructed for states visited is a dotplot . Dotplots are useful for displaying the distribution of relatively small datasets of a quantitative variable.

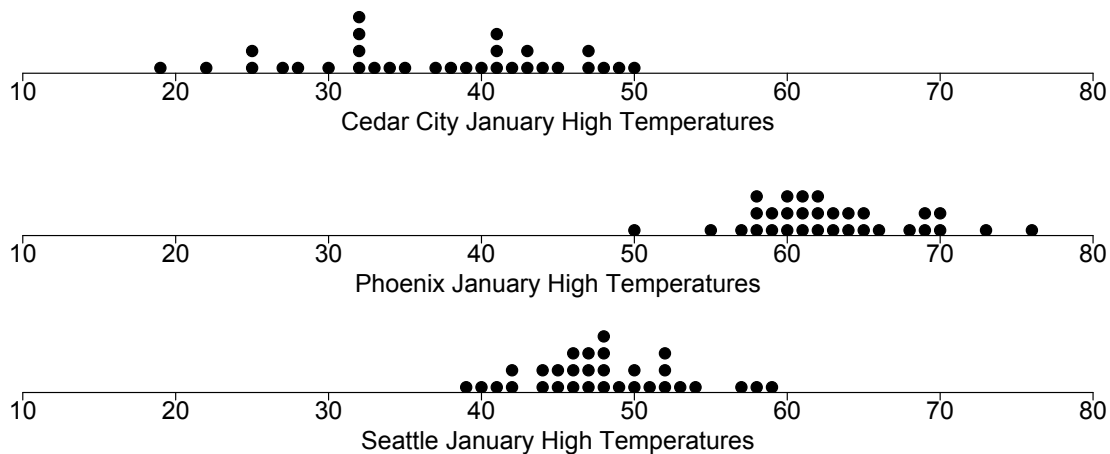
Keep in mind:

- Each dot in a dotplot represents a different observational unit.
 - Develop the habit of always labeling the axis with the name of the variable.
 - Dotplots can be cumbersome to create by hand with larger datasets, so you can often use technology to create the graph.
 - Dotplots pertain to a quantitative variable; bar graphs, to a categorical variable.
- e. Circle your own value on the dotplot you just created. Where do you seem to fall in relation to your peers with regard to number of states visited?
- f. Based on this display, comment on the accuracy of your guesses in part a.
- g. Write a paragraph describing various features of the distribution of number of states visited. Imagine that you are describing this distribution to someone who cannot see the display and has absolutely no idea how many states people visit. [Advice: Here and throughout the course, always relate your comments to the context. Remember that data are not just numbers, but numbers with a context, in this case, the number of states visited by you and your classmates.]

In Topic 1, we mentioned that classifying a variable as categorical or quantitative is extremely important because that classification helps determine the type of statistical analysis to perform. You have already begun to experience this because a bar graph is the simplest graphical display of a categorical variable and a dotplot is the simplest graphical display of a quantitative variable. In other words, by identifying the type of variable, you restrict your choice of graphical displays immediately.

Activity 2-4: January Temperatures

The following dotplots display the distributions of daily high temperatures for the month of January 2023 in Cedar City, Utah; Phoenix, Arizona; and Seattle, Washington.



- a. Which city tended to have the highest temperatures that month? Which city tended to have the lowest temperatures? Explain how you can tell.

Highest:

Lowest:

Explanation:

- b. Which city had the most day-to-day consistency in its high temperatures that month? Which city had the least? Explain how you can tell.

Most:

Least:

Explanation:

When comparing distributions of a quantitative variable, in addition to tendency, look for **consistency**. Tendency refers to the center of the distribution, consistency to the spread (or variability) in the distribution.

Topic 3

Drawing Conclusions from Studies

Lecture

We can divide statistics into two categories: _____.

_____ statistics involve describing a data set:

- We could make a graph of it.
- We can describe its average and how spread out it is.
- We could tell about any interesting features.

However, in descriptive statistics, we limit ourselves to describing the data itself. We do not generalize facts about the data set to a larger group.

If we generalize from a sample to a larger group, we are performing _____ statistics.

The larger group is called a _____. These sorts of generalizations are called inferences, and the investigator is said to “draw inference” on the population, using the **sample**. For example, if we want to learn something about the average height of trees in Rocky Mountain National Park, we can take the average of a sample of 100 trees.

Inferential statistics jargon

A **population** of items is the entire collection of items of interest to the investigator in his or her study. This is the large group we want to know about.

A **sample** is a subset of the population that we actually collect information from.

We have to be careful when generalizing from a sample to a population. The descriptive characteristics of your sample may not accurately reflect the characteristics of the population you are studying.

Two very important questions in statistical inference are:

1. Is my sample large enough?
2. Is my sample representative of the population of interest?

If our sample is small, we can't confidently make inference. For example, if I measure the height of three trees in RMNP, I shouldn't assume that their average height is a good guess for the true average height of all the trees. Maybe I happened to pick three short trees. Or three large trees.

If our sample is not representative of the population of interest, we may introduce _____. For example, if I sample only trees that grow at a high altitude, I shouldn't claim that my sample average is representative of all trees in RMNP, because trees at high altitude tend to be shorter than those at low altitude.

A **parameter** is a numeric characteristic pertaining to a _____. In our example, the parameter of interest is the average height of all trees in RMNP.

Most often, parameters cannot be determined because the entire population cannot be investigated. However, we can still draw inferences about the parameters with statistics. We just can't know what they are with certainty.

A **statistic** is a numeric characteristic pertaining to a _____. Basically, a statistic is anything that you calculate using data. We use statistics (which we calculate) to estimate parameters (which are unknown). In our example, if we use the average height of the trees in our sample to estimate the average height of all the trees in RMNP, then the sample average is our statistic. But remember, the statistic is just an estimate! We can't ever know the true parameter value unless we are able to measure the entire population.

The parameter and statistic will always match up.

If measurements are obtained from every unit in the population, then we have a **census**. Every 10 years, the federal government runs a census, and the goal is to collect information on every single person in the country. In typical investigations we don't get to observe every unit. We aren't going to actually measure every tree in RMNP! If we want to say something about the average height of these trees, we'll need to use _____, and there will always be uncertainty when performing _____.

Example 1:

Suppose CNN takes a poll where they ask viewers to call in and state whether or not they are happy in their marriage. 90% of the call-ins say they are unhappy in their marriage. Do we conclude that among the American population, 90% of all married people are unhappy in their marriage?

- This is an example of a **voluntary response survey**, where only those who are willing to respond do.
- Those who are happy in their marriage are less inclined to call.

Example 2:

The Miami police department wants to know if minority residents of Miami are satisfied with police service in their neighborhood. A questionnaire is prepared. 300 mailing addresses in these neighborhoods are randomly selected, and a uniformed police officer is sent to each address to interview an adult resident. How accurate will this study be?

- This is an example of **interviewer bias**.
- People are usually less apt to say negative things about the police to a uniformed police officer.

Example 3:

A survey was given to SUU students regarding their opinions on possible new businesses to open in the student center. The survey was administered to any student willing to fill it out. Would the responses received be accurate, or somehow biased?

- This is an example of a **convenience sample**.
- Data is collected by choosing people who are easy to interview.

In Example 1 and 3, there are large portions of the populations which were not interviewed.

- Those happy in their marriage.
- Those who do not frequent the student center.

This is a type of **nonresponse**. **Nonresponse bias** often occurs simply because someone refuses to participate or cannot be located.

Activities

Activity 3-1: Puerto Rico Statehood and Alf

Puerto Rico is an American territory and there have been several votes about adding it as the 51st U.S. state, all of which have failed. A famous polling agency called Gallup wants to learn the opinions of all adult Americans about adding Puerto Rico.

Of course they could not ask every adult American this question, since that would be too costly and time consuming, so they sent out a survey to 11,321 Americans and requested the people fill it out and mail it back (the person mailing it had to pay for postage). It turned out that approximately 66% of those that responded favored adding Puerto Rico as a state.

This scenario is very common in statistics: wanting to learn about a large group based on data from a smaller group.

Definitions

The **population** in a study refers to the entire group of people or objects (observational units) of interest. A **sample** is a (typically small) part of the population from whom or which data are gathered to learn about the population as a whole. If the sample is selected carefully, so that it is **representative** of (has similar characteristics to) the population, you can learn useful information. The number of observational units (people or objects) studied in a sample is the **sample size**.

- a. Identify the population and sample in this study about Puerto Rico.

Population:

Sample:

- b. Do you think that 66% accurately reflects the opinions of all Americans on this issue? If not, identify some of the flaws in the sampling method.

In 1936, *Literary Digest* magazine conducted the most extensive (to that date) public opinion poll in history. They mailed out questionnaires to over 10 million people whose names and addresses they had obtained from telephone books and vehicle registration lists. More than 2.4 million people responded, with 57% indicating they would vote for Republican Alf Landon in the upcoming presidential election. (Incumbent Democrat Franklin Roosevelt won the actual election, carrying 63% of the popular vote.)

- c. Identify the population of interest and the sample actually used to study that population in this poll.

Population:

Sample:

Definitions
In selecting a sample from a population, ideally we would start with a sampling frame , an actual list of every member of the population that we want to sample from. In other words, a sampling frame is everyone or everything that could possibly be sampled.

- d. What did the *Literary Digest* use for their sampling frame from which they selected the people to survey?
- e. Explain how *Literary Digest's* prediction could have been so much in error. In particular, comment on why its sampling method made it vulnerable to overestimating support for the Republican candidate.

In both the Puerto Rico study and the Literary Digest presidential election poll, the goal was to learn something about a very large population (all American adults and all American registered voters, respectively) by studying a sample. However, both studies used a poor method of selecting the sample from the population. In neither case was the sample representative of the population; so you could not accurately infer anything about the population of interest from the sample results. This is because the sampling methods used were biased.

Definitions
A sampling procedure is said to have sampling bias if it tends systematically to overrepresent certain segments of the population and to underrepresent others.

These scenarios also indicate some common problems that produce biased samples. Both are **convenience samples** to some extent because they reach those people most readily accessible (e.g., those with an address available to Gallup or listed in the phone book). Another problem is **voluntary response**, which refers to samples collected in such a way that members of the population decide for themselves whether or not to participate in the study. For instance, Gallup asked people to complete the survey if they wanted to participate. The related problem of **nonresponse** can arise even if an unbiased sampling method is used (e.g., those who are not home when the survey is conducted may have longer working hours than those who participate). Furthermore, the **sampling frame** (the list used to select the subjects) in the *Literary Digest* poll

was not representative of the population in 1936 because the wealthier segment of society was more likely to have vehicles and telephones, which overrepresented those who would vote Republican.

Definitions
A parameter is a number that describes a population, whereas a statistic is a number that describes a sample.

Note: To help keep this straight, notice that population and parameter start with the same letter, as do sample and statistic.

- f. Consider the students in your class as a sample from the population of all students at your school. If you want to estimate the proportion of students at your school who use Snapchat on a daily basis, what statistic would you calculate? (Describe it in words.)
- g. Consider the students in your class as a sample from the population of all students at your school. If you calculate the average number of hours that students in your class slept last night, describe (in words) the corresponding parameter of interest.
- h. For part g, why is it incorrect to state that the parameter is “all students at the school” or “how much students at the school slept last night”?
- i. What type of variable leads to a parameter or statistic that is a proportion? What type of variable leads to a parameter or statistic that is an average? [Hint: Review your answers to the last few questions and look for a pattern.]

Proportion:

Average:

Topic 4

Random Sampling

Lecture

We have discussed ways that sampling may lead to bias. Now we look at how we ideally should sample.

The simplest type of sample which avoids sampling bias is known as a **simple random sample** (SRS). A sample is an SRS if each individual in the population has an equal chance of being selected. If we sample any way besides randomly, there is a chance that bias can occur.

Suppose we want to estimate the proportion of SUU students, call it p , who are Utah residents. We can estimate p , the population proportion, by taking a simple random sample of 100 students at the university, and computing the sample proportion, $\hat{p}$, of students who are Utah residents. Suppose we find that 65 of the 100 students are Utah residents. Then the statistic $\hat{p} = 0.65$ is an estimate of the population proportion p .

If we took another sample of 100 students, would we obtain the same sample proportion of 0.65? The fact that we would get different estimates with different samples is known as **sampling variability**. If we take many repeated samples, the resulting statistics would make up the **sampling distribution**.

The **sample size** of a statistical sample is the number of observations that constitute it. We denote sample size with the letter n . Generally, the larger the sample size, the better the sample. By “better”, we mean the larger the sample size, the less variability there is in the sample statistic from sample to sample.

The **sample average** (a.k.a. mean) is easily computed. Simply add together all the observations and divide by the number of observations. We denote the mean using $\bar{x}$ or “x bar”. The sum of all the observations in a sample is denoted by Σx (the symbol “ Σ ” is called sigma, and it means to sum everything up.) So, our formula for the mean is

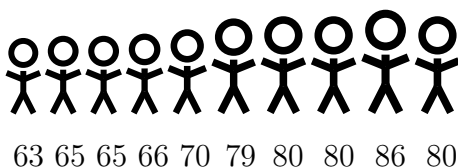
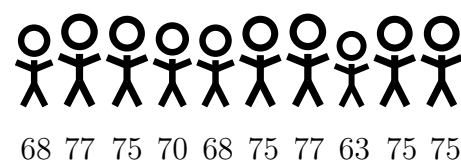
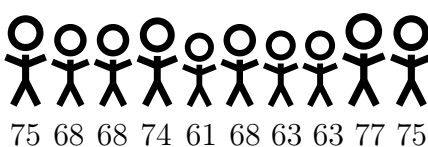
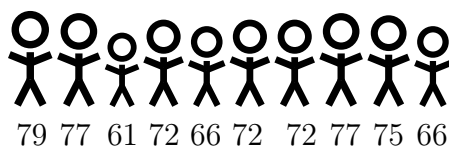
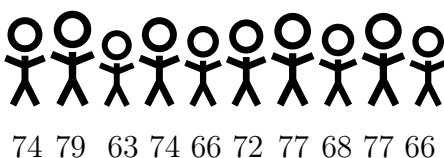
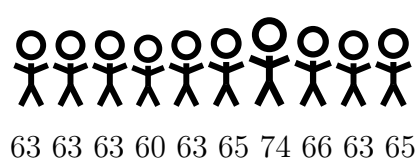
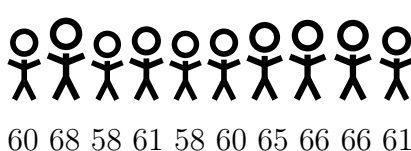
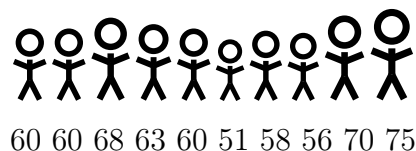
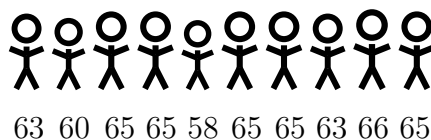
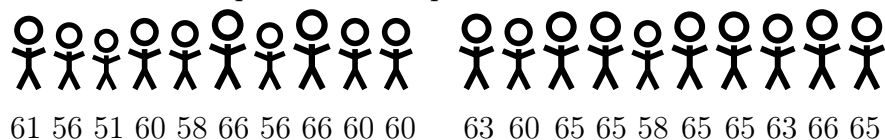
$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

Example: Suppose we are interested in the average number of hours per week that adults in the U.S. spend watching television. We take a sample of $n = 5$ adults and have them track the number of hours they spend watching TV in a week. (Note: this is a very small sample size.) The values are 42, 14, 26, 41, and 30. Then the sample mean is:

Activities

Activity 4-1: Sampling Heights

Below we have 100 people in 10 groups of 10 of various heights. Their heights are given in inches below the picture of the person.



Definitions

One way to avoid a biased sampling method is use **randomization**. The simplest way to do this is to give every member of the population the same chance of being selected for the sample. This kind of sampling is called **simple random sampling**.

- Let's randomly select the heights instead of picking them ourselves. In Excel (or Google Sheets), click on an empty cell and type "`=RANDBETWEEN(1, 100)`". That will generate a random integer between 1 and 100. Then click the square in the bottom right and drag down until you've dragged to at least 10 cells. If you get any repeats, then ignore it and click the square in the bottom right and drag down some more cells. Do this until you get 10 unique numbers. On the first page of this activity, put a box around the corresponding 10 people that were chosen randomly and write down their heights. For example, if you got the random number of 21, then you would put a box

around the first person in the second row with a height of 60 and put 60 in the table below.

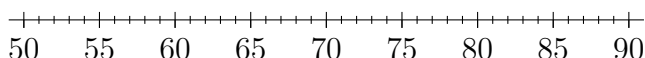
Random number:										
Height										

- b. Create a dotplot of your sample results (heights of those you selected). Label the axis and indicate what the observational units and variable are in this dotplot. Is the variable categorical or quantitative?

Observational units:

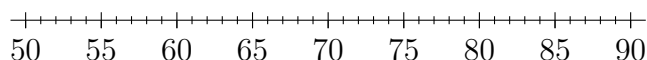
Variable:

Type:



- c. Calculate the average (mean) height in your sample. Is this number a parameter or a statistic? Also identify (in words) the other value. Then add your average to the dotplot on the white board.

- d. Combine your sample average with those of your classmates to produce a dotplot of sample averages. Note: this is no longer a dotplot of heights, it is now a dotplot of average heights. Make sure to label the axis.



- e. Indicate what the observational units and variable are in the dotplot in part d. [Hint: this is not the same as part b.]

Observational units:

Variable:

- f. The average height in the population of all 100 people is 67 inches (5 feet 7 inches).

Mark this value on the dotplot in part d. How many students produced a sample average greater than the actual population average? What proportion of the students does this represent? How many students produced a sample average less than the actual population average? What proportion of the students does this represent?

Greater

Less

- g. Would you say that this sampling method (simple random sampling) is a good sampling method? It is good if it is (roughly) centered at the true value with an (approximately) equal number of averages falling above and below the true value.

This activity shows that random samples (usually) lead to distributions of statistics that are centered at the true value. If we had more students in the class, then our dotplot would be even more centered at the true parameter value of 67 inches. As the number of samples goes to infinity, the distribution would be perfectly centered at 67.

Extra Space for Notes

Topic 5

Designing Experiments

Lecture

When we are designing experiments and studies, we want to be sure we know what it is we want to study or predict and what we are trying to use to predict it.

Response or Dependent Variable: the outcome variable (variable of interest) we are measuring on each subject or individual.

Explanatory or Independent Variable: the variable whose effect on the response variable we are interested in establishing. In the examples of Topic 5, the explanatory variable is categorical, and usually binary.

Example 1: Suppose we want to know if lung cancer and smoking are related. In this example, the response variable is whether an individual gets lung cancer. It's a binary variable (yes or no). The explanatory variable is smoking. It could either be measured as a binary variable (smoker or nonsmoker) or as a quantitative variable (number of cigarettes smoked per day).

Example 2: Does the size of a daycare facility determine how quickly a child develops linguistically. In this example, the response variable is some measure of language development (could be quantitative or categorical) and the explanatory variable is daycare size (likely categorical, and perhaps binary with sizes large and small).

When we talk about using data to learn something about the world, we are usually talking about performing a study or an experiment. These come in two different flavors: observational studies and experiments.

In **observational studies**, variable values are observed and recorded without the researchers getting directly involved. The researchers may ask questions and collect info, but they do not actively split participants into groups.

In a **controlled experiment** the researcher gets to control which members of the study go into which group. So, if we want to determine if a medical treatment is effective, we can _____ people to either receive the treatment (this is the treatment group) or not receive it (this is the control group), then use statistics to see if there is a noticeable difference between the two groups.

The big difference between an observational study and a controlled experiment is that the researcher can try to “_____” for anything that could bias the results by ran-

domizing who goes into which group. For example, if I want to determine whether a weight loss pill is effective, I could use the following observational method: I could get a sample of people who have used the pill and people who have not, and record how much each person's weight has changed in the last 6 months.

Let's suppose I observe that the people who took the pill lost more weight than the people who didn't. Is this solid evidence that the pill works?

It could be that the people who are taking this pill are more motivated to lose weight, and are eating a healthier diet and exercising more than the people who are not on the weight loss pill. In this case, the observed difference in weight loss might not be due to the pill, but to other lifestyle factors.

What should we do instead? To control for these lifestyle factors, we can:

- Randomly sample a group of people (ideally).
- Randomly assign some people to take the pill and the others to not take it.
- Make sure each group had similar diet and exercise.

In a controlled experiment, we can be reasonably comfortable inferring that the treatment applied to the experimental group caused whatever difference we see between the two. But otherwise we must remember that...

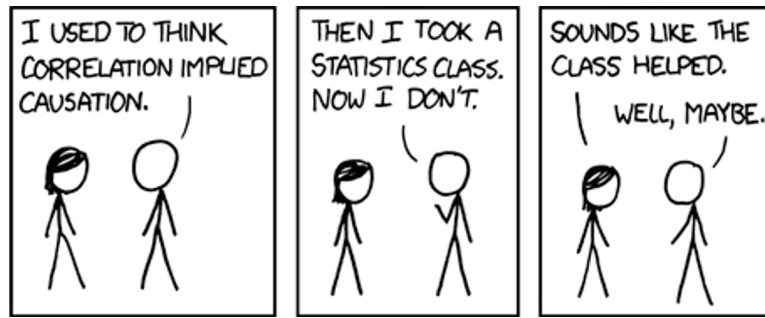
Correlation does not imply _____! If variables are correlated then as one changes, so does the other. This correlation can be positive (as one gets bigger, so does the other) or negative (as one gets bigger, the other gets smaller).

A **confounding variable** is a variable that helps explain the data but is not accounted for in the study. For example: Did you know that the number of shark attacks is directly correlated to the number of ice cream cones bought?

Can you think of another variable that might be related to ice cream sales and shark attacks?

Another example: The size of a child's vocabulary is correlated with that the # of cavities he or she has. Learning how to speak causes cavities!

Another example: This is something to really look out for in politics. It is often implied that, because a politician was in office when something important happened, the politician caused the important thing to happen. Well, maybe and maybe not. If you want to decide whether you believe a politician caused something to happen, you'll need more than statistical correlation in order to make an informed judgment.



Blinding and double blinding are methods that are used to try and eliminate bias that may be produced by the _____ and _____ themselves. It is typically used in experiments that involve human behavior, such as psychological experiments. To motivate this concept, consider the "Pepsi challenge". This is a taste test that was used by the Pepsi corporation, in which participants were asked to compare a cup of Pepsi to a cup of Coca-Cola.

If the participants are handed cups labeled "Pepsi" and "Coke", then their responses may be biased in that they may have preconceived notions of which product they prefer. So, in order to eliminate this potential source of bias, the study should be blinded. A blinded study is one in which the participant does not know which group (a.k.a. treatment) is which. In the context of a taste testing, the study is blinded if the cups are not labeled and the participant does not know which cup contains which product.

However, even if the study is blinded, there could still be a problem. If the experimenter knows which product is in which cup, then he or she could subtly influence the outcome of the experiment. This may or may not be intentional, but to be on the safe side, it is best to make the study double blinded. A double blinded study is one in which neither the participant nor the researcher know which product is which. In our example, the experimenter should not know which cup contains which product.

Activities

Activity 5-1: Acupuncture

Is acupuncture, a technique in which practitioners insert fine needles into the skin, an effective treatment for chronic pain?

- If a friend of yours who receives acupuncture therapy once a week has less pain than another friend who does not, is that compelling evidence that acupuncture works? Why or why not?

Definitions

Anecdotal evidence is based on stories or hearsay rather than scientific research or statistical analysis. Much of the practice of statistics involves designing studies and collecting data so that people do not have to rely on anecdotal evidence.

- b. Now suppose you take a random sample of individuals who receive acupuncture and those that don't. Then compare their pain levels. Identify the explanatory and response variables in this study. Also classify each as categorical (also binary?) or quantitative.

Explanatory:

Type:

Response:

Type:

- c. If the acupuncture group was in less pain, is that compelling evidence that acupuncture works? Why or why not?

The problem with this study, as with all observational studies, is that you do not know whether explanatory variable is the reason the pain levels differ. For example, subjects who choose to go get acupuncture may have an entire health regimen that differs from the people who do not get acupuncture. Or it could be that those who are in more pain go to acupuncture.

When investigating whether one variable causes an effect on another, researchers create a comparison group and assign subjects to the explanatory variable groups.

Definitions

An **experiment** is a study in which the experimenter is directly involved in assigning a **treatment** (explanatory variable group) on the subjects. Ideally, the groups of subjects are similar in all respects other than the explanatory variable so that the researcher can then see the explanatory variable's direct effects on the response variable.

We use **random assignment** to balance out potentially confounding variables in the two groups.

- d. Suppose that instead of observing whether a person goes to get acupuncture or not, you decide to conduct an experiment in which you randomly assign 50 people to get acupuncture and 50 people to not get acupuncture. How is that an improvement over the original design of observing who goes to acupuncture and who doesn't?
- e. Can you identify a flaw in this design that would still prevent researchers from concluding that acupuncture causes a reduction in pain level, even if its group experiences much more pain reduction than the no-treatment group? [Hint: Identify a confounding variable that exists even in this randomized comparative experiment!]

Definitions

The problem here is that the very act of receiving any treatment can have an impact on pain level. This is known as the **placebo effect**. Experimenters can control for this by administering a **placebo** (a fake treatment) to subjects in a **control group** or by comparing a new treatment against a standard treatment. It would be best if subjects are **blind** as to which treatment they actually receive or else that knowledge could affect their responses. When possible, experiments should not know which treatment is which either. That would make the experiment **double-blind**. This can control for implicit biases.

- f. With that in mind, suggest another, even better design to test if acupuncture is helpful in reducing pain level.

Activity 5-2: Memorizing Letters

You were asked to memorize letters for an experiment I conducted. The first group received the letters in this grouping:

USA-CAN-ADA-MEX-ICO-FRA-NCE-AUS-TRA-LIA

and the second received them in this grouping

USAC-ANAD-AMEX-ICOF-RANC-EAU-STRALIA

Each group had 10 seconds to memorize as many letters as possible. In experiments like this, it has been found that arranging letters in chunks that are more familiar leads to more letters memorized.

- a. Explain what makes this an experiment instead of an observational study.

- b. Identify the explanatory and response variables here and classify them.
Explanatory: _____ Type: _____

Response: _____ Type: _____

- c. Explain how random assignment was implemented and why it was important in this study.

- d. Explain how blindness was implemented and why it was important in this study.

- e. Was this study double-blind? Explain.

- f. Create dotplots of the memory scores, comparing the two treatment groups. Label each plot.



- g. Comment on whether these experimental data appear to support the hypothesis those that got the first grouping would memorize more letters.
- h. If the USA group does substantially better than the USAC group, could you reasonably conclude that the grouping of letters caused the higher scores? What if someone were to argue that it may just be that those with better memories were in the USA group? How would you respond to them?

Extra Space for Notes

Chapter 1 Summary

- **Statistics** is the science of data.
- **Data** is information.
- **Observational units** or cases are the actual subjects from whom we are collecting information.
- A **variable** is a characteristic of a subject that can be assigned a number or category. It must be collectible on each observational unit.
- Variables are either **quantitative** or **categorical**.
 - Quantitative variables are numeric. Taking an average makes sense. These can be **continuous** if the variable can take on decimal values or **discrete** if it can only be a whole number.
 - Categorical (or qualitative) variables do not have to be numbers. Taking an average does not make sense. These can be **binary** if it only has two categories or **non-binary** if it has more than two categories.
- The pattern of variation of a variable is known as its **distribution**.
- A **proportion** simply measures a part in comparative relation to the whole. It is like a percentage, but written as a decimal.
- Statistics can be divided into **descriptive statistics** and **inferential statistics**.
- **Observational studies** do not have researchers involved in assigning groups.
 - Descriptive statistics involves simply describing a dataset.
 - Inferential statistics involves generalizing to a larger group.
- The **population** is the large group we want to know something about. Populations are described by **parameters**.
- The **sample** is the smaller group we actually collect information from. Samples are described by **statistics**. Samples should be selected **randomly** to avoid being **biased**.
- **Bias** occurs when the sample is not representative of the population.
- The **response variable** is the one we want to predict.
- The **explanatory variable** is the one we use to predict the response variable.
- **Observational studies**: researchers are not involved in assigning groups.
- **Controlled experiments**: researchers are directly involved in assigning groups.
- A **confounding variable** is one that helps explain the association between variables but is not accounted for in the study.
- A **blinded study** is one where the participants don't know which group they are in. A **double-blinded** study is one where neither the participants nor the researcher knows which group the participants are in.

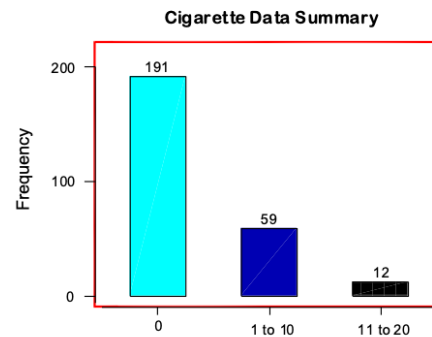
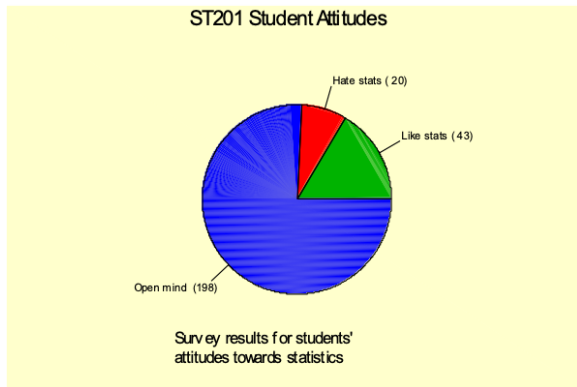
Chapter 2: Descriptive Statistics

Topic 6

Displaying Distributions

Lecture

We will look at pie charts, bar charts, histograms, and box plots.



All of the graphs we will look at show _____ of data. Often this is shortened to just distribution.

A distribution tells you the values your data take on, and the frequency with which those values are taken on.

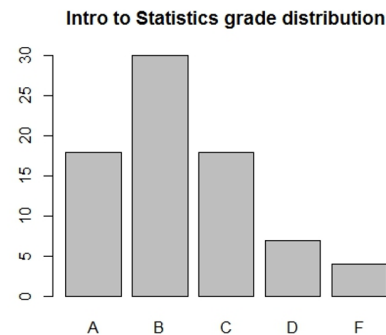
So, if we are looking at the distribution of grades in Math 1040, I could show you the data like this:

B B B B B B A B D F A C B D B C A B B B B C B C C D B F C C A C A C C F D C
D B C A D A B B A B A C B C A D A B C A C B B B C B B B F A B A B A A B A A
C

This is raw data. Or, it can be shown like this:

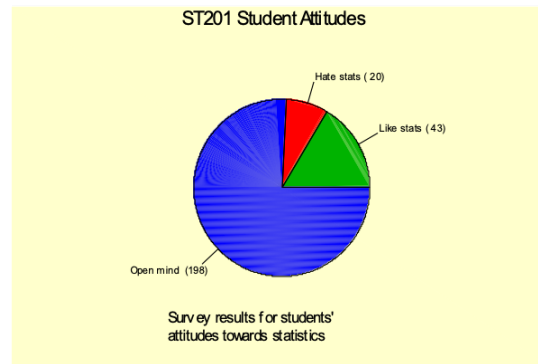
Grade	A	B	C	D	F
# of students	18	30	18	7	4

This is a frequency distribution, because it tells you the different values that the variable "grade" takes on, as well as how often it takes each value on. Or I could show it to you like this:

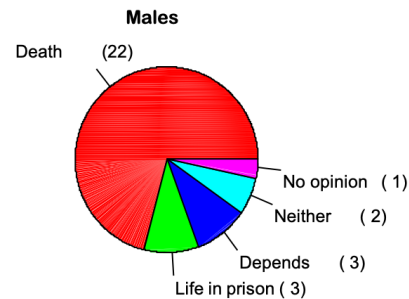
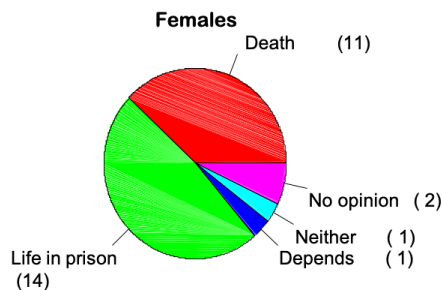


Pie charts

- Can be used to summarize one qualitative variable
- Pie slices represent the proportion of observations in a class
- Sometimes frequency results are also included
- The more categories you have, the more difficult the pie chart will be to read.

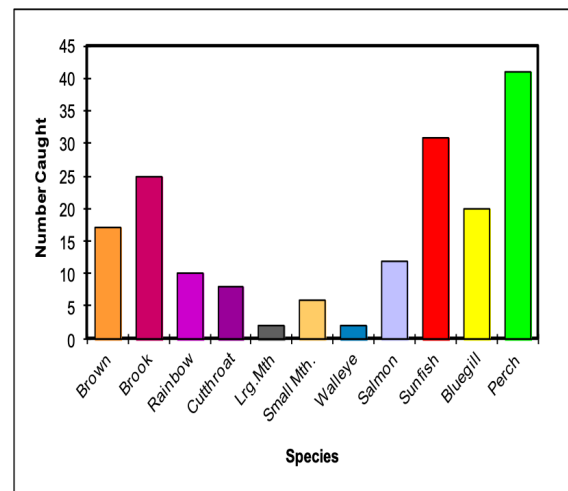


Multiple pie charts can be used to compare two different groups. Here, the pie charts compare attitudes of females and males toward the appropriate punishment for murder.



Bar charts

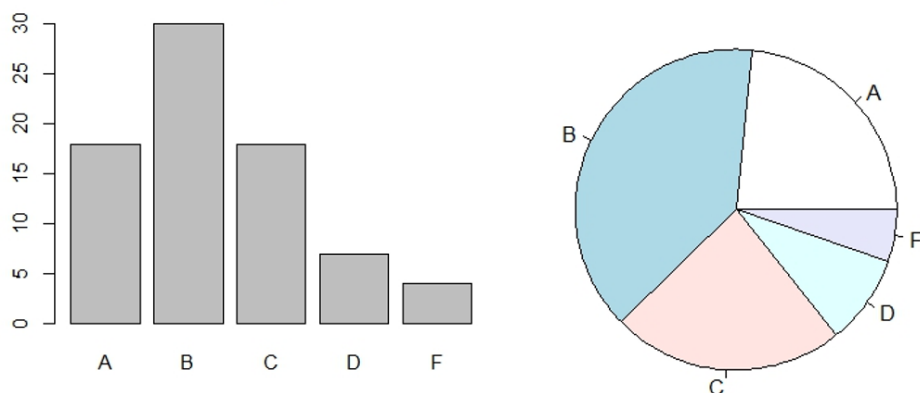
- Bar charts can be used wherever pie charts are used
- Like pie charts, they are used to display qualitative or categorical data
- Each bar in a bar chart shows you the frequency (or count) for the group it is associated with.



This chart shows the frequency of catches for different species of fish.

Bar charts vs pie charts

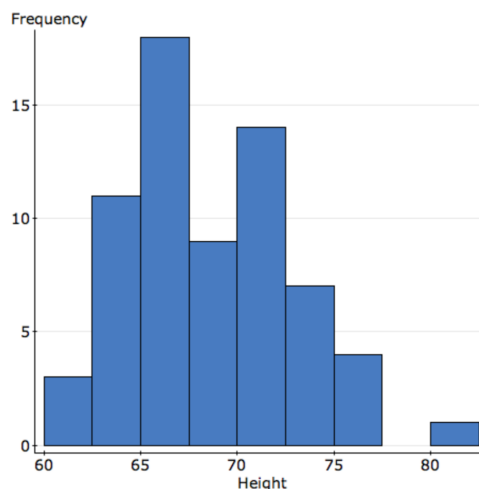
Here is the intro stats grade distribution bar chart from before, alongside a pie chart of the same data.



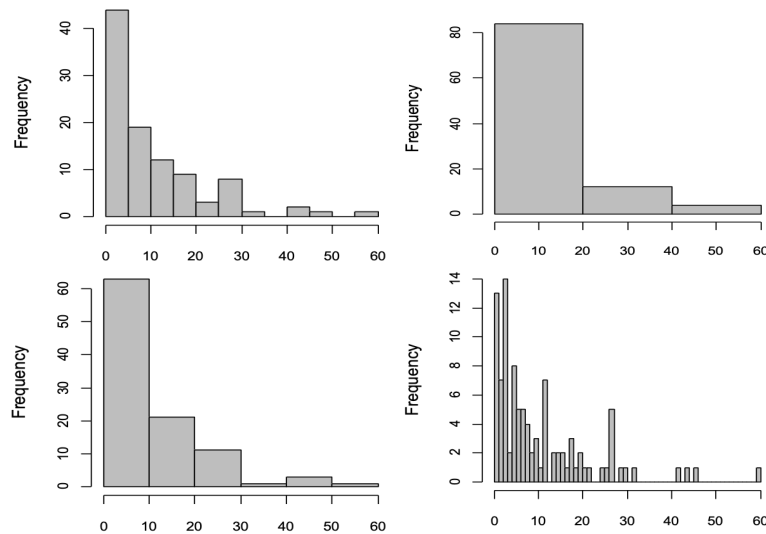
Which one makes comparisons easier?

Histograms

- A histogram is used to show the distribution of _____.
- The difference between a histogram and a bar chart is that bar charts are for qualitative data and histograms are for quantitative data.
- With bar charts, each bar represents a different distinct group. With histograms, each bar represents the number of observations which fall into an interval, also known as a _____.



Note that the number of bins on a histogram is arbitrary, but each binwidth must be equal. Using different numbers of bins can give different looking histograms, even if the underlying data are exactly the same. The following four histograms represent the exact same data:



Which histogram looks best?

Note: If an observation falls directly on a bin endpoint, it is typical to place that observation in the bar to the left of its value. But this is not a hard and fast rule.

Our goal will be to fully describe distributions of variables. For categorical variables, that will involve giving proportions or percentages. For quantitative variables, it involves mentioning all of the following:

Important Characteristics of a quantitative data set:

1. Center: Where is the data set centered along the number line?
2. Spread: (variation): How spread out – dispersed – are the observations?
3. Shape: What is the shape of the distribution of values in the data set?
4. Outliers: Are there any unusual values or features in the data set?

We will go over all of these in some detail.

Activities

Activity 6-1: Cumulative and Relative Frequency

Definitions

The **cumulative frequency** for a category is the sum total of that category and all of the previous ones.

- a. Using the table below, fill in the remaining cumulative frequencies.

Grade	A	B	C	D	F
# of students	18	30	18	7	4
Cumulative frequency	18	48			

- b. What exactly does the cumulative frequency number for the "C" category tell us?

Definitions

Relative frequency gives the proportion of the total that falls in a category. You can find this by dividing the frequency in each category by the total. **Cumulative relative frequency:** The sum of the relative frequencies for each category and all of the previous ones.

- c. Fill in the relative frequency and cumulative relative frequency in the table below.

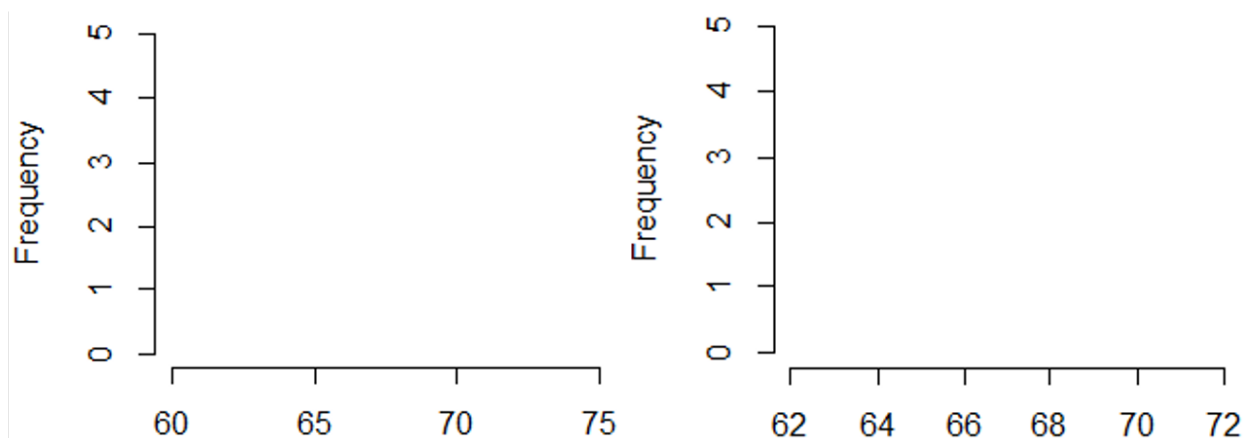
Grade	A	B	C	D	F	Total
# of students	18	30	18	7	4	77
Relative frequency						
Cumulative relative frequency						

- d. What percentage of all students in the class passed the class assuming a grade of a D or F do not count as passing?

Activity 6-2: Creating Histograms

- a. In this table, we have the heights of 10 randomly selected statistics students. By counting the number that fall in each interval, create two histograms: one with a bin width of 5 and the other with a bin width of 2.

Heights	65	67	66	69	66	64	64	63	72	63
---------	----	----	----	----	----	----	----	----	----	----



- b. Which bin size seems like it shows the shape of the distribution better?
- c. Now create a histogram for these data using relative frequency on the y -axis instead of frequency. You will need to create your own y -axis.



Topic 7

Location Statistics, Measures of Center, and Distribution Shape

Lecture

Location Statistics – Measures of Center

(Sample) Mean: the average of the data. For a finite sample of data values $x_1, \dots, x_n$, the sample mean is given by:

$$\bar{x} = \frac{1}{n}(x_1 + \dots + x_n) = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \sum x_i.$$

Example: Consider the heights of students again:

Heights	65	67	66	69	66	64	64	63	72	63
---------	----	----	----	----	----	----	----	----	----	----

1. What is the sample mean?

$$\bar{x} = \frac{65 + 67 + 66 + 69 + 66 + 64 + 64 + 63 + 72 + 63}{10} = 65.9$$

2. What is this mean estimating?

Weighted Average

Consider the following transcript from a student's semester:

How can we find the student's grade point average (GPA)?

Class	Credits	Grade
History of Something	3	A
Introduction to Basic Understanding	4	B
Education 101	3	A
Advanced Dungeons and Dragons	3	C
Learning!	1	B

Generalizing this to finding weighted averages:

1. Find the weights for each observation.
2. Multiply each observation by its weight and add them:
3. Calculate the weighted mean by dividing

(Sample) Median: middle or center of the distribution found by obtaining a number such that 1/2 the values are above, and 1/2 below.

To find the sample median M :

1. Order the data from smallest to largest.
2. Find the index of the median value:
 - Index = $\frac{n+1}{2}$
 - If the index is an integer, then the median is that ordered value in the dataset.
 - If the index is a decimal, the median is the average of the ordered values “around” that index.
3. Write the median.

Example: (n odd), Consider the following table that contains the average number of hours a sample of adults spent watching TV in a week:

Hours	42	14	26	41	30
-------	----	----	----	----	----

Example: (n even). Consider the random heights again that have now been sorted:

Heights	63	63	64	64	65	66	66	67	69	72
---------	----	----	----	----	----	----	----	----	----	----

When should we use the median? The mean?

Location Statistics – Quartiles, Percentiles, and Extremes

Just as the median breaks the data set into two halves, quartiles break the dataset into 4 quarters.

- The lower quartile, Q_1 , is the “median” of all the data below the overall median. Alternatively, we can also find the index of this quartile: $\text{Index} = \frac{n+1}{4}$.
- The upper quartile, Q_3 , is the “median” of all the data above the overall median. $\text{Index} = \frac{3(n+1)}{4}$.
- If the index has a “0.5” in it, take the average of the surrounding ordered values.
- Otherwise, round the index to the nearest integer.

In general, we can find the index of any percentile in our data set with the following formula:

$$\text{Index} = \frac{p(n+1)}{100}$$

Where p is the percentile: the percent of the data that falls below the observation in question.
What percentile is the median? The mean?

We are also often interested in the extremes of a dataset.

These extreme values are referred to as the _____ and the _____.
“Extreme” in this context doesn’t necessarily mean “really big” or “really small”. It just means “the biggest” or “the smallest”.

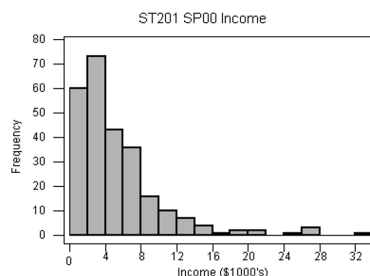
The 5-number summary is often used to summarize a dataset.

This group consists of the minimum, Q_1 , median, Q_3 , and maximum

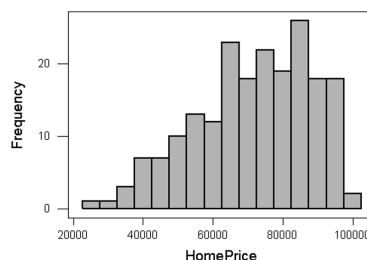
Distribution Shape

The shape of a distribution is best seen using a graphical display such as a histogram.

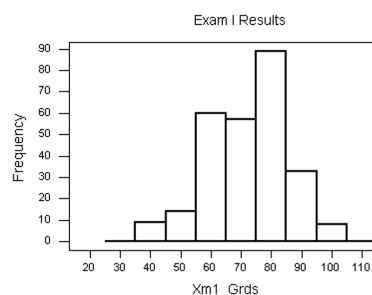
When there are lots of low values and just a few high values the distribution is said to be skewed to the _____, or positively skewed.



When there are lots of high values and just a few low values the distribution is said to be skewed to the _____, or negatively skewed.

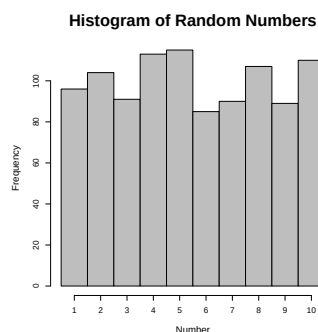


When the two halves of the histogram look approximately like mirror images, the distribution is said to be (almost) _____.



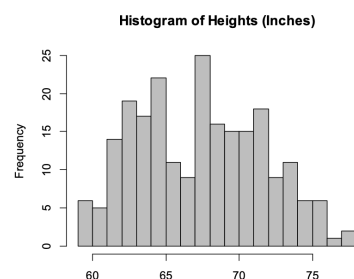
We say "almost" symmetrical because it is rare for a histogram of data to be perfectly symmetrical.

When the height of each bar in the histogram is nearly the same, the distribution is said to be (almost) _____.



Note that uniform distributions are also (almost) symmetrical.

When there are two peaks in a histogram, we say that the distribution is _____.



If there is only one peak, it is called **unimodal**.

Often, bimodality indicates that there are two distinct groups being combined into one dataset.

Activities

Activity 7-1: Calculating Grades

In a certain class similar to the one you are in, the final grade is weighted on 6 grade categories. Homework (10%), attendance (5%), in-class activities (10%), midterm exams (45% at 15% each), the final exam (18%), and the final project (12%).

- If a student gets a 90% on the homework, has perfect attendance, completed all in-class activities, got a score of 76%, 84%, and 82% on the midterm exams, got a score of 79% on the final, and a score of 95% on the final project, what overall score will they get in the class?
- If someone has scores of 90%, 88%, 95%, 92%, 98%, and 100% on the first 6 homework assignments, what score does the student need to get on the last homework assignment (there are seven total) to get a score of at least 9% out of 10% in the homework category?

Activity 7-2: Quartiles

Definitions

The **lower quartile** (or the 25th percentile, or first quartile, abbreviated Q_1) is the value such that 25% of the observations fall below it and 75% above it. The lower quartile is the median of the lower half of the dataset and the index of Q_1 is $(n+1)/4$. The **upper quartile** (or 75th percentile, or third quartile, abbreviated Q_3) is the value such that 75% of the observations fall below it and 25% above it. The upper quartile is the median of the upper half of the dataset and the index of Q_3 is $3(n+1)/4$.

Consider again the small data set given in this table:

Heights	63	63	64	64	65	66	66	67	69	72
---------	----	----	----	----	----	----	----	----	----	----

- a. Find Q_1 and Q_3 by finding the medians of the lower and upper half of the data set, respectively.

- b. Find Q_1 and Q_3 by using the index of each, as is outlined on page 41 of the notes.

Now consider this much larger dataset of size $n = 284$. Not all observations are given, but they are arranged from least to greatest.

Observation	1	2	...	71	72	73	...	142	143	...	213	214	215	...	283	284
Value	51	57	...	71	74	76	...	82	84	...	88	91	97	...	104	104

- c. Report the 5-number summary of this dataset.

Keep in mind: Excel has a couple different functions for finding quartiles. `=QUARTILE.EXC` and `=QUARTILE.INC`. Usually in this class, `=QUARTILE.EXC` is preferred, but you can use either. These use slightly different algorithms for finding the quartiles, so the results may not match exactly with how you did it by hand in this activity.

Activity 7-3: Questionable Conclusions

- A company executive concludes that an accountant must have made a mistake because she prepared a report stating that 90% of the company's employees earn less than the mean salary. Show that the accountant could be right by giving an example with 10 data values illustrating that it is possible for 90% of the employees to earn less than the mean salary.
- A businesswoman calculates that the median cost of the seven business trips she took in a month is \$500 and concludes that the total cost must have been \$3500. Show that she can be wrong by giving an example with seven data values illustrating that it is possible for the median to be \$500 but the total is not \$3500.
- Someone once moved from California to Utah and was told that the mean IQ dropped in both states. Explain how this could happen.

Extra Space for Notes

Topic 8

Measures of Spread

Lecture

The 3 most common measures of spread are:

1. Range
2. Interquartile Range (IQR) (associated with the median)
3. Standard Deviation (associated with the mean)

Range = maximum – minimum value in the data.

- The range is simple to compute, but is heavily influenced by (sensitive to) outliers.

Then the **Interquartile Range (IQR)** is the range between the 25th and 75th percentiles and can be computed as: $IQR = Q_3 - Q_1$. So the middle 50% of the data lie over a distance given by the IQR. In this way, outliers do not play a role in computing the IQR measure of variability.

Standard Deviation

- For data $x_1, \dots, x_n$, the standard deviation is based on looking at all differences or deviations from the sample mean: $x_i - \bar{x}$, $i = 1, \dots, n$.
- Square these deviations, sum them, and divide by $n - 1$ to get the **sample variance** s^2 :

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

- The square root of s^2 is the **sample standard deviation**: $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$.

Why do we divide by $(n - 1)$ instead of n ? Answer: Since the deviations always sum to zero, if we know all but one, we can determine the last one, so that only $n - 1$ deviations are free to vary.

Degrees of freedom: the number of deviations free to vary. (For the standard deviation, the degrees of freedom (df) = $n - 1$).

Example to Calculate s : Data: 36, 29, 13, 24, 18. These data might be the weights of dogs in lbs. $\bar{x} = 24, n = 5$. To compute s directly:

- The sample standard deviation has the same units as the original data.
- Important: The standard deviation s cannot be negative! When is the SD = 0?

Standard Deviation Interpretation

The standard deviation can be interpreted (a bit loosely) as

Quick Aside – Population Parameters

While $\bar{x}$, s , and n denote the sample mean, sample standard deviation, and sample size, respectively, we have separate symbols that are used when we are describing a population.

The population size is usually denoted by N .

The population mean is $\mu = \frac{\sum_{i=1}^N x_i}{N}$.

The population standard deviation is $\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \mu)^2}{N}}$.

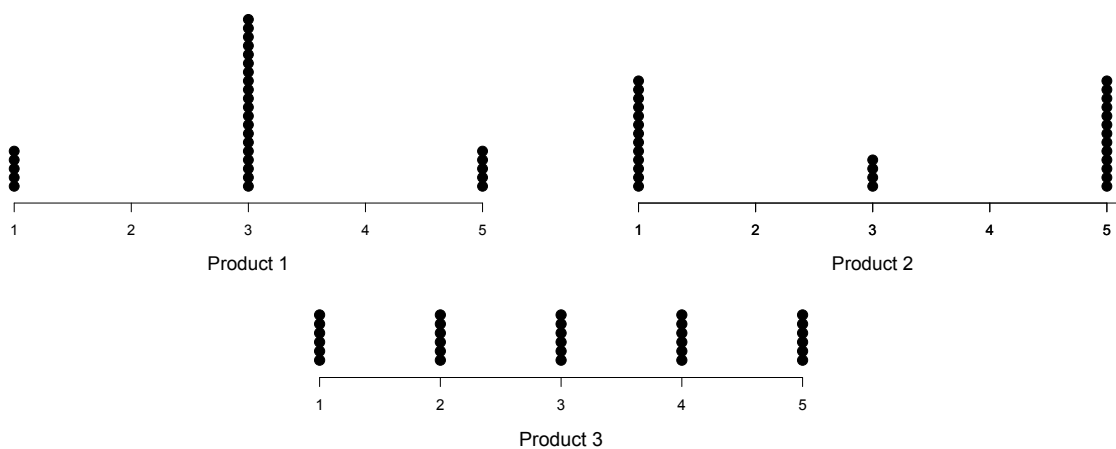
We usually do not calculate these since we rarely know all of the population information.

Activities

Activity 8-1: Product Ratings

Below we have dotplots for the ratings of 3 products from 1 to 5. Each product has 30 ratings. The raw data for each are also given in the following table:

Product 1	1	1	1	1	1	3	3	3	3	3	3	3	3	3
	3	3	3	3	3	3	3	3	3	3	5	5	5	5
Product 2	1	1	1	1	1	1	1	1	1	1	1	1	3	3
	3	3	5	5	5	5	5	5	5	5	5	5	5	5
Product 3	1	1	1	1	1	1	2	2	2	2	2	2	3	3
	3	3	3	4	4	4	4	4	4	5	5	5	5	5



- What are the means and medians of each of the three products' ratings?
- How would you describe the shapes of each distribution? Make sure to mention skewness and number of peaks.

- c. Before calculating any measures of spread, which of the three distributions would you say has the largest spread? How about the smallest? There is no wrong answer here since this is just based on your intuition, but explain your reasoning briefly.

Definitions
A very simple, but not particularly useful, measure of variability is the range, calculated as the difference between the maximum and minimum values in a dataset. $\text{Range} = \max - \min$

- d. Give the range of each of the distributions.

Definitions
Another measure of variability is the interquartile range (IQR), which is the difference between the upper quartile and the lower quartile of a distribution. Thus, the IQR is the length of the middle 50% of the data. $IQR = Q_3 - Q_1$

- e. Find Q_1 and Q_3 for each distribution.

- f. Give the IQR for each distribution. According to the IQR, which produce has the ratings that are most spread out? How about the least?

Definitions

The **standard deviation** (often appreciated SD) is the most widely used measure of spread for quantitative data. It is the square root of the **variance**, which is interpreted as the average squared deviation from the mean. A **deviation** is the difference between a data value and the mean of all the data values. The standard deviation is interpreted as (approximately) the average distances the data values fall from their mean. The sample standard deviation is called s :

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

- g. Since Product 1 has so 5 values of 1, 20 values of 3, and 5 values of 5, we can speed up finding the standard deviation using multiplication. For example, $(1 - 3)^2 + (1 - 3)^2 + (1 - 3)^2 + (1 - 3)^2 + (1 - 3)^2$ is the same as $5 \cdot (1 - 3)^2$. So, we can find its standard deviation using:

$$s_1 = \sqrt{\frac{5 \cdot (1 - 3)^2 + 20 \cdot (3 - 3)^2 + 5 \cdot (5 - 3)^2}{30 - 1}}$$

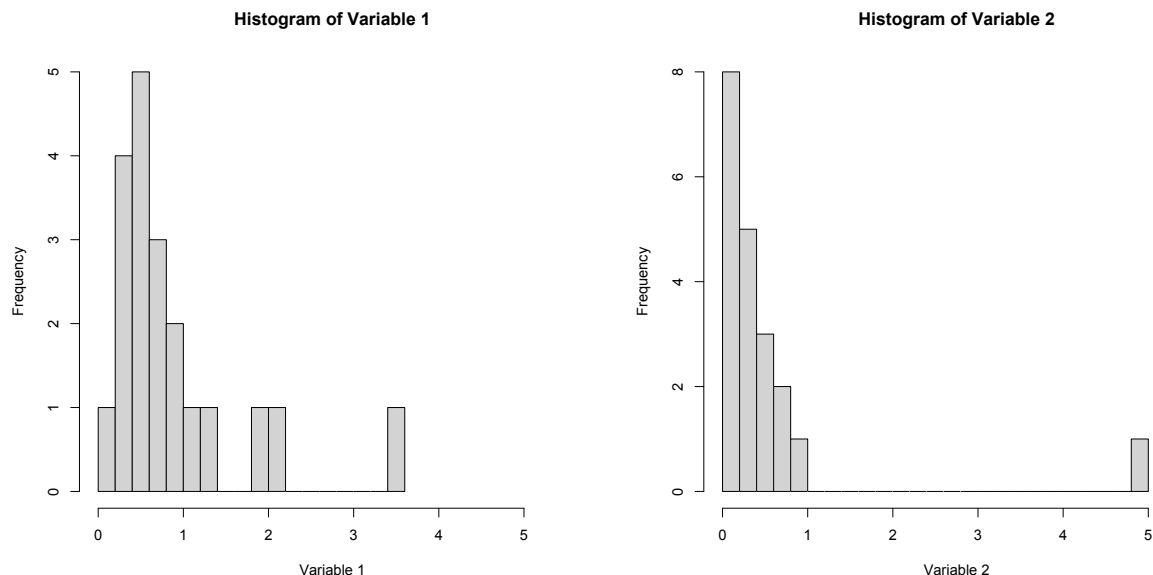
Calculate the standard deviation of Product 1 using the above formula.

- h. Use the `=STDEV.S` (the **S** at the end stands for sample) or the `=STDEV` function in Excel to compute the standard deviation for Products 2 and 3. Report them here.

- i. According to the standard deviation, the ratings of which product are the most spread out? Does that agree with the IQR? How about the range?

Activity 8-2: Skewed Distributions

Consider the histograms for two variables:



- a. Looking at those histograms and before considering the values of the measures of spread given below, which variable would you say is more spread out? Explain.

Some summary statistics for the two distributions:

Variable 1: IQR= 0.558 and $s = 0.820$.

Variable 2: IQR= 0.338 and $s = 1.080$.

The first variable has a larger IQR, but the second variable has a larger standard deviation. Most people would argue that the first distribution tends to be less consistent than the second. This is why the IQR is preferred over the standard deviation when the distribution is skewed.

Topic 9

Z-Scores and the Empirical Rule

Lecture

Standard Scores

The **standard score** or **z-score** measures the position of an observation in terms of the number of standard deviations from the mean. It is calculated using

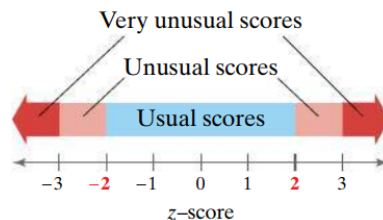
$$z = \frac{x - \mu}{\sigma}.$$

If the population mean and standard deviation are unknown, we can substitute the sample mean and standard deviation:

$$z = \frac{x - \bar{x}}{s}.$$

Interpretation

- The sign of the z -score indicates whether the data point lies _____ the mean.
- Positive z -scores mean that the observation is above the mean and negative z -scores mean that the point is less than the mean.
- The magnitude of the z -score indicates how far away the data point is from the mean in terms of standard deviations. We sometimes say that z -scores are “unitless”, but really their units are # of standard deviations from the mean.



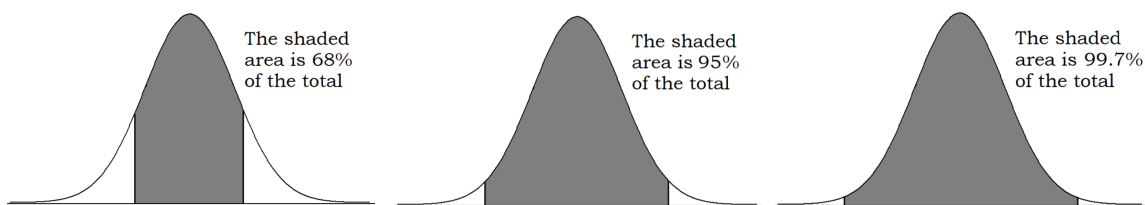
The z -score is especially useful in comparing values that come from different populations since we are standardizing the values into common units.

Example: Two common university admittance tests are the Scholastic Assessment Test (SAT) and the American College Testing (ACT) test. The SAT has a mean of 1,060 and a standard deviation of 210 while the ACT has a mean of 21 and a standard deviation of 5.8. If Andy scored a 1,220 on the SAT and Oscar scored a 26 on the ACT, who scored better relative to the other test takers?

Empirical Rule

If a distribution is fairly bell-shaped (or mound-shaped) and roughly symmetric, then:

1. Approximately 68% of the data fall within 1 standard deviation of the mean.
(i.e.: $(\mu - \sigma, \mu + \sigma)$ contains $\approx 68\%$ of the data).
2. Approximately 95% of the data fall within 2 standard deviations of the mean.
(i.e.: $(\mu - 2\sigma, \mu + 2\sigma)$ contains $\approx 95\%$ of the data).
3. Nearly all (99.7%) of the data fall within 3 standard deviations of the mean.

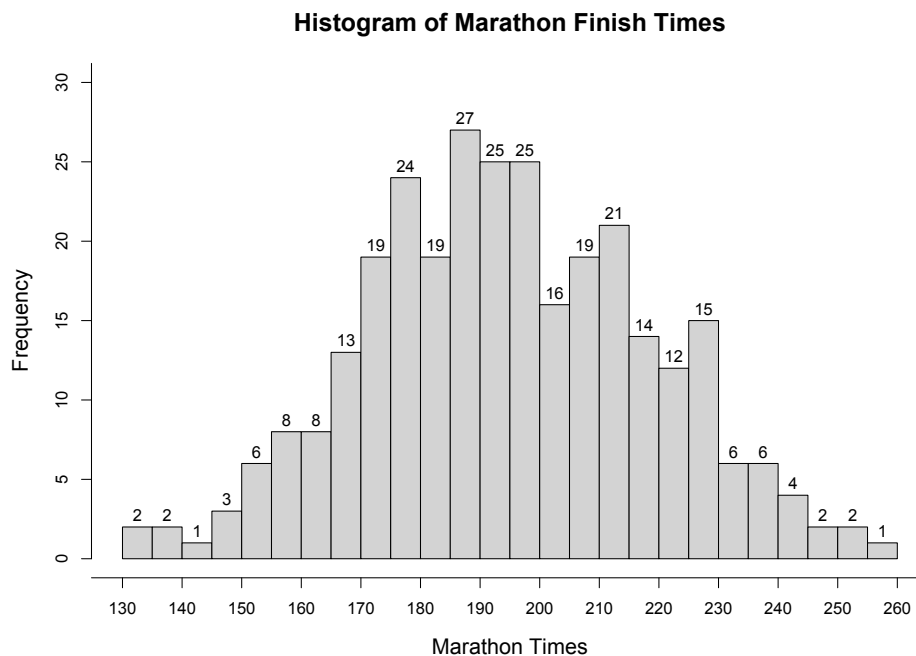


Example: Suppose we know that the distribution of heights of adult females in the U.S. is bell-shaped with a mean of 64 inches and a corresponding standard deviation of 2.75 inches. Using the empirical rule, we can say that about 99.7% of all U.S. females are between what two heights?

Activities

Activity 9-1: Marathon Times

A marathon is a 26.2 mile race that typically takes several hours to complete. The histogram below shows the times it took marathon runners to complete the marathon in which they ran. The average time is 195 minutes with a standard deviation of 25 minutes.



- a. How would you describe the shape of this distribution? Is this the kind of shape for which the Empirical Rule applies?

- b. Approximately what percent of the data, according to the Empirical Rule, should fall within 1, 2, and 3 standard deviations of the mean?

Within 1 standard deviation:

Within 2 standard deviations:

Within 3 standard deviations:

- c. Find the cutoffs for the values that are within 1 standard deviation, 2 standard deviations, and 3 standard deviations.

Lower bound for within 1 SD:

Upper bound for within 1 SD:

Lower bound for within 2 SDs:

Upper bound for within 2 SDs:

Lower bound for within 3 SDs:

Upper bound for within 3 SDs:

- d. Use the histogram to count how many values fall within the bounds for being within 1 standard deviation, within 2 standard deviations, and within 3 standard deviations. Then find the proportion of the data, out of the 300 run times, that fall within those intervals.

Number within 1 standard deviation:

Proportion:

Number within 2 standard deviations:

Proportion:

Number within 3 standard deviations:

Proportion:

- e. Do those proportions approximately agree with what the Empirical Rule said they should be? Why aren't they exactly the same?

One of the most challenging marathons in the world is the Great Wall Marathon where runners run along the Great Wall of China and must run up and down 5,164 steps. The average time for this marathon is about 6 hours with a standard deviation of 1 hour. Assume these times are also approximately bell-shaped.

Definitions

A **z-score** or **standard score** standardizes a value so it can be directly compared to a value from another distribution. This process is called **standardization**. We can calculate a z-score using

$$z = \frac{\text{observation} - \text{mean}}{\text{standard deviation}}$$

These z-scores indicate how many standard deviations above or below the mean a particular value falls.

- f. Calculate the z-score for a runner in a "standard" marathon (the one with an average of 195 minutes and a standard deviation of 25 minutes) who finished with a time of 2.5 hours.

- g. Calculate the z-score for a runner in the Great Wall Marathon who finished in 231 minutes and 30 seconds.

- h. Are either of those marathon finish times considered to be "unusual"? Explain

- i. Which runner ran a better marathon relative to their peers: a runner in the "standard marathon" who finished with a time of 2.5 hours or a runner of the Great Wall Marathon who finished with a time of 231 minutes? Explain.

Topic 10

Outliers, Boxplots, and Describing Distributions

Lecture

Consider the student maximum speed data given by students in a statistics class. There were 42 responses, as given below:

85	188	180	100	100	56	110	100	90	90	90	142	97	105	114	85	105
130	87	110	86	125	150	70	140	85	95	135	100	300	80	110	95	110
100	110	120	110	85	100	111	145									

An **outlier** is an individual observation that falls outside the overall pattern of the data (e.g.: the _____ in the speed data).

Detecting & Displaying Outliers

The IQR is often used to detect outliers in data according to the following **Rule of thumb**: Values at least $1.5 \times \text{IQR}$ greater than Q_3 or $1.5 \times \text{IQR}$ less than Q_1 are said to be **mild outliers**. These lower and upper limits: $(Q_1 - 1.5 * \text{IQR}, Q_3 + 1.5 * \text{IQR})$ are called the lower and upper **fences** of the distribution.

Example::

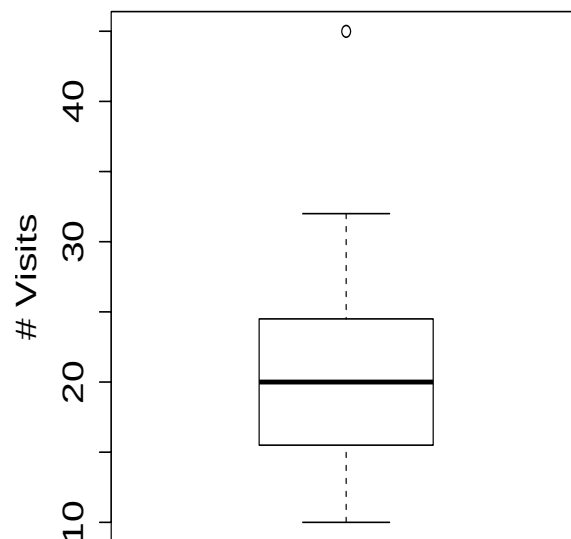
Which values are outliers for the speeds? 5-Number Summary: (56, 90, 102.5, 121.25, 300)

Boxplots

The information in the 5-number summary is often displayed in a **boxplot**.

- Like histograms, boxplots are used to display the distribution of a quantitative variable.
- Can see the center, spread, overall range, and degree of symmetry.
- The shape of the distribution as well as the presence of any possible outliers is easily discerned from the boxplot.
- Boxplots are also great for making multiple comparisons
- The boxplot is sometimes called the box and whiskers plot.
- In a boxplot, half the data lies above the thick black line and half lies below it.
- Half the data lies inside the box, and half lies outside
- The dots are _____.

Boxplot of No. Visits



Boxplots have Two Primary Uses:

- Comparing two or more distributions (as we will see in the next topic).
- Detecting & displaying outliers.

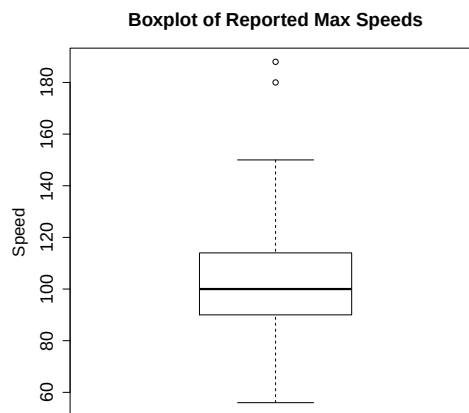
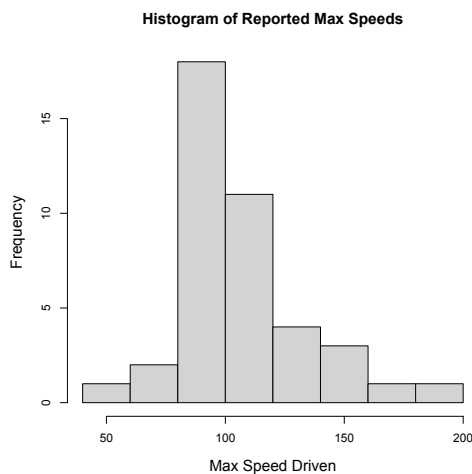
Discerning the shape of a distribution with a boxplot:

Describing Distributions

Important Characteristics of a quantitative data set:

1. Center: Where is the data set “centered” along the number line?
2. Spread: (variation): How spread out – dispersed – are the observations?
3. Shape: What is the shape of the distribution of values in the data set?
4. Outliers: Are there any unusual values or features in the data set?

Example: What do you notice for these data that show the fastest speed driven in a car for some statistics students?



Activities

Activity 10-1: Effects of Outliers

On Canvas, there is a dataset called Top_Speed.xlsx in which the 42 top speeds shown early in this topic are given. Download the file and open it in Excel.

- a. First compute the mean and median of the speeds using only the 41 value that are not considered to be extreme outliers. That is, not including the 300. Report those values here.

- b. Now compute the mean and median of the speeds using all values including the 300. Report the values here. Comment on how the mean and median changed. Which one changed more? Does that make sense? Explain.

- c. Compute the standard deviation and IQR of the 41 values not including the 300. Report the values here. Then compute them using all 42 values including the 300 and report the SD and IQR. Comment on how the standard deviation and IQR changed. Which one changed more? Does that make sense? Explain. Use the =QUARTILE.EXC function to get the quartiles, not =QUARTILE.INC.

- d. Suppose the value of 300 was entered as 30,000. Change the value to 30,000 and repeat parts b and c.

Definitions

A statistic or parameter that is highly affected by outliers is called **sensitive**. If outliers do not change the value of the statistic, then it is said to be **resistant** to outliers. The mean and standard deviation use all values when being computed, so they are sensitive to outliers. The median and IQR are resistant to outliers.

- e. Without actually calculating it, is the range sensitive or resistant to outliers? Explain.

Since the median and IQR are resistant to outliers, it is typically best to use them as the measures of center and spread if outliers are present or if the distribution is skewed. The mean and standard deviation are preferred when there are not any extreme outliers or when the distribution is (approximately) symmetric.

Activity 10-2 Video Game Sales

On Canvas is a dataset called "Video_Game_Sales.xlsx" that contains information about the 100 best selling video games of all time as of December 2016.

- a. Use the *Excel Guide* on Canvas to create a histogram for the Global_Sales variable. Include a rough sketch of the histogram here. Note that the units are in millions here.
- b. What is the shape of this distribution? Include skewness and number of peaks.

- c. Would the mean or median be the more appropriate measure of center? Explain and then use Excel to find the value of both measures of center. Report them both here.

- d. Would the standard deviation or IQR be the more appropriate measure of spread? Explain and then use Excel to find the value of both measures of spread. Report them both here. Use the `=QUARTILE.EXC` function to get the quartiles, not `=QUARTILE.INC`.

- e. Create a boxplot for the `Global_Sales` variable. Based on this, how many outliers are in the data set and what are the sales numbers for those outliers? You do not have to include a sketch of the boxplot.

- f. Using the information from parts a-e, write a few sentences giving a complete description of the global sales variable. Be sure to write in context so that someone unfamiliar with the dataset can understand exactly what you are talking about. Mention the appropriate measure of center (mean or median and indicate which one you are reporting), an appropriate measure of spread (SD or IQR and which one you are reporting), the shape (skewness and number of peaks), and outliers.

Activity 10-3: Comparing Distributions

When comparing distributions, we want to mention all the same key characteristics as we did when describing a single distribution: center (mean or median), spread (standard deviation or IQR), shape (skewness and number of peaks, if we can tell), outliers, and write in context.

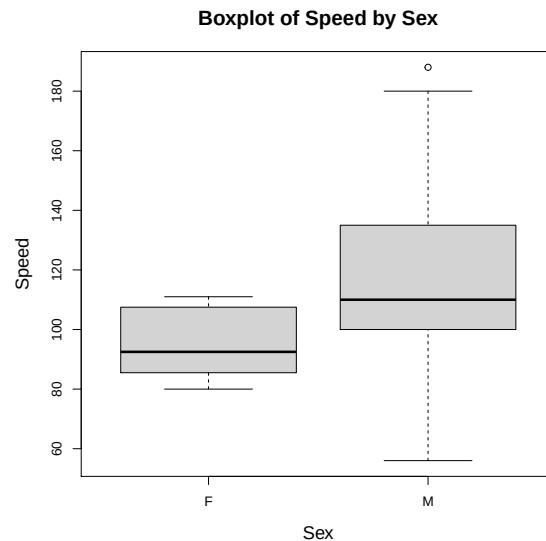
Consider the speed data from the previous example now split between females and males. The 5-number summaries and boxplots are shown below.

5-Number Summary for Females

Min	Q_1	Median	Q_3	Max
80.00	85.75	92.50	106.25	111.00

5-Number Summary for Males

Min	Q_1	Median	Q_3	Max
56	100	110	135	188



- How would you describe the shapes of these distributions? Is the shape the same for both? Explain how you can tell.
- Given the shapes, would the mean or median be a more appropriate measure of center? Report the value of this measure of center.
- Given the shapes, would the standard deviation or IQR be a more appropriate measure of spread? Report the value of this measure of spread.

- d. Report any and all outliers for each distribution. Give the exact value(s).

- e. In the context of the problem, using the information from parts a-d, write a few sentences giving a complete comparison of these distributions. Compare the appropriate centers, spreads, shapes, and mention outliers.

Chapter 2 Summary

1. Data graphics are used to present information from a distribution (or frequency distribution) for all sorts of variables – both categorical and quantitative.
2. The most common graphs for categorical variables are bar charts and pie charts (but pie charts are terrible!).
3. The most common graphs for quantitative variables are dotplots, histograms and boxplots.
4. Proportions (or percentages) are useful when describing categorical variables.
5. When describing a distribution for a quantitative variable, you need to mention **center, spread, shape (including number of peaks), unusual features** (like outliers) and write in context. The order you mention these characteristics does not matter.
6. The most common measures of center are the mean and the median.
7. The most common measures of spread are the standard deviation and the IQR.
8. The most common shapes are skewed left, skewed right, or approximately symmetric.
9. We use the median and IQR when the distribution is skewed and we use the mean and standard deviation when it is symmetric.

Bonus – Symbol Guide

The following table lists symbols and what they are usually denoting. This will be updated as we continue in this class.

Symbol	Description
n	Sample size
N	Population size
x or x_i	Observation
$\bar{x}$	Sample mean
μ	Population mean
s	Sample standard deviation
σ	Population standard deviation
M	Median
Q_1	Lower quartile
Q_3	Upper quartile
IQR	Interquartile range
df	Degrees of freedom
z	Standard score

Chapter 3: Probability

Topic 11

Intro to Probability

Lecture

While we will regularly use descriptive statistics throughout this course, we will begin to shift our focus to inferential statistics. The basis of making inference on a population is **probability**. There are two primary types of probability:

1. Theoretical - list all possible outcomes and determine relative frequency of each.
2. Empirical - probabilities assigned to outcomes of an experiment based on relative frequency of occurrence. (Ex: Flip coin many times.)

Sample Space & Events

An **experiment** is any process whose outcome is unknown before the process is completed. To characterize the **uncertainty** in these outcomes, we will assign them _____ to represent how likely they are to occur.

Definition: The **sample space** for an experiment is the set of all possible **outcomes** of the experiment. Often denoted S .

- Can be specified before the experiment is performed.

Example: Roll a 6-sided die. What is S ?

Example: Observe the test score of a randomly chosen student. What is S ?

Example: Inspect chips from a manufacturing process until the first defective is found. What is S ?

Note: Sample spaces are one of three types:

1. **Finite**: There are a finite number of outcomes in the sample space.
2. **Infinite discrete** or **Countably infinite**: There are an infinite number of outcomes in the sample space, but they can be listed.
3. **Infinite**: Infinite sample spaces are typically intervals of real values.

Note: We will focus on **finite** sample spaces in this chapter and take the classical probability approach where we assume that each of the finite number n possible outcomes has the same probability of occurrence, namely $1/n$.

Example: Toss a fair coin & roll a die. What is S ?

Definition: An event, E , is an outcome with which we can associate a probability. An event is always a subset of the sample space.

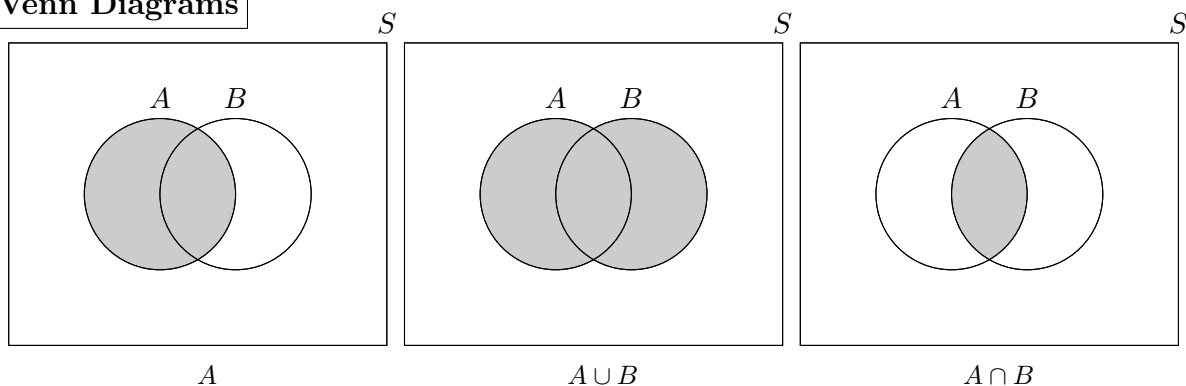
Example: Let $E = \{H2, H4, H6, T2, T4, T6\}$ be an event in S - might be interested in the event that we roll an even #.

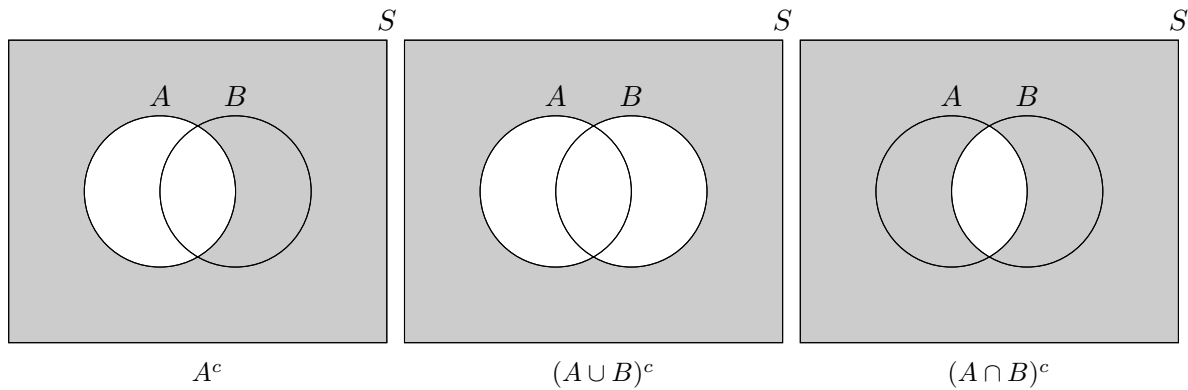
- So E = event that we roll an even #.
- Let F = event that we roll a four or a five, so that $F = \{H4, H5, T4, T5\}$.
- In general, an event E occurs if, when the experiment is performed, the outcome is an element of E .

Probability Notation

1. $P(A)$ is the probability that event A occurs.
2. A^c represents the complement (or opposite) of an event.
3. $P(A^c)$ is the probability of the _____ of event A . In other words, it is the probability that event A does not occur.
4. $P(A^c) =$ _____. For example, if R is the event that it rains tomorrow, and the probability of rain tomorrow is 0.3, then $P(R^c) = 1 - P(R) = 1 - 0.3 = 0.7$.
5. $P(A \text{ and } B)$ is the probability that events A and B both occur. This is often written as $P(A \cap B)$ and referred to as the **intersection** of an event.
6. $P(A \text{ or } B)$ is the probability that either event A or event B (or both) occur. This is often written as $P(A \cup B)$ and is referred to as the **union** of an event.
7. $P(A|B)$ is the probability that event A occurs, given that event B has occurred. This is called "the probability of A , given B ". Sometimes it is also called "the probability of A , conditional upon B ". We refer to statements like this as statements of _____.

Venn Diagrams



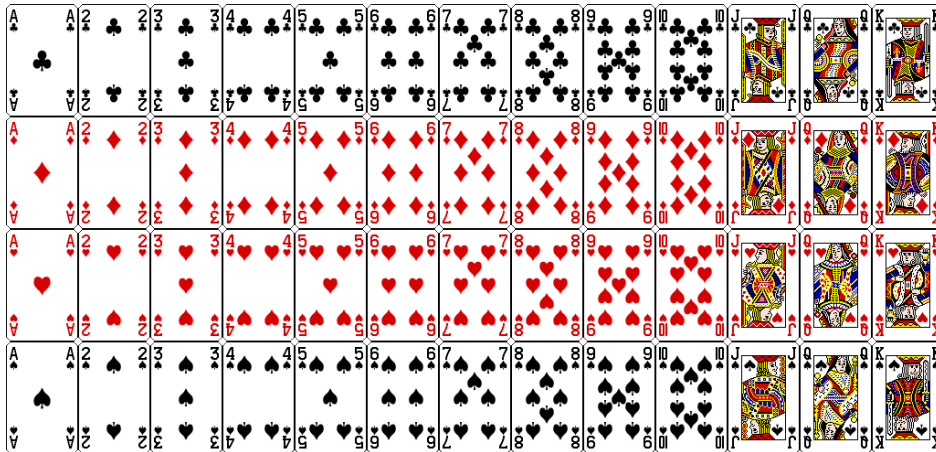


Definition: Two events A & B are mutually exclusive (or disjoint) if they cannot occur at the same time. That is,

Probability Identities

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
2. $P(A|B) = \frac{P(A \cap B)}{P(B)}$, which means $P(A \cap B) = P(A|B)P(B)$.

Example: Deck of cards.



Let "Red" be the event that I draw a red card, "Diamond" be the event that I draw a diamond, and so on.

What is the sample space?

$$P(\text{Red}) = \quad \quad \quad P(\text{Red}^c) =$$

$$P(\text{Diamond}) = \quad \quad \quad P(\text{Diamond}^c) =$$

$$P(\text{Red or Diamond}) =$$

$$P(\text{Red}|\text{Diamond}) =$$

$$P(\text{Diamond}|\text{Red}) =$$

This brings us to a very important fact about probability. In general, $P(A|B) \neq P(B|A)$.

Independence

Independence is a term used in statistics to refer to whether or not two events are “_____” to one another. If knowing that one event has occurred gives you additional _____ about the probability that the other will occur, then the two events are said to be dependent. This term comes from the idea that the probability of one event occurring depends on the outcome of another. If, on the other hand, knowing the outcome of one event gives you no additional information about the other, then the two events are said to be independent.

Formally, events A and B are independent if

$$P(A|B) = P(A) \text{ or } P(B|A) = P(B)$$

The intuition is as follows: If $P(A|B) = P(A)$, then knowing that B has occurred does not change the probability that A will occur. In other words, the occurrence of B has given you no additional information about the probability of A occurring. In this sense, A and B are independent. If, on the other than, $P(A|B) \neq P(A)$, then knowing that has occurred *does* change the probability that A will occur. In other words, the occurrence of B has given you additional information about the probability of A occurring. In this sense, A and B are dependent.

Example: Let’s check to see if the events “draw a diamond” and “draw a red card” are independent or dependent.

$$P(\text{Diamond}) =$$

$$P(\text{Diamond}|\text{Red}) =$$

Are these events independent?

Now try it the other way around, and you should get the same result:

How about the events “draw a king” and “draw a red”? Are they independent?

Another way to check for independence:

Events A and B are independent if $P(A \cap B) = P(A)P(B)$.

So that means:

This gives us another “key word” in probability:

Activities

Activity 11-1: Sample Spaces

For each of the following scenarios, write down the sample space and indicate if each outcome is equally likely or not.

- a. You flip three coins (or one coin thrice).

- b. You roll two four-sided dice.

- c. You have four kids and count how many girls you have.

Activity 11-2: Calculating Probabilities

Suppose you have the following sample space: $S = \{1, 4, 5, 10, 31, 41, 51, 60\}$.

- a. What is the probability that if a random number is selected, that it will be the number 4?
- b. What is the probability that the number will contain the digit "1"?
- c. What is the probability that the number will be at most 10?
- d. What is the probability that the number is 41 given that it contains the digit "1"?
- e. What is the probability that the number is at least 31 or contains the digit "1"?
- f. If you select two numbers (where any number could be selected twice), what is the probability that the first number is 4 and the second number contains the digit "1"? Think about whether those events are independent or not.

Topic 12

IContingency Tables

Lecture

A contingency table is a convenient tool for displaying how two categorical variables relate to one another.

One variable's different categories (often called levels) are listed as the rows; the other's are listed as the columns. The last row and last column show totals for the rows and columns, and the bottom right corner shows the grand total (to which the row and column totals both sum).

Each cell of the contingency table shows how many observations fall into that combination of categories.

Example: A boring contingency table:

		Card Suit				
		Diamond	Heart	Club	Spade	Total
Card Color	Red	13	13	0	0	26
	Black	0	0	13	13	26
	Total	13	13	13	13	52

$$P(\text{Red}) =$$

$$P(\text{Diamond and Red}) =$$

$$P(\text{Diamond}) =$$

$$P(\text{Diamond}|\text{Red}) =$$

$$P(\text{Red or Club}) =$$

$$P(\text{Red}|\text{Diamond}) =$$

Activities

Activity 12-1: A Less Boring Contingency Table

Here is a contingency table showing the relationship between the variables political views and marijuana usage level for 1349 randomly selected individuals.

		Marijuana Usage Level			
		Never	Rarely	Frequently	Total
Political Views	Liberal	479	173	119	771
	Conservative	214	47	15	276
	Other	172	45	85	302
	Total	865	265	219	1349

- Find $P(\text{Frequently})$. Then write a sentence interpreting what this number means.
- Find $P(\text{Never})$.
- Find $P(\text{Rarely}^c)$.
- Find $P(\text{Frequently}|\text{Other})$.
- Find $P(\text{Frequently}^c|\text{Other})$.
- Comment on how the answers to parts d and e are related.

g. Find $P(\text{Liberal}|\text{Never})$.

h. Find $P(\text{Liberal}|\text{Never}^c)$.

i. Are the answers to parts g and h related in the same way that the answers to parts d and e were as you mentioned in part f? Explain.

j. Are the events Liberal and Frequently independent? Be sure to check using a calculation.

k. Are any of these events mutually exclusive? Explain.

Extra Space for Notes

Topic 13

Empirical Probability

Lecture

As was stated earlier, **empirical probability** is probability that is assigned to outcomes of an experiment based on relative frequency of occurrence.

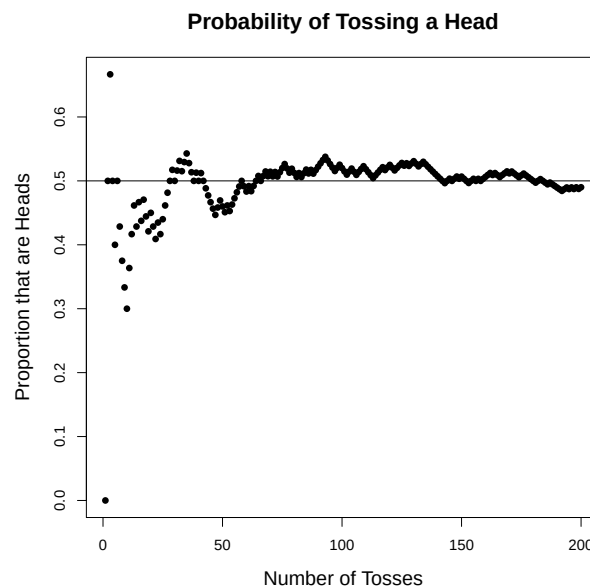
We take

Examples:

- Flipping a coin many times to calculate the probability of flipping heads.
$$\text{Probability} = \frac{\# \text{ of heads}}{\# \text{ of flips}}$$
- A baseball player's batting average is the empirical probability of getting a hit.
$$\text{Probability} = \frac{\# \text{ of hits}}{\# \text{ of at-bats}}$$
- The likelihood that you will get stuck at a red light on your way to class.
$$\text{Probability} = \frac{\# \text{ of times stuck at light}}{\# \text{ of trips to class}}$$

These probabilities will get more and more accurate as you repeat the experiment.

This idea is known as the **law of large numbers**. As an experiment is repeated over and over, the empirical probability of an event approaches the theoretical (actual) probability of the event.



Activities

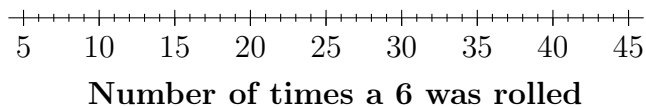
Activity 13-1: Rolling Dice

We are going to test out this empirical probability by rolling 6-sided dice. Suppose we want to estimate the probability of rolling a 6.

- First, what is the exact, theoretical probability of rolling a 6? Give it as a fraction and a decimal.
- If you roll a die 50 times, about how many times would you **expect** a 6 to be rolled?

Note: We are going to formalize this idea of an expected value in Chapter 4. For now, an estimate is fine.

- Roll your die 50 times and keep track of how many times you rolled a 6. You can keep a tally below. Then find the **empirical probability** of rolling a 6 based on your 50 rolls by finding the proportion of rolls that had a value of 6. Then put the number of times a 6 was rolled (not the probability) on the board.
- Create a dotplot of the number of 6's rolled (out of the 50 rolls) using data from the entire class.



- Was this plot centered at (or near) the number you expected from part b?

- f. Take the average of all the numbers you used to make the dotplot in part d. Then divide that average by 50 to get the empirical probability of rolling a 6 using the data from the entire class. Compare this number to the theoretical probability in part a. Is it close? Why is it not exact?
- g. Empirical probabilities can be thought of as estimates for the theoretical probability. How could we make the empirical probability estimate for the probability of rolling a 6 more accurate? Explain.

Extra Space for Notes

Topic 14

Intro to Counting

Lecture

We have seen that we can calculate probabilities by dividing the number of “successes” by the number of total outcomes. Knowing these numbers is not always easy and may require more advanced counting techniques.

Let’s consider an example involving two coins. To find probabilities relating to outcomes of flipping two coins, we need to know the outcomes. The outcomes are:

So we can say that the probability of flipping two heads is:

The probability of flipping one heads and one tails is:

Extending this now to three coins, we list all of the outcomes:

Then we can calculate different probabilities.

It would be more difficult and time consuming to write down every outcome for flipping four or more coins, but we can see a relationship forming.

- Flipping 1 coin has 2 outcomes.
- Flipping 2 coins has 4 outcomes.
- Flipping 3 coins has 8 outcomes.
- Flipping 4 coins has _____ outcomes.
- Flipping 10 coins has _____ outcomes.

This can, of course, be applied to other events.

- Rolling 1 die has 6 outcomes.
- Rolling 2 dice has _____ outcomes.
- Rolling 5 dice has _____ outcomes.

The counting principle tells us how to calculate the results of two experiments performed in succession. Suppose there are n outcomes for the first experiment. If for each outcome of the first experiment, there are m outcomes of the second experiment, then the number of possible outcomes for the two experiments is $n \times m$. This principle extends to any number of such experiments.

Example: How many three-letter codes can we make if the first letter is a vowel (A, E, I, O, or U)? One such code is OXJ.

Example: Utah license plates typically have 6 characters. They begin with a letter, then are followed by three digits between 0 and 9, then have two more letters.

- (a) How many different license plates can be created following this format?

- (b) How many license plates are possible if we insist that on each plate the first digit cannot be a 1 and no numeral can appear more than once?

Applying counting to probabilities

Example: Return to the deck of cards.

- (a) Suppose you draw a card, put it back in the deck, and draw another. What is the probability that the first card is an ace and the second one is a jack? (Assume that the deck is shuffled after the first card is returned to the deck.)

- (b) Suppose you draw a card and, without putting it back, draw another. What is the probability that the first card is an ace and the second one is a jack?

Activities

Activity 14-1: Restaurant Offerings

A restaurant offers 8 appetizers and 14 entrées.

- a. How many choices do you have if you are extra hungry and want to eat both an appetizer and an entrée?

Definitions

That last example showed the **multiplicative principle**, also known as the counting principle. If event A can occur in n ways, and each possibility for A allows for exactly m ways for event B , then the event " A and B " can occur in $n \cdot m$ ways.

There is also the **additive principle**, which states that if event A can occur in n ways and event B can occur in m ways, then the event " A or B " can occur in $n + m$ ways.

- b. How many choices do you have if you want to eat one dish: either an appetizer or an entrée?

Keep in mind: The multiplicative principle works when the events are independent (the selection of the first item has no bearing on the selection of the second). This is again the idea that "and" = multiply when the events are independent. The additive principle works when the events are mutually exclusive (the events cannot occur at the same time). This is the idea that "or" = add when the events are mutually exclusive.

Activity 14-2: Club Election

There are 10 people willing to serve as an officer for a club. It's decided just to put the 10 names in a hat and draw 3 of them out in succession. The first name drawn is declared president, the second name vice president, and the third name treasurer.

- a. How many possible election outcomes are there?

- b. John is a candidate. What is the probability that he will be vice president?

- c. Mary and Anthony are also candidates along with John. What is the probability that all three will be selected to office?

- d. What is the probability that none of the three (Mary, Anthony, and John) will be selected to office?

- e. What is the probability that at least one of the three (Mary, Anthony, and John) will not be selected?

Topic 15

IPermutations and Combinations

Lecture

Definition: A **permutation** of items is an arrangement of the items in a certain **order**. Each item can be used only once in the sequence. We can determine the number of permutations by using the counting principle.

In how many ways can I arrange the four letters A, B, C, D using each letter only once?

The number of permutations of n items is the number of ways to arrange n distinct items without repetition. The Counting Principle shows that this is n factorial, which is defined as

$$n! = n \times (n - 1) \times (n - 2) \times \cdots \times 3 \times 2 \times 1$$

On your calculator:

Example: If there are 10 people waiting to enter a movie theater, how many ways can they enter if they must enter one at a time? What if there are 20 people?

More on arrangements:

Let's now think of this example: We think of selecting 3 lucky winners from a group of 10 people. The first person selected gets first prize, the second gets second prize, and the third gets third prize. How many ways can we select the three winners?

The number of ways to select r items from n distinct items and arrange them in order is called the number of permutations of n items taken r at a time.

The way we have looked at permutations of n items taken r at a time works, but can be tedious when dealing with many items.

For example, consider having 30 books and needed to arrange 12 of them on a shelf in any order. How many different ordering are there?

$$\begin{aligned} & 30 \times 29 \times 28 \times 27 \times 26 \times 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times 19 \\ &= 30 \times 29 \times 28 \times 27 \times 26 \times 25 \times 24 \times 23 \times 22 \times 21 \times 20 \times 19 \end{aligned}$$

So, if we are selecting r items from n items, order matters, and each item can only be selected once, we have

$\text{Permutations of } n \text{ items taken } r \text{ at a time} = {}_nP_r = \frac{n!}{(n-r)!}$
--

On your calculator:

Example: The producer of a talent show has seven slots to fill. Twenty-five acts have requested a slot. Use the permutations formula to find the number of ways to select seven acts and arrange them in a sequence.

Combinations

- A combination of a set of items is very similar to a permutation, except order doesn't matter in this case.
- A combination of a group of items is a selection from that group in which order is not taken into account. Like a permutation, no item can be used more than once in a combination.

We looked at the number of ways to arrange the letters A, B, C, and D. If order mattered, BCAD is different than ADCB. However, those are the same combination.

How many ways can we arrange the letters A, B, C, and D if order doesn't matter?

When thinking about **if order matters or not**: ask yourself if the items are best represented by a list or a group.

Think about it from the perspective of the item being selected.

So, if we are selecting r items from n items, order does not matter, and each item can only be selected once, we have

$$\text{Combinations of } n \text{ items taken } r \text{ at a time} = \binom{n}{r} = {}_nC_r = \frac{n!}{r!(n-r)!}$$

On your calculator:

Example: How many three-person committees can I select from a group of six people? What if they were assigned a position in the committee when they were selected?

Example: In a group of six men and four women, I select a committee of three at random. What is the probability that two of the three committee members are women?

Activities

Activity 15-1: Does Order Matter?

In each of the following scenarios, determine whether the order of selection matters. You do not need to solve the problem.

- a. You are trying to determine how many "words" can be formed by 3 vowels and 6 consonants taken from 5 vowels and 10 consonants.

- b. There are 10 pizza toppings to select from and you need to select 3 or fewer to put on your pizza.

- c. In an attempt to clean up your room, you have purchased a new floating shelf to put some of your 17 books you have stacked in a corner. The new book shelf is large enough to hold 10 of the books and so you selected 10 of the 17 to put on the shelf and want to know how many ways they can be arranged.

- d. You decide to have a dinner party. Even though you are incredibly popular and have 14 different friends, you only have enough chairs to invite 6 of them. How many choices do you have for which 6 friends to invite?

Activity 15-2: Diplomat and Spies

A meeting takes place between a diplomat and fourteen government officials. However, four of the officials are actually spies. The diplomat is going to give secret information to ten of the attendees at random.

- a. Does the order in which the diplomat gives the information matter? Explain.

- b. Can any one person be selected more than once to receive information?

- c. Given the answers to both of the above questions, would a permutation, combination, or neither be appropriate to use here? Explain.

- d. Find the total number of ways the diplomat could give the information to ten of the attendees in this scenario.

- e. Find the number of ways that the diplomat could give the information so that none of the spies receive the info.

- f. Find the probability that no secret information was given to the spies.

Activity 15-3: Combination(?) Locks

A certain digital lock has a 5-number code. Each number must be between 1 and 30, inclusive.

- a. If the numbers can be reused, would a permutation, combination, or neither be appropriate to use here? Explain.

- b. Find the probability that if you randomly guessed a code, it would be correct if the numbers can be reused.

- c. If the numbers cannot be reused, would a permutation, combination, or neither be appropriate to use here? Explain.

- d. Find the probability that if you randomly guessed a code, it would be correct if the numbers cannot be reused.

Keep in mind: So-called combination locks are actually not “combinations” at all. The order in which you enter the numbers matters. Most are not permutations either since usually each number can be used more than once.

Chapter 3 Summary

Summary of Probability Identities

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ in general.
2. $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive.
3. $P(A \cap B) = P(A) \cdot P(B|A)$ in general.
4. $P(A \cap B) = P(A) \cdot P(B)$ if A and B are independent.
5. $P(A \cap B) = 0$ if A and B are mutually exclusive.
6. $P(A|B) = \frac{P(A \cap B)}{P(B)}$ in general.
7. $P(A|B) = P(A)$ if A and B are independent.
8. $P(A^c) = 1 - P(A)$ in general.

Summary of Counting Principles

Principle	Description	Formula
Counting principle	If one event can occur in m ways and a second event can occur in n ways, then the number of ways the two events can occur in sequence is $m \cdot n$. This can be extended to more than two events.	$m \cdot n$
Arrangements	The number of arrangements of n distinct objects.	$n!$
Permutations	The number of permutations of n distinct objects taken r at a time, where $r \leq n$.	${}_nP_r = \frac{n!}{(n-r)!}$
Combinations	The number of combinations of r objects selected from a group of n objects without regard to order, where $r \leq n$.	${}_nC_r = \frac{n!}{r!(n-r)!}$

Chapter 4: Random Variables and Probability Distributions

Topic 16

Intro to Random Variables

Lecture

Often, we are interested in a numeric summary of outcomes of an experiment rather than _____. In a past experiment we had 10 people: (6 men, 4 women) and we looked at choosing a committee of 3.

Let $X = \#$ women chosen for the committee.

- X is an example of a random variable (X is uncertain).

Definition: A **random variable** is a variable that is a numerical description of the outcomes of an experiment.

In other words, a random variable associates a numerical value with each possible experimental outcome.

Like quantitative data, random variable can be either continuous or discrete.

- Like discrete data, a discrete random variable is a random variable that can only take on specific (usually integer) values.
- A continuous random variable may take on any numerical value in an interval. It can be a decimal.

Example: Toss fair coin twice. $S = \{HH, HT, TH, TT\}$.

Outcome X

Let $X = \#$ heads observed.

- What is the probability of the event:
 $\{X = 1\} = \{HT, TH\}$?
 - Concise “random variable” notation is $P(X = 1)$.

$P(X = 0) =$, $P(X = 1) =$, $P(X = 2) =$, $P(X = 7) =$

Probability Distributions

These results can be summarized using a probability distribution:

x	0	1	2
$P(X = x)$			

A **probability distribution** describes how probabilities are distributed over the values of a random variable.

For a discrete random variable X , the probability distribution is defined by a **probability function** or probability mass function (pmf), $P(x) = P(X = x)$, which is the probability that the random variable X takes on a value x .

This probability function provides the probability for each possible value of the random variable.

Properties of Probability Distributions

1. Each probability must be between 0 and 1:
2. The sum of all the probabilities is 1:

Activities

Activity 16-1: Probability Distribution Identification

For each of the following random variables, determine if the distribution given is a valid probability distribution. Explain why it is or isn't for each one.

a.

x	0	1	2	3	4
$P(x)$	0.30	0.35	0.20	0.15	-0.05

b.

y	0	1	2	3	4
$P(y)$	0.30	0.35	0.20	0.15	0

c.

z	0	1	2	3	4
$P(z)$	0.30	0.35	0.20	0.10	0

- d. For the variable that had a valid probability distribution, find the probability that the variable is at least 2.
- e. For the variable that had a valid probability distribution, find the probability that the variable is equal to 0 or equal to 3.

Activity 16-2: Committee Selection

At a certain small college, a math department has 14 math majors, 9 of whom are math education majors and 5 of whom are not education focused. At the end of the year, a committee is chosen to report about the goings-on in the department. This committee must have 3 members. Let X = the number of math education majors on the committee.

- a. What are all the possible values of X ?
- b. Determine the probability that X is equal to each of those values from part a.
- c. Use the information from parts a and b to make a table for the probability distribution of X . This table should look similar to the tables at the top of the previous page.
- d. Is this variable X continuous or discrete? Explain how you know.

Topic 17

Expected Value and Variance

Lecture

The expected value is another name for the mean random variable value of a distribution.

$$\text{Expected value of } X = E(X) = \mu = \sum xP(x)$$

Example: Find the expected value of the probability distribution given in this table:

x	10	15	20	25
$P(x)$	0.4	0.2	0.05	0.35

Example: It costs \$2 to play a certain lottery game and 1,100 tickets are sold. One ticket will win the grand prize of \$500, 10 tickets will win the second prize of \$100, and 50 tickets will win the third prize of \$10. How much do you expect to win?

Variance and Standard Deviation of a Random Variable

We saw that $E(X) = \sum xP(x)$ is the mean of a random variable. Now we should think about the standard deviation since the mean alone is not enough information.

First recall the variance. How is interpreted:

Writing the variance as an expected value: $\text{Var}(X) = \sigma^2 =$

Recall how the standard deviation is related the variance:

So the standard deviation is

$$\text{SD}(X) =$$

Example: Find the standard deviation of the winnings from the previous example.

Winnings ()	498	98	8	-2
$P()$	$\frac{1}{1100}$	$\frac{10}{1100}$	$\frac{50}{1100}$	$\frac{1039}{1100}$

Alternate formula for the variance:

$$\text{Var}(X) = \sigma^2 = \sum x^2 P(x) - \mu^2$$

Check this one:

Activities

Activity 17-1:

Suppose we asked a population of 20 students how many states the students have visited and obtained the results shown in the table below:

States Visited	3	4	5	6
Frequency	8	5	6	1

- Before you actually calculate the average number of states visited from this small sample, first notice that the average of the numbers 3, 4, 5, and 6 is 4.5. Will 4.5 be the average of this dataset? Why or why not?
- Based on the frequencies, would you expect the average to be closer to 4 or to 5? Explain.
- List out the 20 raw data values for the number of states visited and then calculate the average of the states visited variable using the formula $\mu = \frac{\sum x}{N}$. Note: we use the symbol μ and not $\bar{x}$ since we are thinking of this group of 20 people as a population, not a sample.

- d. Now find a weighted average of the number of states visited using the frequencies as the weights. Recall that a weighted average can be found using $\bar{x} = \frac{\sum x \cdot w}{\sum w}$.

- e. Now find the relative frequencies for each of the number of states visited and fill in the values in the table.

States Visited	3	4	5	6
Frequency	8	5	6	1
Relative Frequency				

Keep in mind: Relative frequencies in this case are no different than the probabilities of randomly selecting a person in our sample of 20 that had visited that number of states.

- f. Now find a weighted average of the number of states visited using the relative frequencies (or probabilities) as the weights. What do you notice about the denominator? Will the denominator always be equal to that number when using relative frequencies or probabilities as weights?
- g. Compare the answers in parts c, d, and f. Using this information, explain why it makes sense that the formula $E[X] = \sum xP(x)$ makes sense.

The point of this part of this activity was for you to realize that the expected value formula is nothing more than a weighted average of the values of the states visited variable using the probabilities as the weights. In a similar way, the formulas for the variance and standard deviations of probability distributions are also weighted by probabilities.

- h. Find the variance in the number of states visited using $\text{Var}(X) = \sigma^2 = \sum (x^2 - \mu^2)P(x)$ or $\text{Var}(X) = \sigma^2 = \left(\sum x^2 P(x) \right) - \mu^2$ (either formula will give the same answer, but I recommend you use the second one). What are the units?
- i. Find the standard deviation in the number of states visited. What are the units? Then interpret this value. Think about what a standard deviation represented.

Extra Space for Notes

Topic 18

Bernoulli and Binomial Random Variables

Lecture Part 1

Consider an experiment with two possible outcomes: success or failure. If it's a success, let $X = 1$; if a failure, let $X = 0$. The probability mass function (pmf) of X is:

$$\left\{ \begin{array}{l} P(X = 1) = p \\ P(X = 0) = 1 - p = q \end{array} \right\}, \text{ where } p = \text{prob. of success, } 0 \leq p \leq 1.$$

Definition: Any experiment of this type is known as a **Bernoulli trial**, and X is a **Bernoulli random variable**. *More Info Here*.

Note: The pmf of X can be written: $P(X = x) = \quad \quad \quad$, $x = 0, 1$.

When this occurs, we write: $X \sim \text{Bernoulli}(p)$.

Examples:

- Flipping a coin and it coming up heads.
- Having a baby who is a girl.
- Rolling a die and getting a 6.
- Getting hit in the face with a frisbee.

What if we have more than one trial?

Activities Part 1

Activity 18-1: Frisbee Woes

Suppose I enjoy playing frisbee, but I am woefully uncoordinated and I know that there is a 25% chance that I'll get hit in the face when I play frisbee and I never get any better, so that probability is the same whenever I play (getting hit in the face one day is **independent** of getting hit in the face any other day).

Let's first consider two days where I play frisbee.

- a. What is the probability that I get hit in the face both times I play frisbee?

- b. What is the probability that I do not get hit in the face either time I play?
- c. What is the probability that I get hit in the face exactly one time when I play frisbee twice? Think about every possible way this can happen. [Note: it is possible to find this fairly easily since we know the answers to parts a, b, and c must add to 1. Try to come up with a way to find this answer if you did not know the answer to parts a and b.]

Now let's consider three days where I play frisbee.

- d. What is the probability that I get hit in the face all three times?
- e. What is the probability that I get hit in the face exactly one of the three days? Again, think about every possible way this can happen.

Finally, let's consider four days where I play frisbee.

- f. What is the probability that I get hit in the face all four times?
- g. What is the probability that I get hit in the face exactly one of the four days? Again, think about every possible way this can happen.
- h. What is the probability that I get hit in the face exactly two of the four days? Again, think about every possible way this can happen.

- i. Try to come up with a general formula that takes an input x that is the number of times out of the 4 days I play frisbee that I get hit in the face.

Lecture Part 2

Example: Suppose a couple plans to have a baby. Let $X = 1$ if it's a girl (success(?)), and $X = 0$ if it's not a girl (failure(?)). Assume for this particular family, the gender of each child is independent and: $P(X = 1) = 0.4$.

Suppose the couple has 1 child. What is the probability that child is a girl?

Suppose the couple has 2 children. What is the probability that both children are girls?
What is the probability that one is a girl and one is not?

Suppose the couple has 3 children. What is the probability that all 3 children are girls?
What is the probability that exactly one is a girl?

What about 4 children?

$$P(X = x) =$$

Definition: Let $X = \#$ of successes in n independent Bernoulli trials. Then X is a **Binomial random variable**, and has pmf given by:

$$P(X = x) = {}_nC_x p^x (1 - p)^{n-x} = \frac{n!}{x!(n-x)!} p^x (1 - p)^{n-x}, \quad x = 0, 1, \dots, n.$$

Example: Suppose with 5 children, each child independently is left-handed with probability $p = 1/10$. Let $X = \#$ of children (out of 5) who are left-handed. Pmf of X ?

$$P(X = x) = \frac{5!}{x!(5-x)!} (1/10)^x (9/10)^{5-x} = \begin{matrix} (9/10)^5 \\ 5(1/10)(9/10)^4 \\ \vdots \\ \vdots \end{matrix} = \left\{ \begin{matrix} 0.59049, & x = 0 \\ 0.32805, & x = 1 \\ 0.07290, & x = 2 \\ 0.00810, & x = 3 \\ 0.00045, & x = 4 \\ 0.00001, & x = 5 \end{matrix} \right\}.$$

(a) What is the probability that at least 1 of the 5 is left-handed?

(b) What is $E(X)$?

Properties of a Binomial Process:

1. There are a fixed number of trials, n .
2. There are only two possible outcomes from each trial (success or failure).
3. The probability of success, p , is the same for each trial.
4. The probability of failure, q , is the same for each trial.
5. The n trials are all independent.
6. We are interested in finding the probability regarding the number of successes, x .

Calculator Tips:

Example: Choose 100 cars randomly & count the # of defects on each car. Is this binomial?

Example: Suppose 8% of cars at a certain dealer have at least one minor defect. If you inspect 100 cars, what is the probability at least 10 cars will have a minor defect?

Expected Value and Standard Deviation of Binomial Random Variables

We just saw that for a binomial random variable, X ,

$$E(X) = n \cdot p$$

How about the variance and standard deviation?

$$\text{Var}(X) = \sum x^2 P(x) - \mu^2.$$

Find the variance for the left-handed example with probability distribution

$$\left\{ \begin{array}{ll} 0.59049, & x = 0 \\ 0.32805, & x = 1 \\ 0.07290, & x = 2 \\ 0.00810, & x = 3 \\ 0.00045, & x = 4 \\ 0.00001, & x = 5 \end{array} \right\}.$$

So,

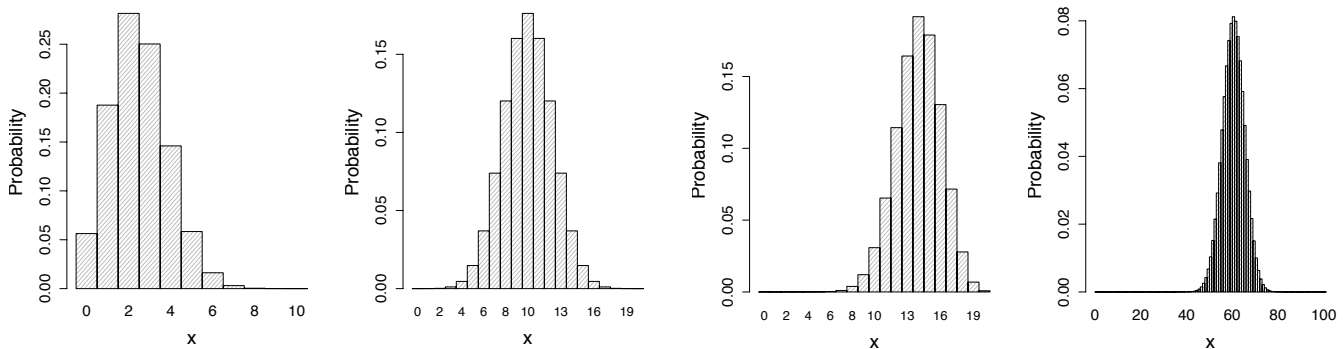
$$\text{Var}(X) = \sigma^2 =$$

and

$$\text{SD}(X) = \sigma =$$

Below are some examples of $\text{Binomial}(n, p)$ pmfs. Can you guess which distributions they are?

Examples of Binomial Probability Mass Functions



Activities Part 2

Activity 18-2: Applying Binomial to Frisbee

Again consider the scenario where there is a 25% chance that I'll get hit in the face when I play frisbee.

- Explain how we can know that the number of days I get hit in the face can be modeled with a binomial random variable in this scenario.
- Find the probability that I get hit in the face exactly one day if I play frisbee on three days using the binomial function or formula. Then compare this to your answer in Activity 18-1 part e.

- c. Find the probability that I get hit in the face on two or fewer days if I play frisbee on 12 days using the binomial function or formula.

- d. In Excel, you can use the =BINOM.DIST function to find binomial probabilities. Repeat parts b and c using Excel and verify that the answers match.

- e. Find the probability that I get hit in the face on at least 40 days if I play frisbee on 120 days.

Activity 18-3: Is it Binomial?

In each of the following situations, determine if the random variable can be modeled with the binomial distribution.

- a. You roll a 6-sided die 5 times and determine the probability of rolling a certain number of 6's.

- b. On any given day, there is a 10% chance that a thingamabob drops from the surface of the ocean down to where a certain mermaid can find it. This mermaid determines the probability that she finds 2 thingamabobs this week.

- c. The average number of dances at Greendale Community College during any school year is 6. The students at the college want to determine the probability that they will have more than 6 dances this year.

Topic 19

Geometric Random Variables

Lecture

In this scenario, instead of counting the number of successes in n Bernoulli trials like we did for the Binomial, we will instead be concerned with the number of trials needed to get one success.

So when $X = \#$ Bernoulli trials required for the first success, X is geometric with prob. p .

What should the pmf of X be in this scenario?

- So: $P(X = x) =$
- And $E(X) =$ and $\text{Var}(X) =$

Calculator Tips

Example: An industrial process manufactures components with a 10% defective rate. Each day the process is subjected to a quality control inspection; products are inspected sequentially until the first defective is obtained.

(a) What is the probability that the first defective will occur in one of the first three components inspected?

(b) What is the probability that more than three components will be inspected before a defective is found?

Activities

Activity 19-1: Memoryless Property

It is known that the probability of getting dealt two pairs in a 5-card poker hand is right around 4.75%. Suppose that you decide that you are going to keep playing poker until you are dealt a hand that has two pairs.

- a. Explain why the number of hands needed until you get a hand with two pairs follows a geometric distribution.

- b. Find the probability that you get a hand with two pairs in the 8th hand you are dealt.

- c. Find the probability that it will take more than 5 hands until you get a hand with two pairs.

- d. Find the probability that it will take more than 12 hands until you get a hand with two pairs **given** that it takes more than 7 hands. Note: this is a conditional probability, so you need to use the conditional probability formula: $P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$.

- e. Compare your answer in part d to your answer in part c. What do you notice?

What we just discovered is known as the **memoryless property**.

Definitions
The memoryless property for a geometric random variable is the fact that $P(X > t + s X > t) = P(X > s)$. That is, if we know that the first success did not occur in the first t trials, then the probability the first success will not occur in the first $t + s$ trials is the same as the probability the first success will not occur in the first s trials. There is essentially no “memory” of the first t trials.

- f. In part d, what were t and s in the definition of the memoryless property?

Topic 20

Poisson Random Variable

Lecture

Poisson Random Variable

Definition: A random variable X is said to be a Poisson random variable with parameter $\mu > 0$ if it has pmf:

$$P(x) = P(X = x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

When do we use the Poisson random variable?

- 1.
2. The probability of the event occurring is the same for each interval.
3. The number of occurrences in one interval is independent of the number of occurrences in other intervals.

Examples of Poisson Random Variables

1. The # of texts received on your cell phone in some time unit.
2. The # of “hits” at a web site on a given day.
3. The # of men killed by horse kicks in the Prussian army in 1 year.
4. The # of misprints on a page of a book.

Calculator Tips

Example: Suppose that recordable injuries at a manufacturing plant are randomly distributed throughout the year, with an average of 1.5 recordable injuries per 30 days.

- (a) Calculate the probability that there are two or more recordable injuries in the next 30 days.
- (b) Calculate the probability that there are at most five recordable injuries in the next 60 days.

Mean & Variance of Poisson Random Variable:

If X is a Poisson random variable with pmf $P(X = x) = \frac{e^{-\mu} \mu^x}{x!}$, then

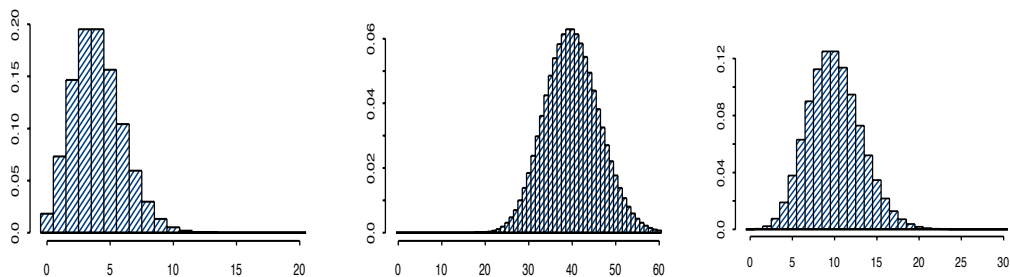
$$\overline{E(X)} =$$

$$\text{Var}(X) =$$

$$\boxed{\text{SD}(X) =}$$

- So the mean and variance are the same for Poisson r.v.'s.
- What do Poisson pmf's look like? Below are four examples of Poisson pmf's.

Examples of Poisson Probability Mass Functions



Activities

Activity 20-1: Smart Phone Life

According to BankMyCell.com, the average life of a smartphone is 2.58 years. Assume that the number of phones a person obtains in a 10-year period follows a Poisson distribution.

- Using the information above, what is the average number of phones a person would obtain in a 10-year period?
- Find the probability that a person would obtain 2 phones or fewer in a 10-year period.
- Find the probability that a person would obtain exactly 3 phones in a 5-year period.
- Find the probability that a person would obtain more than 4 phones in a 5-year period.

Activity 20-2: Determine the Distribution

For each of the scenarios listed below, indicate whether the random variable follows a binomial distribution, a geometric distribution, a Poisson distribution, or none of those. Explain how you know for each.

- a. A casino patron will continue to make \$5 bets on red in roulette until she has won one of these bets. On any bet, she either wins \$5 with probability $18/38$ or loses \$5 with probability $20/38$. What is the distribution of X = the number of bets necessary to win a bet?

- b. A large number of insects are expected to be attracted to a certain variety of rose plant. A commercial insecticide is advertised as being 99% effective. Suppose 2000 insects infest a rose garden where the insecticide has been applied, and let X = the number of surviving insects. What type of random variable is X ?

- c. Suppose I wish to select a committee of size 3 from this class, which has 17 men and 7 women. Let X = the number of men selected. If the selection is random, what type of random variable is X ?

- d. On any given weekday, a 911 dispatcher receives an average of 14 calls. Let X be the number of phone calls received by the dispatcher on a Tuesday. What type of random variable is X ?

Chapter 4 Summary

Distribution	Summary	Formulas
Binomial Distribution	A binomial experiment satisfies these conditions: 1. The experiment has a fixed number of n of independent trials. 2. There are only two possible outcomes for each trial. Each outcome can be classified as a success or a failure. 3. The probability of success p is the same for each trial. 4. We are counting the number of successes. The parameters of a binomial distribution are n and p.	n = the number of trials. x = the number of successes in n trials. p = the probability of success in a single trial. q = the probability of failure in a single trial = $1 - p$. The probability of exactly x successes in n trials is $P(x) = {}_nC_x p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}, x = 0, 1, 2, \dots, n.$ $E(X) = \mu = np$ $\text{Var}(X) = \sigma^2 = np(1-p), \text{SD}(X) = \sigma = \sqrt{np(1-p)}$ $P(X = x) = \text{binompdf}(n, p, x) = \text{BINOM.DIST}(x, n, p, \text{FALSE})$ $P(X \leq x) = \text{binomcdf}(n, p, x) = \text{BINOM.DIST}(x, n, p, \text{TRUE})$
Geometric Distribution	A geometric distribution is a discrete probability distribution of a random variable X that satisfies these conditions. 1. A trial is repeated until a success occurs. 2. The repeated trials are independent of each other. 3. The probability of success p is the same for each trial. 4. The random variable x represents the number of the trial in which the first success occurs. The parameter of a geometric distribution is p.	x = the number of trials in which the first success occurs p = probability of success in a single trial. The probability that the first success occurs on trial number x is $P(x) = (1-p)^{x-1} \cdot p, x = 1, 2, 3, \dots$ $E(X) = 1/p$ $\text{Var}(X) = (1-p)/p^2, \text{SD}(X) = \sqrt{1-p}/p$ $P(X = x) = \text{geompdf}(p, x)$ $P(X \leq x) = \text{geomcdf}(p, x)$
Poisson Distribution	The Poisson distribution is a discrete probability distribution of a random variable X that satisfies these conditions. 1. The experiment consists of counting the number of times x an event occurs over a specified interval of time, area, or volume. 2. The probability of the event occurring is the same for each interval. 3. The number of occurrences in one interval is independent of the number of occurrences in other intervals. The parameter of the Poisson distribution is μ.	x = the number of occurrences in the given interval. μ = the mean number of occurrences in a given interval. The probability of exactly x occurrences in an interval is $P(x) = \frac{e^{-\mu} \mu^x}{x!}, x = 0, 1, 2, \dots$ $E(X) = \mu$ $\text{Var}(X) = \mu, \text{SD}(X) = \sqrt{\mu}$ $P(X = x) = \text{poissonpdf}(\mu, x) = \text{POISSON.DIST}(x, \mu, \text{FALSE})$ $P(X \leq x) = \text{poissoncdf}(\mu, x) = \text{POISSON.DIST}(x, \mu, \text{TRUE})$

For a random variable X with associated probability mass function $P(x)$,

- For $P(x)$ to be valid, $0 \leq P(x) \leq 1$ and $\sum P(x) = 1$.
- $E(X) = \mu = \sum xP(x)$. • $\text{SD}(X) = \sqrt{\text{Var}(X)} = \sqrt{\sum x^2 P(x) - \mu^2}$.
- $\text{Var}(X) = E((x - \mu)^2) = \sigma^2 = \sum (x - \mu)^2 P(x) = \sum x^2 P(x) - \mu^2$.

For discrete random variables (one that can only take on whole number values),

$P(X < x) = P(X \leq x - 1)$	Example: $P(X < 4) = P(X \leq 3)$
$P(X > x) = 1 - P(X \leq x)$	Example: $P(X > 4) = 1 - P(X \leq 4)$
$P(X \geq x) = 1 - P(X \leq x - 1)$	Example: $P(X \geq 4) = 1 - P(X \leq 3)$

Chapter 5: Normal Probability Distributions and the CLT

Topic 21

Continuous Random Variables

Lecture

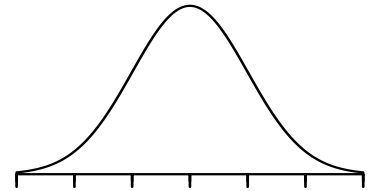
In chapter 4 we learned that a random variable is **discrete** if

Not all random variables are discrete. Some can
E.g.,

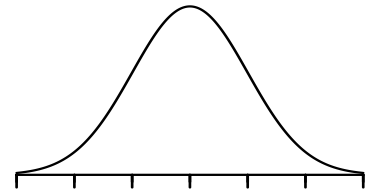
Instead of probability mass functions that describe $P(X = x)$ like discrete distributions have, continuous distributions have probability density functions (**pdfs**) that give an idea of how the values are distributed. **Probabilities** for continuous distributions can be modeled as the area under the curve of the pdf.

Notes:

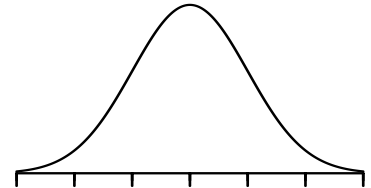
1. $P(a \leq X \leq b)$:



2. $P(-\infty < X < \infty)$:

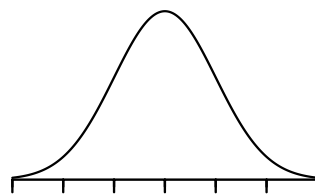
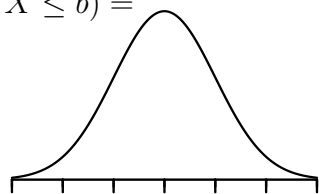


3. $P(X = a) =$



4. $P(X < a) = P(X \leq a)$.

5. $P(a \leq X \leq b) =$



6. $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b).$

Properties of pdfs:

1. All pdf values must be greater than or equal to 0.
2. The total area under a pdf must be 1.

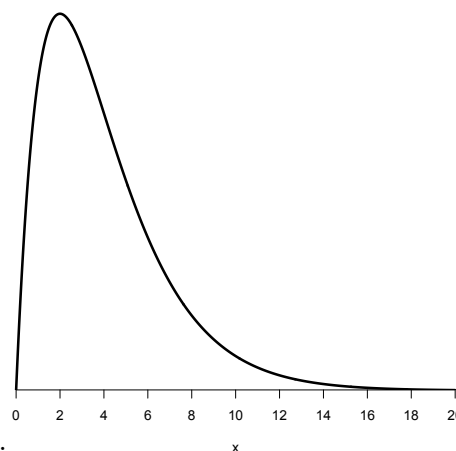
Note: It is okay if any pdf value is greater than 1 as long as both of the above properties hold.

Activities

Activity 21-1: Continuous Distributions

Consider the continuous distribution below. Suppose it is known that $P(X < 2) = 0.2642$ and $P(X \geq 12) = 0.01735$.

- a. Shade in the region of that curve that represents $P(X < 2)$ and shade in the region of the curve that represents $P(X \geq 12)$.
- b. Give the values for $P(X \leq 2)$ and $P(X > 12)$.

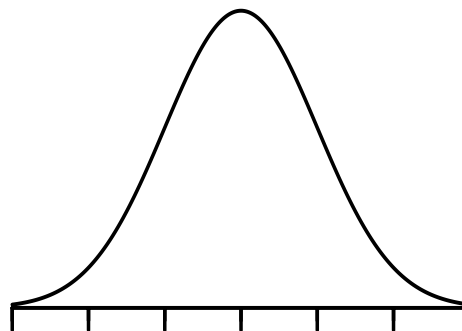


- c. Find $P(X > 2)$, $P(X \geq 2)$, $P(X < 12)$, and $P(X \leq 12)$.
- d. Find $P(2 < X < 12)$.

The Normal Distribution

Lecture

Normal distributions all have the same bell-shape, as shown to the right. A particular normal pdf is specified by giving its mean μ and standard deviation σ . The mean occurs at the peak of the curve. The standard deviation is the distance between the mean and the first tick mark next to it.



Definition: A random variable X has a **normal (Gaussian) distribution** with parameters μ, σ^2 if X has probability density function (pdf):

$$f(x) = \frac{1}{\sqrt{2\pi} \cdot \sigma} \exp \left\{ -\frac{(x - \mu)^2}{2\sigma^2} \right\}, \quad -\infty < x < \infty, \quad -\infty < \mu < \infty, \quad \sigma^2 > 0.$$

- In this case, we say that X is normally distributed with mean μ and variance σ^2 , and write $\boxed{X \sim N(\mu, \sigma^2)}$.

Properties of a Normal Distribution

1. Normal densities are always bell-shaped.
2. Normal densities are symmetric about the mean μ .
3. The normal distribution follows the empirical rule.

Why are normal random variables so important?

1. Many random phenomena are approximately normally distributed.
(Examples: Heights of men or women in this class, arsenic concentrations in groundwater, diameter of a tree, annual snowfall, etc.)

2. Normal probabilities can be used to approximate binomial probabilities. This was the original use of the normal distribution, and the result governing this approximation is the **DeMoivre-Laplace Theorem** (1812).
3. Sample means drawn from virtually any population are approximately normal! This result is one of the most important results in statistics and is known as the **Central Limit Theorem**.

Two ways to obtain normal probabilities:

1. Use z -score(s) and table.

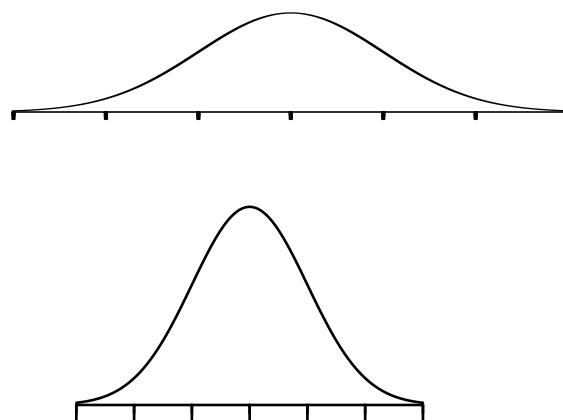
Recall that when we have a value that comes from a distribution with mean μ and standard deviation σ , a z -score or standard score is given by $z = \frac{x - \mu}{\sigma}$. This is commonly used to compare values from different distributions.

Steps to compute normal probabilities:

- (1) Convert the value or values in question to z -scores.
- (2) Look up the z -score in the Z table to find the area to the left.

Example: Suppose $X \sim N(\mu = 2, \sigma^2 = 16)$. Find $P(X < 3)$.

$P(X < 3)$ =



2. Use calculator or Excel.

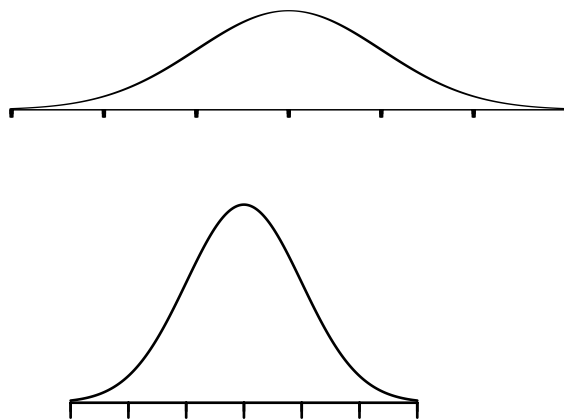
Two ways to obtain normal percentiles:

1. Use table and z -scores

Steps to compute normal percentiles:

- (1) Look up the area to the left in the Z table to find the z -score. Convert the value or values in question to z -scores.
- (2) Reverse the z -score formula to find x :

Example: Suppose $X \sim N(\mu = 2, \sigma^2 = 16)$. Find the value that separates the lowest 5% of the data from the rest.



2. Use calculator or Excel.

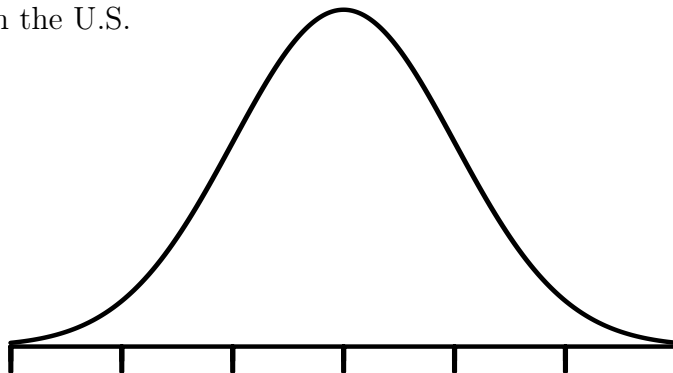
Example: If the amount of liquid in a can of soda is normally distributed with a mean of 12 oz and a standard deviation of 0.13 oz, what is the probability that a can will have between 11.61 and 12.39 oz of liquid? 95% of the cans have between what two amounts of liquid?

Activities

Activity 22-1: Baby Weights

Birth weights of babies in the United States can be modeled by a normal distribution with mean 3300 grams (about 7.3 pounds) and standard deviation 570 grams (about 1.3 pounds). Babies weighing less than 2495 grams (about 5.8 pounds) are considered to be of low birth weight.

- a. On the graph, fill in the missing numbers on the x axis to represent the weights of newborn babies in the U.S.



- b. On the graph in part a, shade in the region whose area corresponds to the probability that a randomly selected baby will have a low birth weight.
- c. Based on this shaded region, make an educated guess as to the proportion of babies born having a low birth weight. Remember that the total area under the normal curve is equal to 1.
- d. Calculate the z-score for a birth weight of 2495 grams.
- e. Use a z-table to find the probability (or proportion) of babies born having a low birth weight. That will give us the value for $P(Z < z)$ where z is the z-score you computed in part d. That will also give the value for $P(X < 2495)$, where X is the weight of a newborn baby.

- f. Use the `normalcdf` function in the calculator or the `=NORM.DIST` function in Excel to find the probability that a baby will be low weight (less than 2495). Compare your answer to part e.
- g. Now find $P(X > 4536)$ using technology.
- h. Now find $P(3000 < X < 4000)$ using technology.
- i. If we had a sample of 2,500,000 babies, how many of them would you expect to be low weight? [Hint: use your answer from parts e or f to help you.]
- j. How little would a baby have to weigh to be among the lightest 2.5% of all newborns? Use the `invNorm` function in your calculator or the `=NORM.INV` function in Excel to find this answer.

Definitions

A **percentile** is defined by how much data falls below the value in question. If you find a value that is the cutoff for the lowest 2.5% of the data, then that value is the 2.5th percentile.

- k. Now find the 2.5th percentile using a z -table and z -score. This is the last time you'll have to use the Z-table. I recommend you just use Excel or the calculator from now on.
- l. How much would a baby have to weigh to be among the heaviest 10% of all newborns? You can use any method to find this. [Hint: think about what percentile this would be.]

Extra Space for Notes

Topic 23

The Central Limit Theorem

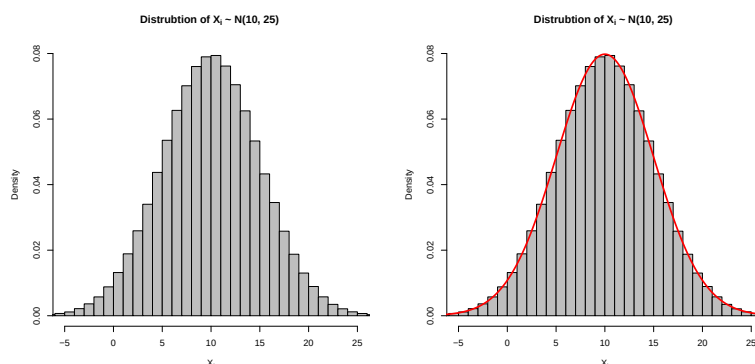
Lecture

Remember that the sample mean is $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} = \frac{1}{n} \sum_{i=1}^n X_i$.

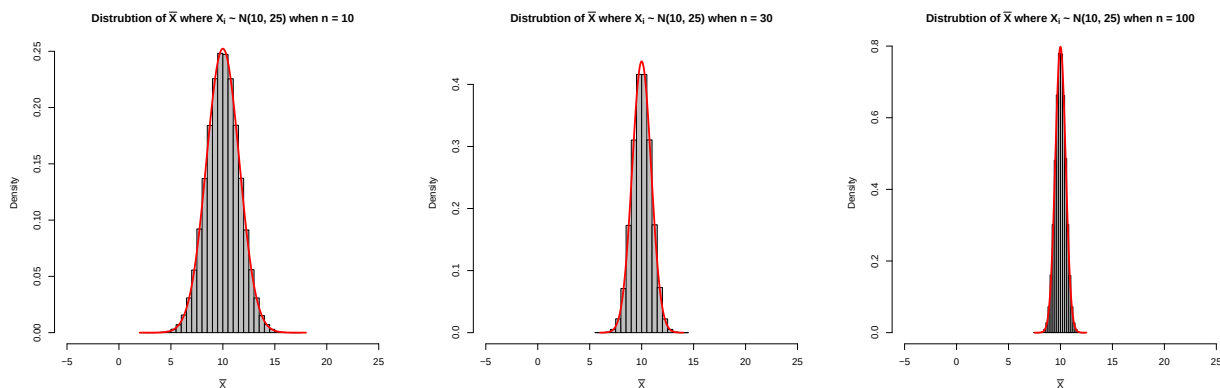
We can think of each observation X_i as a random variable with its own distribution. For example, it might be the case that each observation comes from a normal distribution: $X_i \sim N(\mu, \sigma^2)$.

We can also think of the sample mean as a random variable with its own distribution. Our goal is to figure out what that distribution will be.

Let's suppose we have random variables each independently drawn from a normal distribution with mean 10 and variance 25 (so standard deviation is 5). We can say $X_i \sim N(10, 25)$. If we generate samples of this random variable and plot a histogram, we get

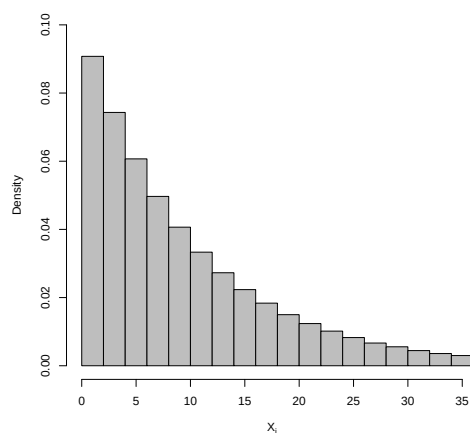


Now, let's take samples of size n for different n values: 10, 30, and 100. If we take many samples of size n and find the sample means for each of those samples, then plot the histogram of the sample means we get the following plots.



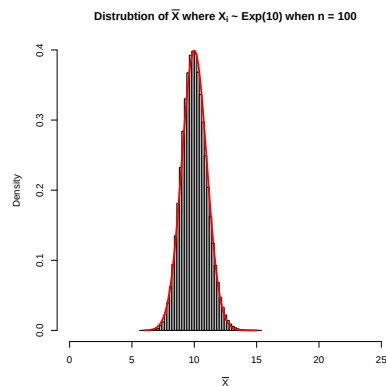
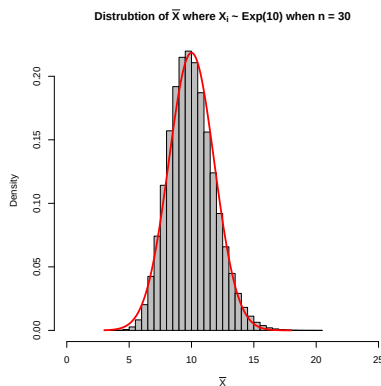
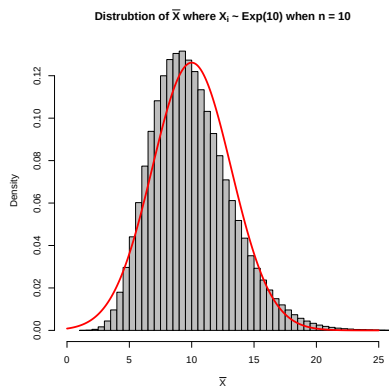
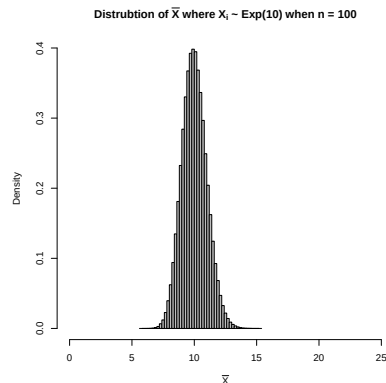
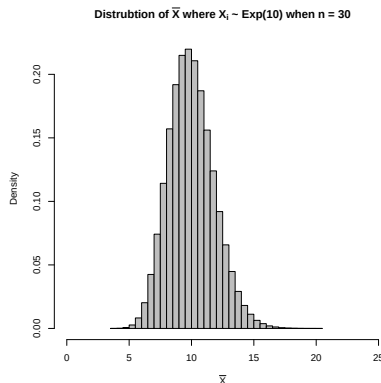
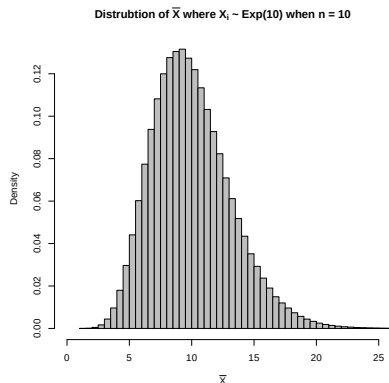
What is happening?

Distribution of $X_i \sim \text{Exp}(10)$



What if X_i is not normally distributed? Suppose $X_i \sim \text{exp}(\mu)$. What are some properties of X_i ?

Below are some distributions of $\bar{X}$ when $X_i \sim \text{exp}(10)$.



Now what is happening?

Central Limit Theorem Statement:

Under certain conditions, when $X_1, \dots, X_n$ are a random sample from *any* distribution with mean μ and variance $\sigma^2 < \infty$, then the sample mean: $\bar{X} \dot{\sim} N(\mu, \sigma^2/n)$.

Conditions?

Example: Suppose that from past experience a professor knows that the test score of a student taking her final examination is a random variable with mean 70 and standard deviation 8. What is the approximate probability that the class average is above 73 if there are 60 students in the class?

Activities

Activity 23-1: Penny Ages

The following histogram displays the distribution of ages (in years) for a population of 1000 pennies in circulation. Some summary info for this distribution of ages is also given.

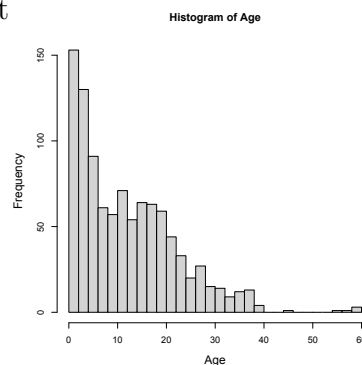
Number of Pennies	Mean	Standard Deviation	Min	Q_1	Median	Q_3	Max
1000	12.44	10.12	0	4	11	19	60

- a. Identify the observational units and variable of interest here. Is this variable quantitative or categorical?

Observational units:

Variable:

Quantitative or categorical?



- b. If you regard these 1000 pennies as a population from which you can select samples, are the values in the previous table parameters or statistics? What symbols represent the mean and standard deviation?
- c. Does this population of coin ages roughly follow a normal distribution? If not, what shape does it have?

Instead of taking a sample from the 1000 coins by hand, below is a table of the population of 1000 coins that shows their ages and the frequencies of those ages. We can think of each coin as being assigned an ID number as well. Then you can randomly generate numbers between 1 and 1000 to get a random sample of these coins. For example, if you randomly generate the number 921, then your coin would have an age of 28 since 921 falls between 916 and 927.

Age	ID	Age	ID	Age	ID
0	1-45	17	682-715	34	964-965
1	46-101	18	716-744	35	966-970
2	102-153	19	745-772	36	971-977
3	154-234	20	773-803	37	978-986
4	235-283	21	804-827	38	987-990
5	284-341	22	828-847	39	991
6	342-374	23	848-864	40	992-994
7	375-404	24	865-880	46	995
8	405-435	25	881-890	55	996
9	436-459	26	891-900	58	997
10	460-492	27	901-915	59	998
11	493-532	28	916-927	60	999-1000
12	533-563	29	928-932		
13	564-592	30	933-942		
14	593-617	31	943-949		
15	618-647	32	950-956		
16	648-681	33	957-963		

- d. Randomly sample 10 coins from the table and record their ages.
- e. Calculate the sample mean age of your 10 coins.
- f. Take two more samples of size 10 and find the sample mean of each of those. Then record the three sample means in the table below.

Sample	1	2	3
Sample Mean ($\bar{x}$)			

- g. Did you get the same value for the sample mean all three times?

Definitions

The phenomenon in which different samples will yield different statistics is called **sampling variability**. The sample mean varies from sample to sample according to a predictable long-term pattern known as a **sampling distribution**. The **Central Limit Theorem (CLT)** tells us the the sampling distribution of the sample mean will be approximately normal with a mean equal to μ and a standard deviation equal to $\sigma/\sqrt{n}$.

- h. Find the mean and standard deviations of the three sample mean values.

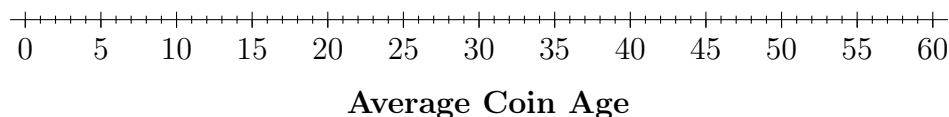
Mean of $\bar{x}$ values:

SD of $\bar{x}$ values:

- i. Is the mean of the $\bar{x}$ values reasonably close to the population mean of $\mu = 12.44$? Is the standard deviation greater than, less than, or about equal to the population standard deviation of $\sigma = 10.12$?

Now consider taking a random sample of 30 coins from this same population. By taking 3 samples of 10 pennies each, you have essentially done this already. Consider all of your observations as a random sample of size 30. (Note that you are ignoring the small possibility that a coin could be repeated in your sample of 30.) The sample mean of this sample of 30 coins is exactly the mean of your 3 sample means recorded in part h.

- j. Now produce a dot plot of all the sample means from you and all of your classmates.



- k. Does this distribution appear to be centered at the population mean ($\mu = 12.44$)?
- l. Do the values appear to be more or less spread out than either the population distribution or the distribution of your three sample means of size 10? Remember that the histogram of the original population is at the beginning of this activity.
- m. Does this distribution appear to be closer to a normal (bell) shape than the distribution of ages in the original population?
- n. Use the CLT to find the (approximate) probability that any sample mean would be greater than 15 years. Compare this to the proportion of means in the dotplot in part j that are above 15.

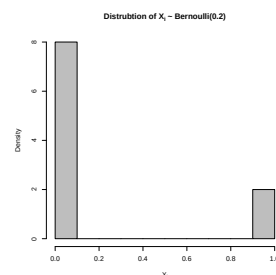
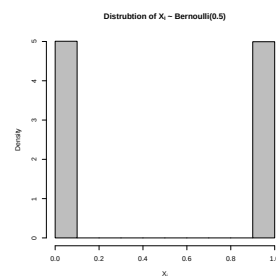
Topic 24

The Central Limit Theorem for Proportions

Lecture

As stated above, the CLT holds for any distribution. Consider Bernoulli trials. Let X_i be i.i.d. Bernoulli trials with $P(X_i = 1) = p$. Then, in this case,

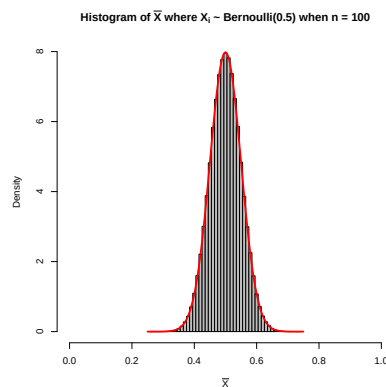
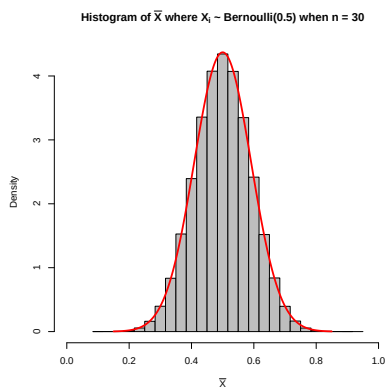
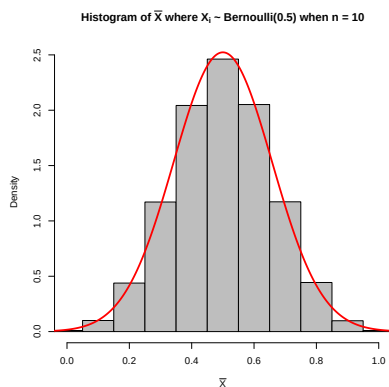
$$\bar{X} = \frac{X_1 + X_2 + \cdots + X_n}{n} =$$



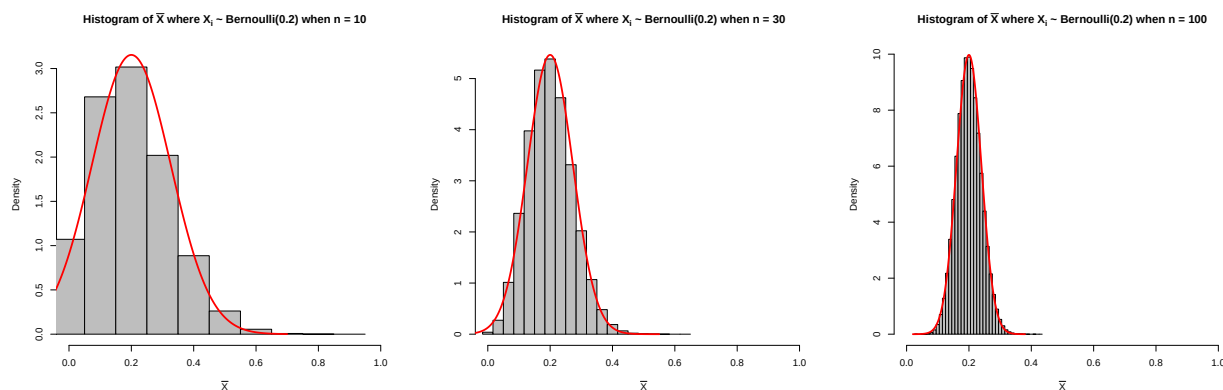
For a sample proportion, $\hat{p} \stackrel{\cdot}{\sim} N\left(p, \frac{p(1-p)}{n}\right)$.

How large should n be now?

Consider when $p = 0.5$.



Now when $p = 0.2$.



Rule of thumb:

Example: It is known that 83% of the population enjoys the movie *High School Musical*. If you take a random sample of 72 people from that population, what is the approximate probability that fewer than 80% of those in the sample will enjoy *High School Musical*?

Definitions

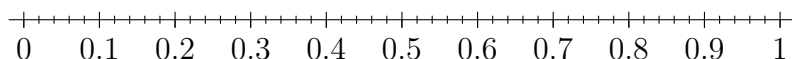
The distribution of the sample proportions from sample to sample is called the **sampling distribution** of the sample proportions. These simulated sample proportions approximate the theoretical sampling distribution derived from all possible random samples. Increasing the number of samples in the simulation improves this approximation and helps you recognize the longterm pattern of the results.

Activities

Activity 24-1: Flipping Coins

In this activity, we will show that proportions follow a distribution that is approximately normal.

- a. Take a coin and flip it 20 times. Record the number coin flips that came up heads and then find the proportion of your coin flips that were heads.
- b. Using information from the entire class, create a dot plot of the proportion of the 20 coin flips that came up heads.



Proportion of Heads

- c. According to the central limit theorem, where is the theoretical mean of the sampling distribution of the proportion of heads?
- d. According to the central limit theorem, where is the theoretical standard deviation of the sampling distribution of the proportion of heads?
- e. What percentage of the sample proportion of heads from the class were below 0.4?
- f. According to the central limit theorem, what is the probability of getting a sample proportion less than 0.4 supposed to be in this instance? Compare this answer to the answer in part e and comment on whether this seems to be a good approximation.

Chapter 5 Summary

1. Continuous distributions have pdfs instead of pmfs that describe how the probabilities are distributed.
2. All pdf values must be at least zero and the total area under a pdf must be 1.
3. For all continuous distributions, $P(X < x) = P(X \leq x)$, $P(X > x) = P(X \geq x)$, and $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$.
4. A normal distribution is bell-shaped and centered at its mean.
5. To find probabilities for a variable X that is normally distributed, use a z -score and table with $z = \frac{x-\mu}{\sigma}$ or a calculator with `normalcdf(lower, upper,  $\mu$ ,  $\sigma$ )`.
6. To find percentiles for a variable X that is normally distributed, use a table and inverse z -score formula $x = \mu + \sigma z$ or a calculator with `invnorm(area left,  $\mu$ ,  $\sigma$ )`.
7. To find probabilities for a sample mean $\bar{X}$ when n is large enough, use a z -score and table with $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ or a calculator with `normalcdf(lower, upper,  $\mu$ ,  $\sigma/\sqrt{n}$ )`.
8. To find percentiles for a sample mean $\bar{X}$ when n is large enough, use a table and inverse z -score formula $\bar{x} = \mu + z \cdot \sigma/\sqrt{n}$ or a calculator with `invnorm(area left,  $\mu$ ,  $\sigma/\sqrt{n}$ )`.
9. To find probabilities for a sample proportion $\hat{p}$ when n is large enough, use a z -score and table with $z = \frac{\hat{p}-p}{\sqrt{p(1-p)/n}}$ or a calculator with `normalcdf(low, up,  $p$ ,  $\sqrt{p(1-p)/n}$ )`.
10. To find percentiles for a sample proportion $\hat{p}$ when n is large enough, use a table and inverse z -score formula $\hat{p} = p + z\sqrt{p(1-p)/n}$ or a calculator with `invnorm(area left,  $p$ ,  $\sqrt{p(1-p)/n}$ )`.
11. In Excel, you can find the following functions:
 - $P(X < x) = \text{NORM.DIST}(x, \text{mean}, \text{standard deviation}, \text{TRUE})$
 - $P(X > x) = 1 - \text{NORM.DIST}(x, \text{mean}, \text{standard deviation}, \text{TRUE})$
 - $P(a < X < b) = \text{NORM.DIST}(b, \mu, \sigma, \text{TRUE}) - \text{NORM.DIST}(a, \mu, \sigma, \text{TRUE})$
 - You can find the cutoff value by using `=NORM.INV(area left, mean, SD)`
12. Central Limit Theorem Statements:
 - (1) $\bar{x}$ is approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.
 - (2) $\hat{p}$ is approximately normally distributed with mean p and standard deviation $\sqrt{\frac{p(1-p)}{n}}$.

Chapter 6: Confidence Intervals

Topic 25

Intro to Confidence Intervals and CIs for Proportions

Lecture

Now that we have learned about probability, we can return to inferential statistics. First, an updated symbol table:

Inferential Statistics Overview Review

Symbol	Description
n	Sample size
N	Population size
x or x_i	Observation
$\bar{x}$	Sample mean
μ	Population mean
s	Sample standard deviation
σ	Population standard deviation
M	Median
Q_1	Lower quartile
Q_3	Upper quartile
IQR	Interquartile range
df	Degrees of freedom
z	Standard score or z -score
x	Number of "successes"
$\hat{p}$	Sample proportion
p	Population proportion or probability of success
q	$1 - p$ or probability of failure

Throughout these notes, it will help to reference the "Inference Summary Sheet" handout on Canvas.

A point estimate is a single value estimate for a population parameter.

- The point estimate for the population mean μ is
- The point estimate for the population standard deviation σ is
- The point estimate for the population proportion p is

One issue we face is that our estimates are nearly always wrong! They are wrong not because we make mistakes (though this is possible), but because they are based on limited, random samples. A sample mean, for instance, is random due to the sample itself being random. If a new sample is taken, a different mean will be calculated. This is known as sampling variability.

We deal with the fact that we are wrong by determining how wrong we may be. We quantify this by calculating a margin of error, which we add and subtract from our point estimate in order to form a confidence interval (CI).

This way, we have not only our “best guess” for the value of the unknown parameter, but we have an idea of the extent to which our best guess may be in error.

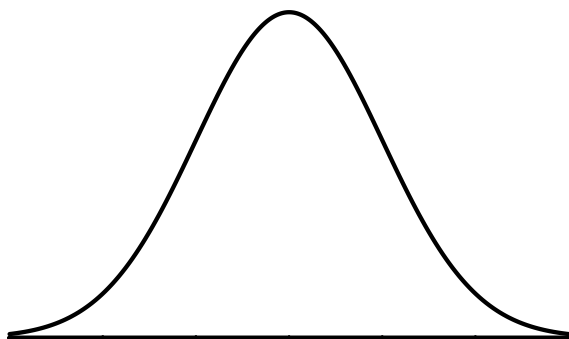
Confidence Interval for a Single Population Proportion, p

The formula for a confidence interval for a population proportion (p) is

$$\hat{p} \pm z_c \cdot \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where z_c is known as the **critical value** that corresponds to a certain **confidence level**. The **margin of error** is the part of the formula that follows the “ $\pm$ ”. This confidence interval is only valid if our sample size is large enough. **Must check** that $n\hat{p} \geq 5$ and $n\hat{q} \geq 5$.

How to determine z_c :



In a calculator/Excel:

Confidence Level	z_c value
90%	1.645
95%	1.96
98%	2.326
99%	2.576

Example: Suppose a political polling firm would like to determine whether or not a majority of registered voters in a certain congressional district plan to vote for the incumbent congressional representative. The firm conducts a poll in which 300 registered voters are surveyed. Of these 300 voters, 159 state that they are planning on voting for the incumbent candidate. Construct a 95% CI for the proportion of voters in this district who plan to vote for the incumbent representative.

Writing a confidence interval

The way I would like the confidence intervals written in this class is in this double inequality form:

$$\boxed{\text{Lower Bound of CI} < \text{Parameter of Interest} < \text{Upper Bound of CI}}$$

This not only is easy to read, it also lets me know that you know the parameter for which you are finding a confidence interval.

Interpreting a confidence interval

Once the CI is constructed, we need to give an English interpretation. The general forms of some acceptable interpretations are:

1. "We are (confidence level)% confident that the true (parameter) lies between (lower bound) and (upper bound)."
2. "Under repeated sampling, (confidence level)% of similarly constructed intervals will contain the true (parameter)."

Note that we do **not** say, "There is a (confidence level)% chance that the true (parameter) lies between (lower) and (upper)."

This is because the value of the parameter is fixed (but unknown). Once we have constructed a CI, the parameter either is or is not contained in this interval. We just don't get to know whether it is or is not.

The key here is that the CI is the thing that is random, not the parameter. We can talk about the probability that a random CI will contain the parameter, not the probability that a specific CI contains the parameter; it either does or does not.

The correct interpretations above are in very basic form. We will always need to include context in our interpretations as well.

Interpret the confidence interval from the example on the previous page:

Activities

Activity 25-1: Kissing Couples

To avoid bumping noses, people typically lean their heads either to the left or the right when kissing their partners. The question we will attempt to answer is whether people are more likely to lean to the right or to the left when kissing. Suppose a random sample of 154 kissing couples was selected and it was found that 92 of them leaned to the right when kissing.

- a. Identify the observational units and variable in this study.

- b. Describe the sample data numerically with a proportion and graphically with a bar chart.

- c. Is the proportion you found in part b a parameter or a statistic? How do you know? What symbol do we use to denote it?
- d. Describe in words the relevant population parameter of interest in this study. What symbol would you use to represent it?
- e. Is the Central Limit Theorem for a sample proportion valid in this context? If so, what does it predict for the shape, center, and spread of the sampling distribution of the sample proportion? You can either write those out or include a sketch of this distribution, indicating with symbols the mean and standard deviation.

Since we do not know the true population parameter here, we will need to estimate it with the sample statistic.

Definitions

The estimated standard deviation of the sample statistic $\hat{p}$ is called the **standard error** of $\hat{p}$. The true standard deviation of $\hat{p}$ is

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$$

and the standard error of $\hat{p}$ is

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}.$$

- f. Calculate the standard error of $\hat{p}$ for the proportion of kissing couples who turn their heads to the right.
- g. Now multiply that standard error by 1.96 (the critical value for a 95% confidence interval) to obtain the margin of error for the 95% confidence interval for the true proportion of kissing couples who lean their heads to the right.
- h. Now take this margin of error and subtract it from $\hat{p}$ to get the lower bound of the 95% confidence interval. Then take the margin of error and add it to $\hat{p}$ to get the upper bound of the 95% confidence interval. Then write the 95% confidence interval in double-inequality form.
- i. Do you know for sure whether or not the actual value of the population proportion is contained in this interval?
- j. Interpret this interval in the context of the problem. Begin by saying "We are 95% confident that..."
- k. Based on this interval, do we have evidence kissing couple more often lean to the right? Explain.

- l. Now compute a 99% confidence interval for the true proportion of kissing couple that lean to the right.
- m. Find the midpoint of the 95% confidence interval and then find the midpoint of the 99% confidence interval. Compare these numbers.
- n. Find the width of the 95% confidence interval and the width of the 99% confidence interval. Compare them. Which one is larger? Why is that the case?
- o. In part g, you found the margin of error for the 95% confidence interval. How does this compare to the width of the 95% confidence interval? How exactly are they related?
- p. Find the margin of error for the 95% confidence interval if $\hat{p}$ is the same but $n = 1540$ instead of 154. How does increasing the sample size affect the margin of error?

Activity 25-2: Critical Values

Before calculating a confidence interval, you need to be able to determine the critical value, z_c . As an example, suppose you want to find the value of z_c for a 97% confidence interval. In other words, find the value z_c such that the area under the standard normal curve between $-z_c$ and z_c is equal to 0.97 by following these steps:

- a. According to the table of critical values on the inference summary sheet, between what two values do we know the critical value for a 97% confidence interval must lie?
- b. Draw a sketch of the standard normal curve and shade the area corresponding to the middle 97% of the distribution. Also indicate roughly where $-z_c$ and z_c fall in your sketch.
- c. Based on your sketch, what is the total area under the standard normal curve to the left of the value z_c ?
- d. Find the value (which is z_c) that has this area that you answered in part c to its left under the standard normal curve. You can use your calculator or Excel to find this.
- e. For the value you found in part d, find $P(-z_c < Z < z_c)$ for a standard normal random variable Z and show that it is approximately equal to 0.97 (since this is the critical value for a 97% confidence interval).

f. Repeat this process to find the critical value for a 99.9% confidence interval.

g. What is the critical value for a 100% confidence interval? Explain.

Keep in mind: The more confident we want to be, the larger the critical value becomes, which means the confidence interval will be wider. If we want to be 100% confident that the true proportion is in our interval, we could say the interval is $0 < p < 1$, but that would not be at all useful.

Extra Space for Notes

Topic 26

The t Distribution and Confidence Intervals for Means

Lecture

If σ is known, then the formula for a confidence interval for a population mean (μ) is

$$\bar{x} \pm z_c \cdot \sigma / \sqrt{n}$$

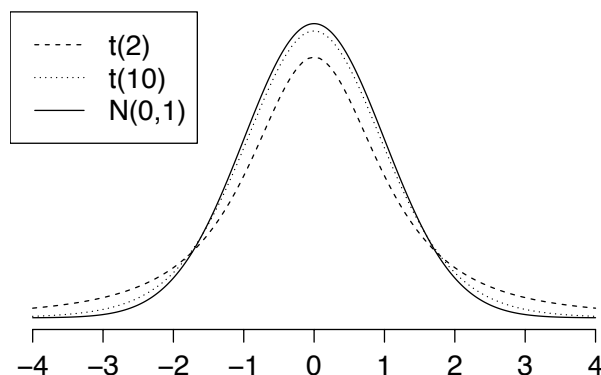
Issue with this:

The t -distribution

The t -distribution is actually an entire *family* of distributions with a single parameter called the degrees of freedom (d.f.).

This family of distributions has the following properties:

1. They are symmetric about zero and mound-shaped (like the standard normal distribution).
2. The t -distribution has wider tails than the normal distribution to account for the extra variability in estimating σ .
3. The degrees of freedom completely specify which t -distribution to use.



4. As the degrees of freedom increase, the t -distribution gets closer and closer to the standard normal distribution.
5. We can “convert to t ” in a similar way that we converted to z before:

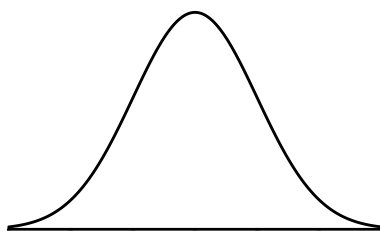
$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

The t -distribution allows us to account for the fact that we also have to estimate the standard deviation. That gives us the following formula for a confidence interval for population mean (μ):

$$\boxed{\bar{x} \pm t_c \cdot s/\sqrt{n}; \quad df = n - 1}$$

where t_c is known as the **critical value** that corresponds to a certain **confidence level**. Each t_c value will have associate degrees of freedom, $df = n - 1$. The **margin of error** is the part of the formula that follows the “ $\pm$ ”. This confidence interval is only valid if our sample size is large enough. **Must check** that the sample is random and $n \geq 30$ or data come from a normal distribution.

How to determine t_c :



In the calculator/Excel:

Example: An education researcher is interested in determining whether high school students give correct answers to SAT reading comprehension questions that refer to passages they have not read, at a rate that differs from what would be expected by random chance. 12 high school students are each given 100 SAT reading comprehension questions to answer. None of the students have read the relevant passage before answering the questions. If the students were just randomly guessing, then they should on average answer 20 correctly just by chance.

The average number of correct answers given by the 12 students is 33.7, with a standard deviation of 10.2. Construct a 99% CI for the average number of correct answers given by all students in this scenario.

Activities

Activity 26-1: t vs z

Suppose you need to find the critical value t_c for a 95% confidence interval based on a sample size of $n = 10$.

- Draw a rough sketch of the t -distribution with 9 degrees of freedom. Note that the degrees of freedom is equal to $n - 1$ for a confidence interval for a mean. [Hint: It should look very much like a standard normal curve.]

- b. The critical value t_c for a 95% confidence interval is the value such that 95% of the area under the curve is between $-t_c$ and t_c . Shade this area on your sketch.
- c. What is the area to the right of t_c under the curve? [Hint: This area is not .05.]
- d. Use the `invT` function in your calculator or the `T.INV` function in Excel to find t_c .
- e. Now use the t-table to find the critical value for a 95% confidence interval when we have 9 degrees of freedom. Verify that it matches your answer from part d.
- f. Confirm that the critical value you found in parts d and e is correct by using the `tcdf` function or the `=T.DIST` function to find $P(-t_c < T < t_c)$, where T follows a t distribution with 9 degrees of freedom, and show that the probability is approximately equal to 0.95.
- g. Write down what the critical value would be for a 95% z interval. Compare your t critical value for a 95% confidence interval to this z critical value. Which is larger? Does that make sense?
- h. Now find the t critical value for a 95% confidence interval when the degrees of freedom value is 90 (so the sample size is 91). How does this compare to the t critical value when $df = 9$?

Activity 26-2: M&M Selection

Brian Wansink, author of the popular book *Mindless Eating*, conducted research into factors that influence people to eat without conscious regard for how much they are eating. One of his studies involved a sample of 20 undergraduates from the University of Illinois at Urbana-Champaign. Participants were allowed to take as many M&M candies as they wished from a 24-ounce bowl for consumption during a lab study session that day. After they selected the M&Ms individually, the researcher determined how many M&Ms each person had selected. Suppose you plan to offer a study session and need to know how many candies to bring and therefore would like to estimate how many candies students tend to take on average. This data can be found in the **m&m.xlsx** file on Canvas.

- a. Is the variable *number of M&Ms selected* categorical or quantitative?

- b. Use Excel to produce either a boxplot or histogram of the this variable. Then draw a quick sketch of that plot here.

- c. Does it seem like the number of M&Ms selected is approximately normally distributed? If not, is our sample size large enough for the CLT to apply here?

Note: if the original data are not normally distributed and the sample size is less than 30, then the confidence interval may not be valid. While it is important to keep this in mind, we will continue in this activity and make the interval anyway.

- d. Use Excel to find the mean and standard deviation of the values in this data set and report them here with the proper symbols. Are the mean and standard deviation you calculated here parameters or statistics?

As you have seen, you can form an interval estimate of a parameter by starting with the sample statistic and going a certain number (determined by your confidence level) of standard deviations to either side of it. The Central Limit Theorem for a sample mean tells you the standard deviation of a sample mean $\bar{x}$ is $\sigma/\sqrt{n}$, where σ represents the population standard deviation and n the sample size.

- e. Explain what σ is in your own words. Do we know the value of σ for this M&M study?

When the population standard deviation σ is known, a confidence interval for a population mean is given by $\bar{x} \pm z_c \cdot \sigma/\sqrt{n}$. However, a major drawback to using this procedure is that it requires you to know the value of the population standard deviation σ , which you almost never know if you don't know the population mean μ .

Since we do not know the true population parameter here, we will need to estimate it with the sample statistic.

Definitions
The estimated standard deviation of the sample statistic $\bar{x}$ is called the standard error of $\bar{x}$. The true standard deviation of $\bar{x}$ is $\sigma_{\bar{x}} = \sigma/\sqrt{n}$ and the standard error of $\bar{x}$ is $SE_{\bar{x}} = s/\sqrt{n}$ where s is the sample standard deviation.

Now that we are estimating σ with s , we cannot use the standard normal distribution for critical values. Instead, we need to use the t distribution. Therefore, the formula for the confidence interval becomes $\bar{x} \pm t_c \cdot s/\sqrt{n}$.

- f. Compute a 95% confidence interval for the true mean number of M&Ms selected using the formula and then write it in double inequality form. Be sure to find t_c using either the t table, a calculator, or Excel (but you should know how to use either the t table or the calculator for exams).

- g. If you have a TI 83/84 calculator, use the **TInterval** function to find the 95% confidence interval and confirm it matches your answer to part f. If not, you can skip this problem.
- h. Interpret the interval you computed in parts f (and g if you did that part).
- i. Suppose the sample size had been 200 instead of 20, but the sample statistics turned out exactly the same. How would you expect a 95% confidence interval to differ in this case from the 95% confidence interval in part f (or g)? Explain.
- j. Suppose the sample size had been 200 but the sample distribution of number of candies had been skewed even more to the right. Would this t-interval calculation still be valid? Explain.

The χ^2 Distribution and Confidence Intervals for Standard Deviations and Variances

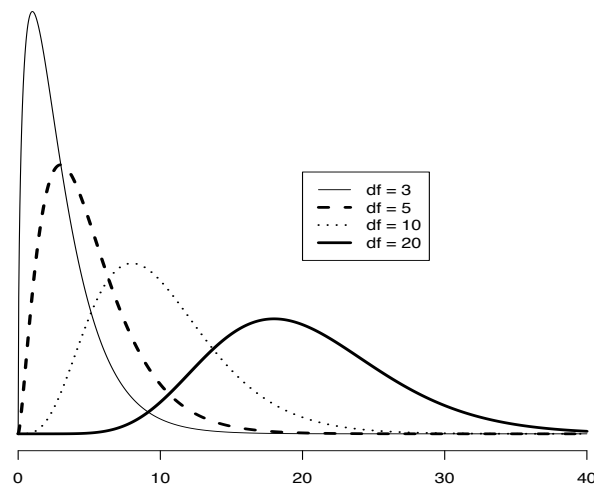
Lecture

Just like the population mean (μ) or the population proportion (p), the population variance and standard deviation are typically not known, yet it is often of interest to have an idea of how much our distribution varies. In this section, we will discuss how to create confidence intervals for the population variance and standard deviation.

The χ^2 distribution

Just as a special distribution was needed to get the confidence interval for a mean when σ was unknown, we must use what is known as a χ^2 distribution to obtain critical values for the confidence interval for σ^2 and σ . We cannot estimate these with a normal distribution.

Like a t -distribution, a χ^2 distribution is a family of distributions that are characterized by their degrees of freedom.



This family of distributions has the following properties:

1. They are not symmetric, they are skewed right.
2. The degrees of freedom completely specify which χ^2 -distribution to use since the mean is equal to the df and the standard deviation is equal to $\sqrt{2 \cdot df}$.
3. All χ^2 values are greater than 0.
4. As the degrees of freedom increase, the χ^2 -distribution gets more and more symmetrical.

5. We can “convert to χ^2 ” in a similar way that we converted to z and t before:

$$\chi^2 = \frac{(n-1)s^2}{\sigma^2}$$

From this last equation, we have:

The confidence intervals are given by:

$$\frac{(n-1)s^2}{\chi_R^2} < \sigma^2 < \frac{(n-1)s^2}{\chi_L^2}; \quad df = n - 1$$

and

$$\sqrt{\frac{(n-1)s^2}{\chi_R^2}} < \sigma < \sqrt{\frac{(n-1)s^2}{\chi_L^2}}; \quad df = n - 1$$

These χ_L^2 and χ_R^2 values are critical values. We need two different critical values since the χ^2 distribution is not symmetric. Original data must be normal.

We can obtain these critical values using either a table or Excel.

In Excel:

Example: A manufacturing plant quality control department wants to make sure that the assembly line is producing lead pipes with very little dispersion from the promised size of at least 10 inches. The QA manager samples 36 pipes and finds the average is 10.5 inches with a standard deviation of 0.2 inches. Assume the data are normally distributed. Calculate a 95% confidence interval for the mean and a 95% confidence interval for the standard deviation.

Activities

Activity 27-1: χ^2 Critical Values

TI calculators do not have a built in function to find χ^2 critical values, so let's try to determine critical values for the χ^2 distribution using the excel function =CHISQ.INV.

- Suppose we have $df = 36$ and we want to find the two χ^2 values for a 90% confidence interval for a standard deviation. Use the Excel Guide on Canvas to look up how to find the two values.
- Repeat part a if we want the critical values for a 99% interval.
- Use the $\chi^2\text{cdf}$ function in the calculator and put in the two critical values you found in part b as the lower and upper bounds, then put in 36 for the degrees of freedom. This is finding $P(\chi_L^2 < X < \chi_R^2)$ where X follows a χ^2 distribution with $df = 36$. Verify this probability is around 0.99.

Activity 27-2: M&M Spread

Recall the M&M selection from Activity 27-2. The standard deviation for these data was $s = 30.143$ and $n = 20$.

- a. What are the conditions for this interval to be valid?

Note that again the normality condition may be violated here, but we will compute the interval anyway.

- b. Find the values of χ_L^2 and χ_R^2 for a 95% confidence level using Excel.
- c. Find the 95% confidence interval for the population **variance** of number of M&Ms selected. Write it in double inequality form.
- d. Interpret the interval from part c.
- e. Find the 95% confidence interval for the population **standard deviation** of number of M&Ms selected. Write it in double inequality form.
- f. Interpret the interval from part e.

Chapter 6 Summary

Confidence intervals both give us an idea of an interval of plausible values for our unknown population parameter and help us quantify how uncertain we are in our estimates.

Constructing a confidence interval for a population proportion, p :

1. Identify the statistics $\hat{p}$ and n or calculate the sample proportion as $\hat{p} = \frac{x}{n}$ where x is the number of “successes.”
2. Verify that the sampling distribution of $\hat{p}$ can be approximated by a normal distribution by checking that $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$.
3. Find the critical value z_c that corresponds to the given level of confidence.
4. Find the left and right endpoints by calculating $\hat{p} \pm z_c \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$.
5. Write the confidence interval and provide an interpretation.

Note: Steps 3 and 4 can be done with the `1-PropZInt` command in the calculator.

Constructing a confidence interval for a population mean, μ :

1. Identify the statistics $\bar{x}$, s , and n or calculate them.
2. Verify that the sample is random and that $n \geq 30$ or the data come from a normal distribution.
3. Find the critical value t_c that corresponds to the confidence level with $df = n - 1$.
4. Find the left and right endpoints by calculating $\bar{x} \pm t_c \cdot s / \sqrt{n}$.
5. Write the confidence interval and provide an interpretation.

Note: Steps 3 and 4 can be done with the `TInterval` command in the calculator.

Constructing a confidence interval for a population variance and standard deviation, σ^2 and σ :

1. Identify the statistic s or s^2 or calculate them.
2. Verify that the sample is random and that the data come from a normal distribution.
3. Find the critical values χ_L^2 and χ_R^2 that correspond to the given level of confidence with $df = n - 1$ using Excel.
4. Find the left endpoint by calculating $\frac{(n-1)s^2}{\chi_R^2}$ for the variance or $\sqrt{\frac{(n-1)s^2}{\chi_R^2}}$ for the SD.
5. Find the right endpoint by calculating $\frac{(n-1)s^2}{\chi_L^2}$ for the variance or $\sqrt{\frac{(n-1)s^2}{\chi_L^2}}$ for the SD.
6. Write the confidence interval and provide an interpretation.

Chapter 7: Hypothesis Testing

Topic 28

Introduction to Hypothesis Testing

Lecture Part 1

There are two primary procedures we use in inferential statistics to draw inference on a population:

1. Confidence intervals (introduced in chapter 6)
2. Hypothesis tests (which we are introducing now!)

With confidence intervals, we set up an interval that contains bounds on what we believe are the plausible values of our parameter based on our sample.

Hypothesis tests are a bit different. Instead of obtaining bounds where we believe the parameter lies, we instead test to see if we have sufficient evidence to reject a null hypothesis (H_0) in favor of an alternative hypothesis (H_A).

Activities

Activity 28-1: High School Musical Proportion

Let's say that we want to know if the proportion of all students who have seen *High School Musical* has increased since Disney+ launched. Suppose we (somehow) know that 30% of students had seen the movie before Disney+. Suppose we take a completely unbiased, random sample of 100 students and calculate the proportion of them who have seen *High School Musical* to help answer my research question.

- a. What is the variable of interest in this situation?
- b. What is the parameter of interest?
- c. If my sample proportion from the 100 students is 90% ($\hat{p} = 0.9$), would you say this gives enough evidence that the true proportion of **all** students who have seen the movie has increased from 30%? Explain.

- d. If my sample proportion from the 100 students is 50% ($\hat{p} = 0.5$), would you say this gives enough evidence that the true proportion of **all** students who have seen the movie has increased from 30%? Explain.
- e. If my sample proportion from the 100 students is 35% ($\hat{p} = 0.35$), would you say this gives enough evidence that the true proportion of **all** students who have seen the movie has increased from 30%? Explain.
- f. If my sample proportion is 31% ($\hat{p} = 0.31$), would you say there is enough evidence that the proportion of **all** students who have seen the movie has increased from 30%?

This part of the activity is supposed to show that there must be some cutoff for $\hat{p}$ such that if $\hat{p}$ falls above this cutoff, then we would have sufficient evidence to suggest the true proportion has indeed increased and if $\hat{p}$ falls below this cutoff, then we do not have enough evidence to suggest the true proportion has increased.

- g. Suppose that in our sample of size 100, $\hat{p}$ really was equal to 0.35. So far, our only inference tool we have discussed is a confidence interval. Construct the 95% confidence interval for p in this scenario. From that confidence interval, can we be confident that the true proportion has increased from 30%?
- h. Using confidence intervals (and perhaps a bit of guessing and checking), come up with the smallest value for $\hat{p}$ that would give us evidence, at the 95% confidence level, that p has increased from 0.30.

Lecture Part 2

What is the cutoff for our sample proportion to be large enough to conclude that the true proportion has actually changed?

We determine this cutoff by assuming the true proportion is still 30% ($p = 0.30$) and then computing a test statistic from a set of data. We can use this test statistic to compute a p -value, which will tell us the probability of obtaining a test statistic at least as large as that which we obtained under the scenario that the null hypothesis is true. If this p -value is small enough, we reject that the true proportion is 0.30, otherwise we fail to reject that the true proportion is 0.30.

Hypothesis Testing: The Six Steps**Lecture****Hypothesis testing outline**

The hypothesis testing procedure can be performed in 6 steps. Note that how these steps are defined is subjective; other instructors and textbooks will define them differently, but the outcome will be the same.

1. State the parameter(s) of interest.
2. Set up the null & alternative hypotheses.
3. State the significance level (if we have one) and the corresponding critical value.
4. Compute the test statistic and p -value.
5. Make the statistical decision.
6. Interpret the results.

Example 1: Suppose a political polling firm would like to determine whether or not a majority of registered voters in a certain congressional district plan to vote for the incumbent congressional representative. The firm conducts a poll in which 300 registered voters are surveyed. Of these 300 voters, 159 state that they are planning on voting for the incumbent candidate. Do the results of this survey provide enough evidence to conclude that a majority of voters do plan to vote for the incumbent?

Example 2: An education researcher is interested in determining whether high school students give correct answers to SAT reading comprehension questions that refer to passages they have not read, at a rate that differs from what would be expected by random chance. 12 high school students are each given 100 SAT reading comprehension questions to answer. None of the students have read the relevant passage before answering the questions. If the students were just randomly guessing, then they should, on average, answer 20 correctly just by chance. The average number of correct answers given by the 12 students is 33.7, with a standard deviation of 10.2. Is this significantly different from what we would expect from just random guessing alone? Use $\alpha = 0.01$.

We will not work through these examples yet. First, let's walk through the steps of hypothesis tests.

Step 1: State the parameter(s) of interest.

The parameter of interest is a single value that summarizes what we want to know about the population. It is crucial that we know what this is and its symbol so we can set up our hypotheses correctly. For now, this will be either a population mean (____), proportion (____), variance (____), or standard deviation (____).

Step 2: Set up the null and alternative hypotheses.

In hypothesis testing, we usually make the null hypothesis (H_0) the proposal we would like to _____.

This proposal can be stated as a comparison between some unknown parameter (such as a true population mean or proportion) and some hypothesized value.

If we reject H_0 , we do so in favor of the alternative hypothesis, H_A . Thus if you are trying to find evidence in favor of a proposal, that proposal goes in H_A and its opposite goes in H_0 .

Two tailed or one tailed test?

When setting up the null and alternative hypotheses, we must determine which of three scenarios makes the most sense for the question at hand: a left-tailed test, a right-tailed test, or a two-tailed test.

In every case, the notion of equality goes in the _____ hypothesis and the notion of non-equality goes in the _____ hypothesis.

Two-tailed test

If the null hypothesis that we are trying to reject is that a population parameter is _____ to some value, and the alternative hypothesis is that this population parameter is _____ to this value, then the test is a two-tailed test.

In a two tailed test, we are willing to reject the null hypothesis if we find evidence that the unknown parameter is either less than or greater than the hypothesized value, and we do not specify ahead of time which it should be.

So, if the question is whether or not the true unknown parameter is "different from" or "not equal to" some specified value, then the test should be a two-tailed test.

One-tailed test

On the other hand, we sometimes are only interested in testing to see if the true unknown parameter is _____ than some specified value, or _____ than some specified value, but not in testing for both possibilities at once.

In this case, the test will be either "_____ tailed" (if our alternative hypothesis is that the parameter is less than the hypothesized value) or "_____ tailed" (greater than the hypothesized value).

Example 1: Steps 1 and 2. We would like to see if we have enough evidence to conclude that the majority of voters support the incumbent. In this scenario, the parameter is the true proportion of voters who support the incumbent. When the parameter is a proportion, we denote it p . Also, since we are testing to see if a "majority" support the incumbent, this will be a _____ test.

We can now write the parameter of interest and the null and alternative hypotheses:

Example 2: Steps 1 and 2. We would like to see if the true mean number of correct answers differs from what we would expect to see if the students were simply guessing randomly. When the parameter is a mean, we denote it μ . Also, since we are testing to see if the mean “differs” from some value, this will be a _____ test.

We can now write the parameter of interest and the null and alternative hypotheses:

Step 3: State the significance level and the corresponding critical value
--

The **significance level** (α) is the probability we are willing to accept of rejecting a true null hypothesis. If this is not given to you, the rule of thumb is to use $\alpha = 0.05$.

The **critical value** is the value of the test statistic such that α is the area remaining in the tail of the sampling distribution of the test statistic under H_0 .

The critical value is the “cutoff” that distinguishes values of the test statistic that result in rejecting H_0 from values that result in failing to reject H_0 .

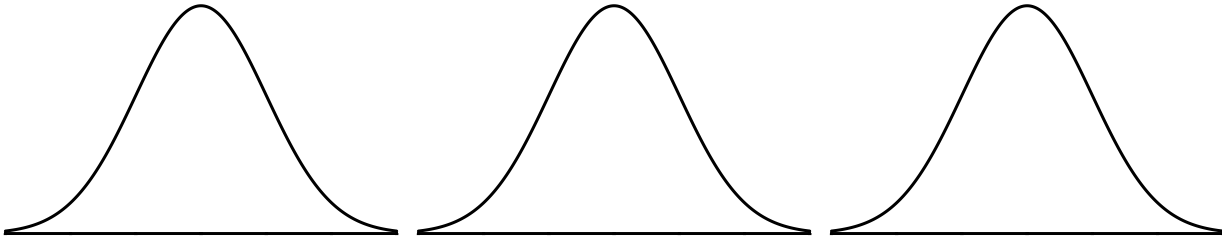
The sampling distribution of the test statistic under H_0 shows the distribution of values of the test statistic that we would expect to obtain if H_0 were true. Values in the middle of the distribution are likely to occur when H_0 is true; values in the tail(s) of the distribution are unlikely to occur when H_0 is true.

For two-tailed tests, we will reject H_0 if we obtain a test statistic in either the left or right tail of this distribution. For one-tailed tests, we will reject H_0 only if we obtain a test statistic in the relevant tail (either the left or the right).

The area in the tail of the distribution next to the critical value will be α .

For one-tailed tests, this area is contained only in the relevant tail. For two-tailed tests, this area is split between both tails, so that the area in each tail next to the critical value is $\alpha/2$.

The critical values will always be z values for proportions. For means, it depends if we know σ . If σ is known, the critical value is a z . If σ is unknown, the critical value is a t with $df = n - 1$. We can use the t table to get critical values for z and t .

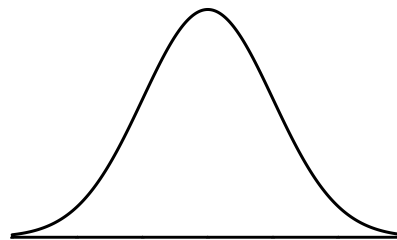


In calculator and Excel:

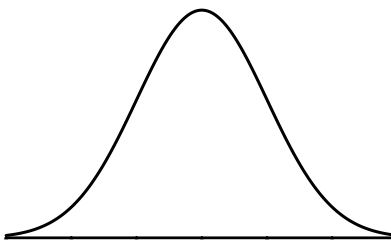
Test type:	Left Tailed	Right Tailed	Two Tailed
z_0	<code>invNorm(α, 0, 1)</code> <code>=NORM.INV(α, 0, 1)</code>	<code>invNorm($1 - \alpha$, 0, 1)</code> <code>=NORM.INV($1 - \alpha$, 0, 1)</code>	$\pm \text{invNorm}(1 - \alpha/2, 0, 1)$ $\pm \text{=NORM.INV}(1 - \alpha/2, 0, 1)$
t_0	<code>invT(α, df)</code> <code>=T.INV(α, df)</code>	<code>invT($1 - \alpha$, df)</code> <code>=T.INV($1 - \alpha$, df)</code>	$\pm \text{invT}(1 - \alpha/2, \text{df})$ $\pm \text{=T.INV}(1 - \alpha, \text{df})$
χ_0^2	<code>=CHISQ.INV(α, df)</code>	<code>=CHISQ.INV($1 - \alpha$, df)</code>	<code>=CHISQ.INV($\alpha/2$, df)</code> <code>=CHISQ.INV($1 - \alpha/2$, df)</code>

Example 1: Step 3

	1000	100	10	5	2
z critical values	∞	1.645	1.96	2.33	2.58
α for 2-tailed tests		0.10	0.05	0.02	0.01
α for 1-tailed tests		0.05	0.025	0.01	0.005



Example 2: Step 3



Step 4 Part 1: Compute the test statistic

The test statistic tells us how much evidence we have against H_0 . The bigger the test statistic, the _____ the evidence.

The test statistic we use will depend on what kind of parameter we are dealing with. You can refer to your Inference Summary Sheet to see which test statistic is used for which parameter.

For a single proportion and a single mean, the test statistics are, respectively,

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \quad \text{and} \quad t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}; \quad df = n - 1$$

Step 4 Part 2: Compute the p -value

The p -value is the area in the tail(s) of the distribution next to the test statistic. You need your calculator or Excel to find this.

If the test statistic is a z value, you can use `normalcdf` or `=NORM.DIST`. If the test statistic is a t value, you can use `tcdf` or `=T.DIST`. The bounds are determined by what type of test we have.

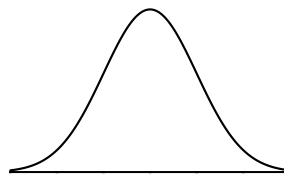
This again gives evidence against the null hypothesis. We can interpret it in one of a couple ways:

For left-tailed tests: The p -value is the probability of obtaining a sample statistic equal to the one we observed or smaller if the null hypothesis is true.

For right-tailed tests: The p -value is the probability of obtaining a sample statistic equal to the one we observed or larger if the null hypothesis is true.

For two-tailed tests: The p -value is the probability of obtaining a sample statistic at least as unusual as the one we observed if the null hypothesis is true.

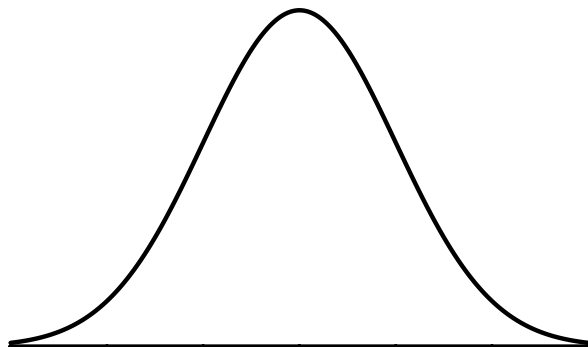
Important: if we have a two-tailed test, we must multiply the probability we get from our calculator by 2 to get our p -value. Why?



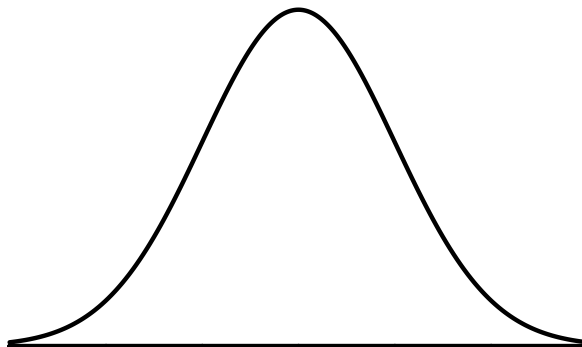
To find p -values using technology:

Test type	Left Tailed	Right Tailed	Two Tailed
z test stat	<code>normalcdf(-1000, z, 0, 1)</code> <code>=NORM.DIST(z, 0, 1, TRUE)</code>	<code>normalcdf(z, 1000, 0, 1)</code> <code>=1-NORM.DIST(z, 0, 1, TRUE)</code>	If $z < 0$: double left tailed If $z > 0$: double right tailed
t test stat	<code>tcdf(-1000, t, df)</code> <code>=T.DIST(t, df, TRUE)</code>	<code>tcdf(t, 1000, df)</code> <code>=1-T.DIST(t, df, TRUE)</code>	If $t < 0$: double left tailed If $t > 0$: double right tailed
χ^2 test stat	<code>χ^2cdf(-1000, χ^2, df)</code> <code>=CHISQ.DIST(χ^2, df, TRUE)</code>	<code>χ^2cdf(χ^2, 1000, df)</code> <code>=1-CHISQ.DIST(χ^2, df, TRUE)</code>	If $\chi^2 < df$: double left tailed If $\chi^2 > df$: double right tailed

Example 1: Step 4. The parameter we are drawing inference on is p , which in this case is the true (i.e. population) proportion of the population who plan to vote for the incumbent candidate. The test statistic is:



Example 2: Step 4. The parameter we are drawing inference on is μ , which in this case is the true (i.e. population) mean number of SAT reading comprehension questions answered correctly by students who have not read the relevant passage. The test statistic is:



Step 5: Make the statistical decision

There are two ways to make the statistical decision about whether we want to reject the null hypothesis or not.

1. Compare our p -value to our level of significance (α).

If the p -value $< \alpha$, we reject H_0 .

If the p -value $> \alpha$, we fail to reject H_0 .

2. Compare our test statistic to our critical value. This is known as using rejection regions.

For left-tailed test:

If the test statistic $<$ critical value, we reject H_0 .

For right-tailed test:

If the test statistic $>$ critical value, we reject H_0 .

For two-tailed test:

If the test statistic is not between the two critical values, we reject H_0 .

Step 6: Interpret the results

Once we have made the statistical decision, we should state what our results mean in plain English. In other words, relate the statistical results back to the original question of interest. We will interpret our results in terms of the alternative hypothesis. We will simply say whether we do have enough evidence (if we reject H_0) or do not have enough evidence (if we fail to reject H_0) to suggest the alternative hypothesis is true. That is:

- If we reject H_0 in step 5, then we have evidence for the alternative hypothesis.
- If we fail to reject H_0 in step 5, then we do not have evidence for the alternative hypothesis.

In general, results are significant if they are unlikely to come about as the result of chance alone. That is, if the p -value is small.

Example 1: Steps 5 and 6.

Example 2: Steps 5 and 6.

These two examples demonstrate two kinds of hypothesis tests. In practice, there are hundreds (maybe thousands) of different kinds of hypothesis tests. We will only look at a handful of them in this class.

What all hypothesis tests have in common is that they identify null and alternative hypotheses, and they utilize observed data to answer the question “how likely would it be to obtain results like this if the null hypothesis was true?”

We can think of hypothesis tests similar to the creed of “innocent until proven guilty”. That is, we will always go with the null hypothesis (innocent) unless there is enough evidence to reject it (guilty).

Activities

Activity 29-1: Left, Right, or Two-Tailed Test?

In each of the follow scenarios, write down the null and alternative hypotheses, H_0 and H_a , using symbols and then indicate whether the test is left tailed, right tailed, or two tailed and explain how you know. You do not need to perform the test.

- a. In the population, the average IQ is 100 with a standard deviation of 15. A team of scientists want to test a new medication to see if it has either a positive or negative effect on intelligence, or no effect at all. A sample of 30 participants who have taken the medication has a mean of 140. Did the medication affect intelligence?

- b. In a packaging plant, a machine packs cartons with jars. It is supposed that a new machine would pack faster on average than the machine currently used, which takes, on average, 2 minutes to pack a carton. To test the hypothesis, the time it takes each machine to pack ten cartons are recorded.

- c. A sample of 20 students were selected and given a diagnostic module prior to studying for a test and then they were given the exam after completing the module. The usual pass rate for this exam is 60%, and we want to see if there is significant improvement in the student’s performance (pass rate) due to this test prep method.

Activity 29-2: Critical Values for Hypothesis Tests

For each of the following, find the critical value or values. Be sure to label the critical values as either z_0 or t_0 depending on whether a z or t test is appropriate.

- a. $\alpha = 0.05$; $H_0: p \leq 0.7$; $H_a: p > 0.7$.

- b. $\alpha = 0.01$; $n = 61$; $H_0: \mu = 105$; $H_a: \mu \neq 105$.

- c. $\alpha = 0.10$; $n = 400$; $H_0: p \geq 0.34$; $H_a: p < 0.34$.

Activity 29-3: Find the p -value

For each of the following tests, find the p -value. Of course, be sure to note whether the test statistic is a z or a t to determine if `normalcdf` or `tcdf` should be used.

- a. $H_0: \mu \leq 551$
 $H_a: \mu > 551$
Test statistic: $t = 2.15$ with a sample size of 74.

- b. $H_0: p = 0.15$
 $H_a: p \neq 0.15$
Test statistic: $z = 1.02$ with a sample size of 463.

- c. $H_0: \mu \geq 0.67$
 $H_a: \mu < 0.67$
Test statistic: $t = -0.56$ with a sample size of 96.

Activity 29-4: Making Statistical Decisions and Interpreting Results

For each of the following, determine whether we should reject the null hypothesis or whether we should fail to reject it. Then interpret the results in terms of the alternative hypothesis and in the context of the problem.

- a. Parameter: p = true proportion of all students who have ever watched a *Land Before Time* movie.
 $H_0: p = 0.20$
 $H_a: p \neq 0.20$
 $\alpha = 0.05$
 $p\text{-value} = 0.0421$.

- b. Parameter: μ = true average number of Brussels sprouts eaten per month.

$$H_0: \mu \leq 2.1$$

$$H_a: \mu > 2.1$$

$$\alpha = 0.10, t_0 = 1.782$$

$$\text{Test statistic: } t = 1.68.$$

In addition to stating the decision and interpreting, draw a picture of the distribution here, label where the rejection region is, and draw the test statistic on the graph.

Topic 30**Hypothesis Testing: Complete Example****Lecture**

Example: A statistician is researching the proportion of students that have taken a statistics class at SUU. She believes that less than 30% of students have taken a statistics class. Out of her sample of 119 SUU students, she found that 25 of them had taken a statistics class. Test her claim at the 0.01 significance level.

Activities

Activity 30-1: GRE

It has been reported that the average score on the graduate record examinations (GRE) in 1997 was 558 out of a possible 800 points. After taking your GRE test, you believe that the average GRE score has changed since 1997. You obtain a random sample of 101 students who have taken the GRE and find the average is 584 with a standard deviation of 120.

- a. Are those values, the 584 and the 120, parameters or statistics?

- b. **Step 1:** Write out the parameter of interest in this situation. Include its symbol.

- c. Are the necessary conditions for this test met? Comment on each one.

- d. **Step 2:** Write down the null and alternative hypotheses using symbols:

H_0 :

H_a :

- e. Is this a left-, right-, or two-tailed test? Explain how you know.

- f. **Step 3:** Write down the significance level that we will be using here and report the critical value(s).

- g. **Step 4 Part 1:** Write down the formula for the test statistic and then use it to find the test statistic for this hypothesis test.
- h. **Step 4 Part 2:** Calculate the p -value for this test. Note that you can determine which function to use, `normalcdf` or `=NORM.DIST` vs `tcdf` or `=T.DIST`, by realizing if the test statistic is a z or a t value.
- i. Interpret the p -value in the context of the problem. The general template for interpreting a p -value for a two-tailed test is as follows: *If the null hypothesis is true, the probability that we would obtain a sample result as or more unusual than the one we observed is (p -value)*. Be sure to add context.

Note that interpreting the p -value is not one of the six steps, so you will not need to interpret unless specifically asked, but knowing how to interpreting the p -value is important and is something you should know how to do.

- j. **Step 5:** State the statistical decision (should we reject H_0 or fail to reject it)? Explain why using both the p -value and the rejection region.
- k. **Step 6:** Interpret the results in terms of the alternative and in the context of the problem.

1. Construct a 95% confidence interval for the true average GRE score and then interpret the interval. Do the results agree with the hypothesis test?

Relationship between a hypothesis test and a confidence interval

A confidence interval can be thought of as an “inverted” two-tailed hypothesis test. What is meant by this is that a CI contains all the possible null values that would result in failing to reject a null hypothesis.

This should make intuitive sense: if we consider a value for an unknown parameter plausible, then we would not want to reject it. However, if a value is outside the range of what we consider plausible, then we would want to reject it.

Note that this relationship holds for two-tailed hypothesis tests. You can see this on your t -table; the 95% confidence level column corresponds to a two-tailed test at $\alpha = 0.05$; the 99% confidence level column corresponds to a two-tailed test at $\alpha = 0.01$, and so on.

For a one-tailed hypothesis test, it is possible to reject a null value that falls inside a corresponding confidence interval.

Topic 31

Hypothesis Testing for the Variance and Standard Deviation

Lecture

Just like with confidence intervals, the hypothesis tests for a standard deviation do not use z or t values, they use χ^2 values instead. Therefore, the test statistic is

$$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}; \quad df = n - 1$$

The p -value can be found using $\chi^2\text{cdf}$ on a calculator or `=CHISQ.DIST` in Excel (see page 160 or the summary page). Otherwise, the 6 hypothesis testing steps still apply.

Note: hypothesis test about a standard deviation and a variance are equivalent. So, we do not have to test both.

Example: A company claims that the standard deviation of the call wait times is less than 1.8 minutes. A random sample of 25 incoming telephone calls has a standard deviation of 1.5 minutes. At $\alpha = 0.10$, is there enough evidence to support the company's claim? Assume the population is normally distributed.

Activities

Activity 31-1: χ^2 Critical Values for Hypothesis Tests

Find the critical value(s) for each of the following hypothesis tests. The Excel guide on Canvas will be helpful here.

- a. $\alpha = 0.02$; $n = 50$; $H_0: \sigma \geq 10$; $H_a: \sigma < 10$.

- b. $\alpha = 0.05$; $n = 200$; $H_0: \sigma^2 = 144$; $H_a: \sigma^2 \neq 144$.

Activity 31-2:

The annual salaries (in dollars) of randomly chosen fire fighters are listed below and can be found in the **firefighters.xlsx** on Canvas.

- a. Compute the sample standard deviation in Excel and report it here.

- b. At $\alpha = 0.05$, is there enough evidence to support the claim that the standard deviation of the annual salaries is different from \$5400? Assume the population is normally distributed. Perform a full hypothesis test showing all 6 steps.

Topic 32

Hypothesis Testing Errors

Lecture

When doing a hypothesis test, there are two types of errors that can come about.

1. The first type, which we call a Type I error, is when we reject the null hypothesis when the null hypothesis is actually true.
2. A Type II error is failing to reject the null hypothesis when it is actually false.

Usually, Type I errors are regarded as worse. If we make a Type II error, then that just means that we say we have no additional information than what we started off with. If we make a Type I error, we say things have changed when they actually haven't. Because a Type I is worse, that is the one we control with our significance level:

α = probability of committing a Type I error. We often denote the probability of committing a Type II error with β .

Continuing with our courtroom metaphor, a Type I error is saying someone is guilty when they are actually innocent and a Type II error is saying someone is innocent when they are actually guilty.

Example: A manufacturer of compact fluorescent light (CFL) bulbs claims the average lifetime of its CFL bulbs is greater than the current industry standard of 8000 hours.

Decision	Null Hypothesis	
	True	False
Reject H_0	Type I Error	Correct Decision
Fail to Reject H_0	Correct Decision	Type II Error

What is a Type I error in the context of this problem?

What is a Type II error in the context of this problem?

Activities

Activity 32-1: Testing Errors

In each of the following situations, write down the null and alternative hypotheses and then write in words and in the context of the problem what the type I and type II errors would be.

- A furniture store believes that at least 40% of its new customers will return to buy their next piece of furniture.
- A local chess club claims that the length of time to play a game has a standard deviation of more than 14 minutes.

Chapter 7 Summary

Hypothesis testing is another statistical inference procedure. In each hypothesis test, we assume the null hypothesis is true and then gather evidence that it is not based on a sample. Each hypothesis test follows the following 6 steps:

1. State the parameter(s) of interest.
2. Set up the null & alternative hypotheses.
3. State the significance level and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p -value (if we are basing the decision on p -values).
5. Make the statistical decision.
6. Interpret the results.

The test statistic's form depends on what population parameter we are testing. Refer to the Inference Summary Sheet for information about the test statistic for each parameter.

Every hypothesis test will be a left-tailed, right-tailed, or two-tailed test. Pay special attention to what the claim is and remember that the statement of equality always goes in the null hypothesis.

To find **critical values**, use the t-table or use technology:

Test type:	Left Tailed	Right Tailed	Two Tailed
z_0	<code>invNorm(α, 0, 1)</code> <code>=NORM.INV(α, 0, 1)</code>	<code>invNorm($1 - \alpha$, 0, 1)</code> <code>=NORM.INV($1 - \alpha$, 0, 1)</code>	<code>±invNorm($1 - \alpha/2$, 0, 1)</code> <code>±=NORM.INV($1 - \alpha/2$, 0, 1)</code>
t_0	<code>invT(α, df)</code> <code>=T.INV(α, df)</code>	<code>invT($1 - \alpha$, df)</code> <code>=T.INV($1 - \alpha$, df)</code>	<code>±invT($1 - \alpha/2$, df)</code> <code>±=T.INV($1 - \alpha$, df)</code>
χ_0^2	<code>=CHISQ.INV(α, df)</code>	<code>=CHISQ.INV($1 - \alpha$, df)</code>	<code>=CHISQ.INV($\alpha/2$, df)</code> <code>=CHISQ.INV($1 - \alpha/2$, df)</code>

To find **p-values**:

Test type:	Left Tailed	Right Tailed	Two Tailed
z test stat	<code>normalcdf(-1000, z, 0, 1)</code> <code>=NORM.DIST(z, 0, 1, TRUE)</code>	<code>normalcdf(z, 1000, 0, 1)</code> <code>=1-NORM.DIST(z, 0, 1, TRUE)</code>	If $z < 0$: double left tailed If $z > 0$: double right tailed
t test stat	<code>tcdf(-1000, t, df)</code> <code>=T.DIST(t, df, TRUE)</code>	<code>tcdf(t, 1000, df)</code> <code>=1-T.DIST(t, df, TRUE)</code>	If $t < 0$: double left tailed If $t > 0$: double right tailed
χ^2 test stat	<code>χ^2cdf(-1000, χ^2, df)</code> <code>=CHISQ.DIST(χ^2, df, TRUE)</code>	<code>χ^2cdf(χ^2, 1000, df)</code> <code>=1-CHISQ.DIST(χ^2, df, TRUE)</code>	If $\chi^2 < df$: double left tailed If $\chi^2 > df$: double right tailed

Chapter 8: Inference with Two Samples

Topic 33

Inference for the Difference in Proportions

Lecture

In chapter 6, we were introduced to the idea of confidence intervals and in chapter 7, we were introduced to the idea hypothesis testing. All of the inferential procedures we've dealt with so far involved only a single parameter from one population: a proportion (p), a mean (μ), a variance (σ^2), or a standard deviation (σ).

In practice, it is common to want to investigate how two parameters differ from one another or if they differ at all, so we will now extend these topics to include dealing with two parameters – either two proportions or two means. The idea of confidence intervals is still the same: they give us information on where we believe the true parameter lies and hypothesis tests will still follow the same hypothesis testing outline:

Hypothesis testing outline

1. State the parameter(s) of interest.
2. Set up the null & alternative hypotheses.
3. State the significance level and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p -value (if we are basing the decision on p -values).
5. Make the statistical decision.
6. Interpret the results.

Difference in Proportion

It is often of interest to know whether population proportions differ. We will only consider the case where the samples are independent in these notes. That is, we are assuming the samples are not affected by one another. This is often referred to as a **two-proportion z test**. When testing for the difference in proportion, the alternative hypothesis are usually one of the following:

$$H_A : p_1 - p_2 > 0;$$

$$H_A : p_1 - p_2 < 0;$$

$$H_A : p_1 - p_2 \neq 0$$

The formula for the test statistic that compares two sample proportions is:

$$z = \frac{\hat{p}_1 - \hat{p}_2 - (p_1 - p_2)}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

Here, the subscripts refer to sample 1 and sample 2 and $p_1 - p_2$ is the hypothesized difference between the means. This is almost always zero, because the null hypothesis is almost always that the means are equal. Therefore, the test statistic usually simplifies to

$$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$$

where $\bar{p} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{\text{successes}_1 + \text{successes}_2}{n_1 + n_2}$ and $\bar{q} = 1 - \bar{p}$. Of course, this is only valid when we have two random, independent samples and the sample size is large enough in each: $n_1\hat{p}_1 \geq 5, n_1\hat{q}_1 \geq 5, n_2\hat{p}_2 \geq 5$, and $n_2\hat{q}_2 \geq 5$.

Example: Suppose the Acme Drug Company develops a new drug, designed to prevent colds. The company states that the drug is equally effective for men and women. To test this claim, they choose a simple random sample of 100 women and 200 men from a population of 100,000 volunteers. At the end of the study, 38% of the women caught a cold; and 51% of the men caught a cold. Based on these findings, can we reject the company's claim that the drug is equally effective for men and women? Use a 0.02 level of significance.

We can also construct confidence intervals for the difference in proportions. The formula is

$$\hat{p}_1 - \hat{p}_2 \pm z_c \cdot \sqrt{\frac{\hat{p}_1 \hat{q}_1}{n_1} + \frac{\hat{p}_2 \hat{q}_2}{n_2}}$$

Example: Construct a 98% confidence interval for the difference in the proportions of men and women who get a cold using the information from the last example. Are the results the same?

Activities

Activity 33-1: Cigarette Tax

Time magazine reported the result of a telephone poll of 800 adult Americans. The question posed of the Americans who were surveyed was: “Should the federal tax on cigarettes be raised to pay for health care reform?” The results of the survey were:

Non-Smokers	Smokers
$n_1 = 605$	$n_2 = 195$
351 said “yes”	41 said “yes”

- Is there sufficient evidence at the $\alpha = 0.05$ level to conclude that the two populations – smokers and non-smokers – differ significantly with respect to their opinions? Conduct a full hypothesis test showing all six steps. You can do this on the next page.

- b. Answer whether there is sufficient evidence at the $\alpha = 0.05$ level to conclude that the two populations – smokers and non-smokers – differ significantly with respect to their opinions by constructing a 95% confidence interval instead of a test. Be sure to interpret your interval.

Topic 34

Inference for the Difference in Independent Means

Lecture

For the difference in means, there are two types of two-sample tests. The distinction has to do with whether the samples are independent or dependent. We will begin with independent samples.

When we have two distinct populations, each with their own mean, we are dealing with an independent samples hypothesis test. This is also known as a **two-sample t test**. In this situation, the alternative hypothesis are usually one of the following:

$$H_A : \mu_1 - \mu_2 > 0$$

$$H_A : \mu_1 - \mu_2 < 0$$

$$H_A : \mu_1 - \mu_2 \neq 0$$

The test statistic for a two-sample t test is

$$t = \frac{\bar{x}_1 - \bar{x}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

Here, the subscripts refer to sample 1 and sample 2 and $\mu_1 - \mu_2$ is the hypothesized difference between the means. This is almost always zero, because the null hypothesis is almost always that the means are equal. Therefore, the test statistic usually simplifies to

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}; df = \text{smaller of } n_1 - 1 \text{ and } n_2 - 1$$

Of course, this is only valid when we have random, independent samples and the populations are normal or each sample size is at least 30.

Note: this form of the t test statistic is assuming the variances of the populations are not equal. This is known as **not pooling**. If the variances are equal (or if we have reason to believe they are), then there is a different test statistic. We will not be dealing with that here because we usually do not know if the variances are equal or not.

Additionally, the degrees of freedom given here is not technically correct. It is a simplification to prevent us from having to deal with a complicated formula for calculating the degrees of freedom. Therefore, the critical values and p -value may differ from what you would get when you use a calculator or Excel.

Example: A researcher who is studying the relationship between concentration and balance conducts an experiment in which nine elderly subjects and eight young subjects each stood on a “force platform” and were asked to maintain a stable upright position and to react as quickly as possible to an unpredictable noise by pressing a handheld button. The noise came randomly and the subject concentrated on reacting as quickly as possible. The researcher would like to test to see if the younger participants had quicker reaction times in regard to how well they can maintain balance in this situation. Assume the original populations follow normal distributions. The data of reaction times (in ms) are provided in the following tables:

Table 1: Elderly Subjects

Time	14	41	18	11	16	24	18	21	37
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Table 2: Young Subjects

Time	17	10	16	22	12	14	12	18
------	----	----	----	----	----	----	----	----

We can also calculate confidence intervals for the difference of two means. The formula for this is

$$\bar{x}_1 - \bar{x}_2 \pm t_c \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}; df = \min(n_1 - 1, n_2 - 1)$$

Example: Construct a 95% confidence interval for the difference in mean reaction time between the elderly and young. Does this agree with your hypothesis test conclusion?

Activities

Activity 34-1: Effective Teaching

Within a school district, students were randomly assigned to one of two Math teachers – Mrs. Smith and Mrs. Jones. After the assignment, Mrs. Smith had 30 students, and Mrs. Jones had 32 students. At the end of the year, each class took the same standardized test. Mrs. Smith’s students had an average test score of 78, with a standard deviation of 10; and Mrs. Jones’ students had an average test score of 85, with a standard deviation of 15. Test the hypothesis that Mrs. Smith and Mrs. Jones are equally effective teachers. Use a 0.10 level of significance. (Assume that student performance is approximately normal.)

Topic 35

Inference for the Difference in Dependent Means

Lecture

We now turn to dealing with dependent samples. Sometimes, when testing for a difference in means, we are able to measure the same subjects _____ and test for differences in the two measurements. Examples include “before” and “after” type studies (e.g. compare blood pressure before being put on a drug to after being on the drug), or studies where each subject can be measured under two different treatments, or a treatment and a control (e.g. conduct a vision test with your right eye, then with your left eye, and compare the results). These “**paired**” studies have an advantage over studies using independent samples:

_____.

For example: if you are comparing the average blood pressure of people in a “treatment” group who are given a real drug to the average blood pressure of people in a “control” group who are given a placebo, there is always the possibility that the difference you observe is due not to the drug, but due instead to the fact that you have _____. Different people have different blood pressure, and all kinds of things affect blood pressure. If you observe differences in average blood pressure between the two groups, it might be due to this inherent variability rather than to the drug you are testing.

Using paired observations, i.e. multiple observations on the same subjects, addresses this problem directly. The sources of variability in what is being measured (in our example, blood pressure) that are attributable to the fact that we have different people in each group go away. There will still be some sources of inherent variability; for instance there may be measurement error, and an individual’s blood pressure changes by some amount on a day to day or week to week basis. But the amount of inherent variability will be greatly reduced if you can measure the same people twice. In this case, measuring the blood pressure of subjects before and after taking the drug, then comparing these differences, would be an example of observing “paired” data.

The formula for a paired t-test is essentially identical to the formula for a t-test for a single mean. The only difference is that the data we use is not raw observations of a variable; rather it is the differences in the paired observations. The average of these differences is known as $\bar{d}$. Otherwise the formula is identical to the one for testing a single mean:

$$t = \frac{\bar{d} - \mu_{d0}}{s_d / \sqrt{n}}$$
$$df = n - 1$$

- $\bar{d}$ is the sample average of the differences.
- s_d is the sample standard deviation of the differences.
- μ_{d0} is the hypothesized mean of the differences in the paired data in the population, which is usually zero in practice and for us it will always be zero.

Plugging in 0 for the hypothesized mean difference, the equation will simplify to

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}; df = n - 1$$

The confidence interval for the difference in means for paired data has the same format as the confidence interval for a single mean, except we use the differences in our sample:

$$\bar{d} \pm t_c \cdot s_d/\sqrt{n}; df = n - 1$$

When dealing with paired or dependent means, we don't define μ_1 and μ_2 separately. Instead, we define μ_d = the true average of the differences.

Example: An instructor wants to use two exams in her classes next year. This year, she gives both exams to the students. She wants to know if there is evidence at the 0.02 significance level that one is significantly more difficult than the other. Assume the test scores are normally distributed.

Student	Exam 1	Exam 2	Difference
1	74	70	
2	81	82	
3	64	51	
4	97	94	
5	71	71	
6	86	95	

Activities

Activity 35-1: Exercise Regime

A physical fitness program is designed to increase a person's upper body strength. To determine the effectiveness of this program a simple random sample of 9 members of a health club was selected and each member was asked to do as many push-ups as possible in 1 minute. After 1 month on the program the participants were once again asked to do as many push-ups as possible in 1 minute. These values were recorded and the difference (After - Before) was computed.

Subject	Before	After	Difference
1	28	32	
2	34	32	
3	28	42	
4	60	64	
5	20	41	
6	25	33	
7	32	49	
8	19	32	
9	29	50	

- Explain why this is a situation where dependent means should be used. That is, explain why this is a paired data set.
- Compute all of the differences in the number of push-ups these subjects did. In general, it does not matter if we subtract the before from the after or vice-versa, but in this case, the problem tells us that we should take the after values and subtract the before values. Fill in the table above with these differences.

Instead of defining μ_1 and μ_2 like we do for the difference in independent means, with paired data, we instead define μ_d , which is the true average in the differences. It can be helpful to think of $\mu_d = \mu_1 - \mu_2$, but we will not define μ_1 and μ_2 individually here.

- Define the parameter of interest, μ_d , in the context of the problem. Be sure that in your definition, it is clear whether the before or the after was subtracted from the other.

- d. Use Excel to find $\bar{d}$ and s_d using the differences you computed in part b. Report these statistics here. Note that while it is true that $\bar{d} = \bar{x}_1 - \bar{x}_2$, it is **not** true that $s_d = s_1 - s_2$. So performing the calculations on the differences is crucial.
- e. Test to see if the program is effective in increasing upper body strength as defined by the “number of push-ups in one minute” metric, using $\alpha = 0.01$. Show all 6 steps.

Typically, if we were to construct a confidence interval that would match (or “invert”) the result of the test, we would use a $(1 - 0.01) \times 100\% = 99\%$ confidence interval. However, since this was a one-tailed test, we need to use a different confidence level to “invert” this test. The confidence level will be in the same column of the t table that was used to find the critical value for our one-tailed test.

- f. What confidence level is needed to “invert” the hypothesis test in part e?
- g. Construct the confidence interval and interpret it.

Chapter 8 Summary

The formulas for hypothesis testing and confidence intervals will be different depending on what kind of parameter is being investigated, but the overall process is always the same. Refer to the Inference Summary Sheet for information about the test statistic for each parameter.

Here is an updated symbol table.

Symbol	Description	Symbol	Description
n, n_1, n_2	Sample sizes	s, s_1, s_2	Sample standard deviations
N, N_1, N_2	Population sizes	$\sigma, \sigma_1, \sigma_2$	Population standard deviations
df	Degrees of freedom	s^2, s_1^2, s_2^2	Sample variances
E	Margin of error	$\sigma^2, \sigma_1^2, \sigma_2^2$	Population variances
x, x_1, x_2	Number of "successes"	χ^2	Test statistic for variance/standard deviation
$\hat{p}, \hat{p}_1, \hat{p}_2$	Sample proportion	χ_L^2	Left (smaller) chi-square critical value
p, p_1, p_2	Population proportions	χ_R^2	Right (larger) chi-square critical value
$\bar{p}$	Proportion of all successes from both samples	$\bar{d}$	Sample mean of the differences
z	Test statistic for proportion(s)	μ_d	Population mean of the differences
z_c	z critical value for confidence intervals	s_d	Standard deviation of the differences
z_0	z critical value for hypothesis tests	H_0	Null hypothesis
$\bar{x}, \bar{x}_1, \bar{x}_2$	Sample means	H_A	Alternative hypothesis
μ, μ_1, μ_2	Population means	α	Significance level
t	Test statistic for mean(s)	μ_0	Hypothesized mean
t_c	t critical value for confidence intervals	p_0	Hypothesized proportion
t_0	t critical value for hypothesis tests	σ_0	Hypothesized standard deviation
$q, \hat{q}, q_0, \bar{q}$	$1 - p, 1 - \hat{p}, 1 - q_0, 1 - \bar{p}$	σ_0^2	Hypothesized variance

Chapter 9: Correlation and Regression

Topic 36

Scatter Plots and Describing Association

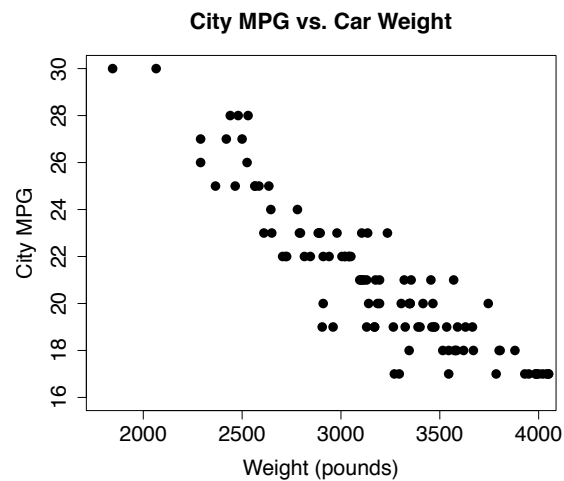
Lecture

Now that we have a solid background in confidence intervals and hypothesis tests, we turn to other kinds of statistical analysis. First, we will look at how to describe the relationship between two quantitative variables.

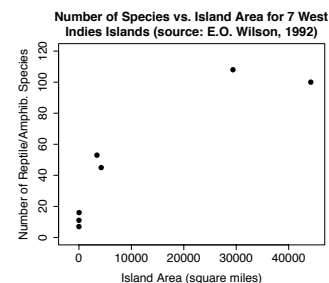
The most common display for examining the **association** (or relationship) between two *quantitative* variables is the **scatter plot**. A scatter plot simply assigns one axis to one variable, and the second axis to the other, and plots the values against each other.

Examples:

1. Is there a relationship between the mileage (MPG) and weight of cars, and if so, what is the nature of the relationship?



2. Is there an association between island size and the number of reptile or amphibian species an island can support? How would you describe this relationship?



Recall when we were describing distributions of single variables. We looked for center, spread, shape, unusual features, and context. When describing the relationship between variables, we also want to be sure to mention a few key aspects.

Describing Association: 4 Things to Look For:

1. **Direction**: In what direction do the values of one variable change with increases in the values of the other variable?
 - (a) **Positive Association**: Two variables are said to have positive association if the values of one variable tend to *increase* with the values of the second variable. So larger values of one variable generally correspond to larger values of the second variable. Examples?
 - (b) **Negative Association**: Two variables are said to have negative association if the values of one variable tend to increase as the values of the second variable tend to *decrease*. So larger values of one variable generally correspond to smaller values of the second variable. Examples?
 - (c) **Other Types of Association**: Two variables may be associated in both a positive and negative fashion over different ranges of values for the two variables. Examples?
 - (d) **No Association**: Two variables are not associated if there is no apparent association (positive, negative, or otherwise) present in the scatterplot. Examples?
2. **Form**: In what form, or how quickly do the values of one variable change with the values of the other variable?
 - (a) **Linear**: The relationship between two quantitative variables is said to be *linear* if they exhibit a straight-line pattern in their scatterplot.
 - (b) **Nonlinear**: If the relationship is not linear, we say it is nonlinear. Common forms of nonlinear relationship are *exponential* or *quadratic*.

3. **Strength**: How much variability or spread is present in the scatterplot?

- Lower scatter (stronger association) indicates clearer trends or patterns from which estimation is more precise. If there is too much scatter, we may not be able to discern any relationship between the two variables.

4. **Outliers**: You should note any points which do not fit the overall relationship.

Activities

Activity 36-1: Marriage Ages

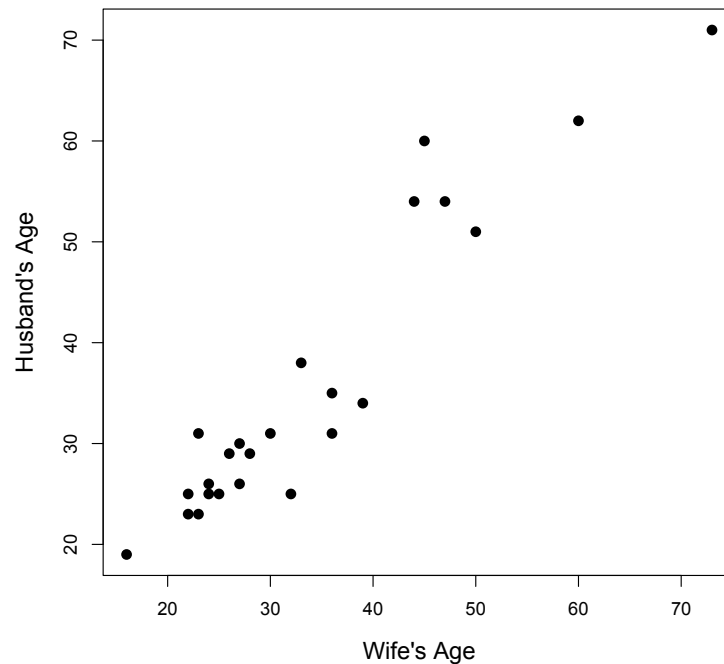
The data in the table below contains the ages (in years) at marriage for a sample of 24 heterosexual couples who obtained their marriage licenses in Cumberland County, Pennsylvania, in 1993.

Couple #	Husband Age	Wife Age	Couple #	Husband Age	Wife Age
1	25	22	13	25	24
2	25	32	14	23	22
3	51	50	15	19	16
4	25	25	16	71	73
5	38	33	17	26	27
6	30	27	18	29	26
7	60	45	19	31	36
8	54	47	20	26	24
9	31	30	21	62	60
10	54	44	22	31	23
11	23	23	23	29	28
12	34	39	24	35	36

a. What are the observational units here?

- b. How many variables are reported in the table for each observational unit? What type (categorical or quantitative) is each variable?

Consider a scatter plot for these data.

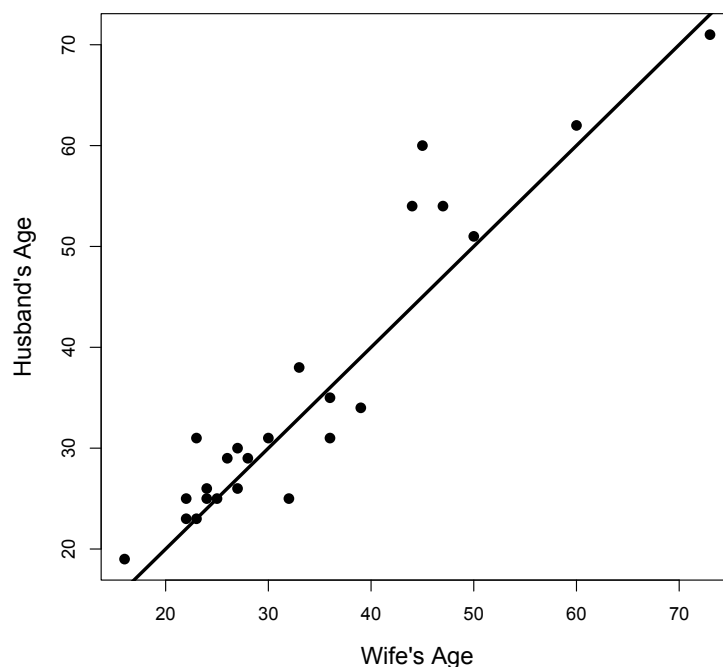


- c. Circle the point in the scatter plot that observation that belongs to couple #12.
- d. Does the scatter plot reveal any relationship between husband's age and wife's age when getting married? In other words, does knowing a wife's age provide any useful information about the husband's age (or vice versa)? Write a sentence or two describing the relationship between the two variables. Be sure to mention, direction, form, strength, and any outliers.

- e. Find an example of two couples where the husband's age for the first couple is younger than the husband's age for the second couple and the wife's age for the first couple is older than the wife's age for the second couple. Write down the couple # for each couple.

The concept of association is another example of a statistical tendency. It is *not always* the case that an older wife means the husband is also older, but older wives *tend* to marry older husbands.

Now the scatter plot is shown with a 45° line (or $y = x$ line) going through the data. That line is showing where the point would be if the husband's age was equal to the wife's age. Note that this is not the line of best fit that we will discuss in Topic 38.



- f. Refer to the original listing of the data to determine how many of the 24 couples' ages fall exactly on the line. In other words, how many couples listed the same age for both husband and wife on their marriage license?
- g. Again looking back at the data, for how many couples is the husband older than the wife? Do these couples fall above or below the line drawn in the scatter plot?

- h. For how many couples is the husband younger than the wife? Do these couples fall above or below the line drawn in the scatter plot?

- i. Summarize what you can learn about the ages of marrying couples by noting that the majority of couples produce points that fall above the $y = x$ line.

Activity 36-2: Home Prices

On June 1, 2023, information about homes in the Cedar City area was collected from Zillow.com. Information collected included what type of home it was (single family, duplex, or townhome/condo), the square footage of the home, and the listing price of the home (in thousands rounded to the nearest thousand). The data can be found on Canvas in the **homeprice.xlsx** file.

- a. Create a scatter plot with square footage on the x axis and listing price on the y axis using all of the homes in the data set. Either include that image or create a rough sketch of it here.

- b. Describe the association between square footage and home price mentioning all of the key attributes.

c. Now create a scatter plot using only the duplex homes, include or sketch that here, then describe the association.

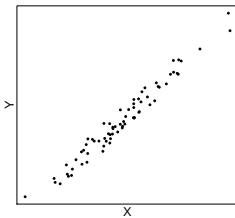
d. Now create a scatter plot using only the single family homes, include or sketch that here, then describe the association.

e. Now create a scatter plot using only the townhomes, include or sketch that here, then describe the association.

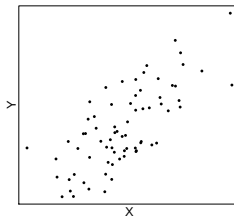
Correlation

Lecture

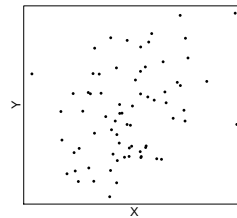
Scatter plots are great ways to visualize correlation between two variables. We have already briefly discussed correlation in Chapter 1. Correlation describes the *linear* relationship between two quantitative variables. There could be an association between variables that is not linear, and we would say there is not a strong correlation.



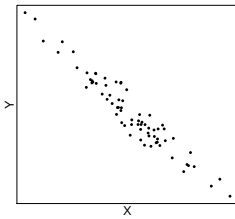
Strong Positive



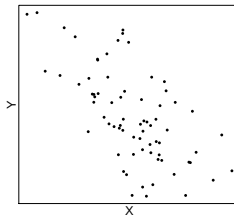
Moderate Positive



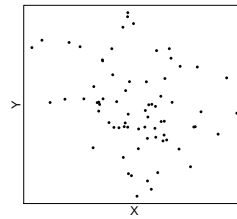
Weak Positive



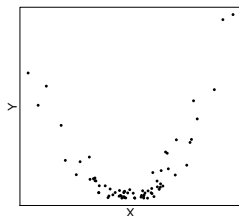
Strong Negative



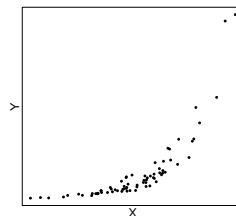
Moderate Negative



Weak Negative



Strong Quadratic



Strong Exponential

Simply looking at a scatterplot to obtain correlation can be subjective, however. There is an objective way to measure correlation by using the **correlation coefficient**. This is also known as the Pearson correlation coefficient.

$$r = \frac{1}{n-1} \sum_{i=1}^n \left[\left(\frac{x_i - \bar{x}}{s_x} \right) \cdot \left(\frac{y_i - \bar{y}}{s_y} \right) \right]$$

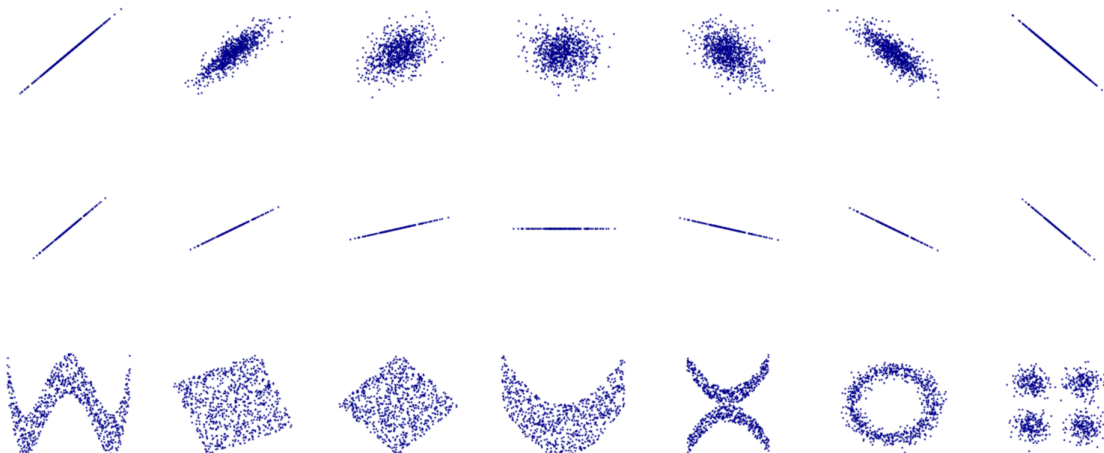
This is a statistic that shows how much linear correlation there is between the variables X and Y . The correlation coefficient is always a number between -1 and 1 .

- If $r = 1$, that means the variables X and Y are perfectly positively correlated.
- If $r = -1$, that means the variables X and Y are perfectly negatively correlated.
- If $r = 0$, that means the variables X and Y are not correlated at all.

So, the farther from zero the correlation coefficient is, either in the positive or negative direction,

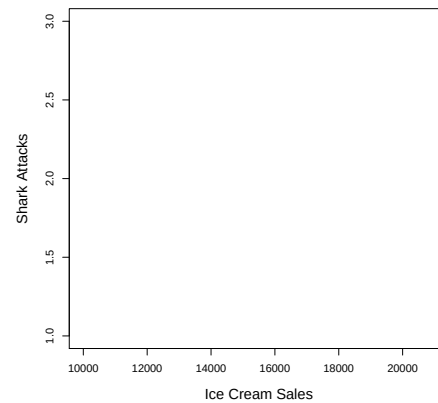
Correlation interpretation: A very general rule of thumb for interpreting the correlation coefficient is:

If $|r| = 1$, then the correlation is “perfect”.
 If $0.9 \leq |r| < 1$, then the correlation is “strong”.
 If $0.6 \leq |r| < 0.9$, then the correlation is “moderate”.
 If $0 < |r| < 0.6$, then the correlation is “weak”.
 If $|r| = 0$, then there is no correlation.



Example: We stated in Chapter 1 that ice cream sales and shark attacks are correlated. Calculate the correlation coefficient between this sample of ice cream sales from several different years. Note that the number of ice cream cones sold in this sample has a mean of 14,600 and a standard deviation of 4,393 while shark attacks has a mean of 1.6 and a standard deviation of 0.89.

Ice Cream Cones Sold	10,000	12,000	13,000	17,000	21,000
Shark Attacks	1	1	2	1	3



What is the most important thing we should always remember about correlation?

Check this out!

Hypothesis test for the population correlation coefficient, ρ

The correlation coefficient, r , describes a sample, so it is a _____. This means it is an estimate for the **population correlation coefficient**, ρ .

$$\rho = \frac{1}{N} \sum_{i=1}^N \left[\left(\frac{x_i - \mu_x}{\sigma_x} \right) \cdot \left(\frac{y_i - \mu_y}{\sigma_y} \right) \right]$$

Of course, we will almost never be able to directly calculate ρ since we usually do not have access to all data points in our population. We can conduct a hypothesis test for ρ based on our sample correlation coefficient, r .

The steps for this test are the same as we have done before. When testing for the population correlation coefficient, we usually want to know if it is different than zero. Why?

So, our alternative hypothesis will be one of the following:

$$H_A : \rho > 0$$

$$H_A : \rho < 0$$

$$H_A : \rho \neq 0$$

The test statistic for testing the population correlation coefficient is:

$$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}; \quad df = n - 2$$

Example: Test whether the correlation between all ice cream sales and all shark attacks is greater than zero based on the results in the previous example.

Activities

Activity 37-1: Guess the Correlation

In this activity we are going to play a game about guessing correlations of scatter plots. Go to one of these website: Rossman/Chance Applet or <https://www.guessthecorrelation.com>. Either one is fine. The second one will require you to log in with a Google account. If you don't want to do that, go to the first one or do a web search for "Rossman Chance guess the correlation". If you use the Rossman/Chance applet, I recommend you change the number of points to at least 50 and click on the Track Performance option.

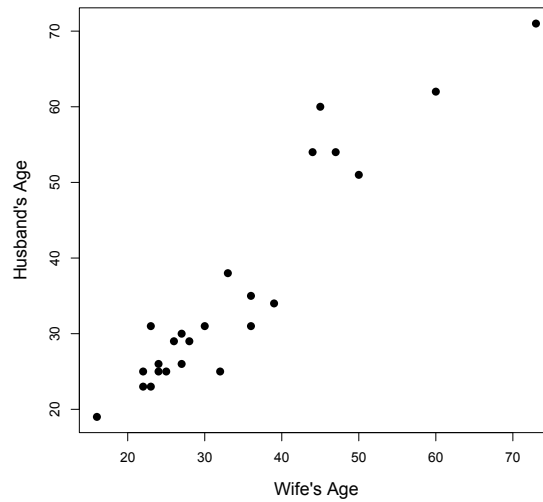
- a. Take turns playing this as a group until everyone has had a chance to guess the correlation at least 10 times. Be sure to notice if the correlation is positive or negative. For each of the times you played, write down both the guess and the correct correlation.

True Correlation	Correlation Guess	Difference (True - Guess)

- b. Take the average of the differences between the true correlations and the guessed correlations. Compare this to the other members of your group (for bragging rights, of course).
- c. Did you tend to get better in your guesses as you got more experience?

Activity 37-2: Return to Marriage Ages

Recall Activity 36-1 that had data regarding the ages at which 24 husbands and wives got married. The scatter plot of those ages is given again here:



The correlation coefficient for these variables is 0.9436. Conduct a full hypothesis test (showing all 6 steps) to test whether husband's age and wife's age are significantly correlated.

Topic 38

Introduction to Simple Linear Regression

Lecture

Identifying correlation between two variables is useful for knowing if they are dependent or not. We can take this a step further, though. If there exists correlation between variables, we can garner more information about how connected the variables are by conducting a **linear regression analysis**, or just a regression analysis.

There are two primary goals of regression:

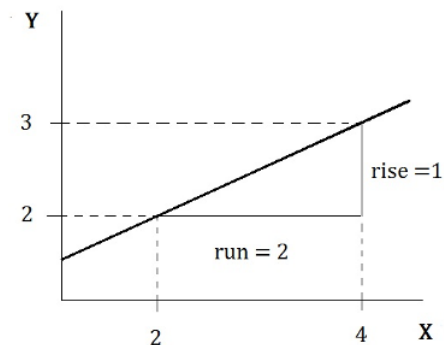
1. To predict values of the Y variable using values of the X variable(s).
2. To quantify the nature and magnitude of relationships between multiple quantitative variables.

Simple linear regression

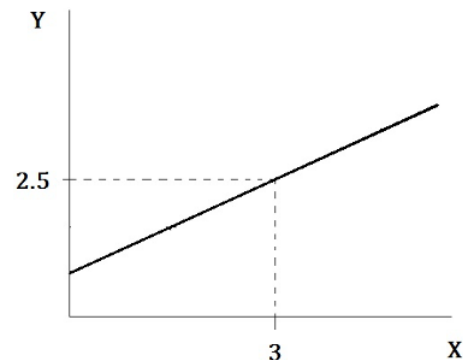
The most basic kind of linear regression involves making a two dimensional scatter plot of data, and then drawing a straight line through this scatter plot in order to show the overall trend in the data. We do this when we have two variables, one for each axis of the plot.

Recall the equation of a line:

$$y = mx + b$$



You can plug in any value for x and calculate the value for y :



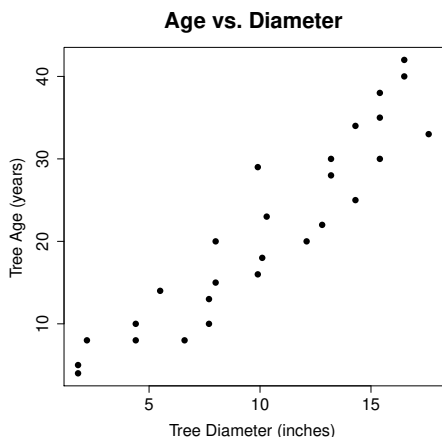
In linear regression, we use our data and some formulas (which we will not consider in depth) in order to calculate the slope and intercept of the "line of best fit" (LOBF).

That will give us an equation for our regression analysis, known as the **regression model** or **regression line**:

$$\hat{y} = mx + b$$

The Y variable that we are predicting is known as the **response variable** or dependent variable. A value of the Y variable will be denoted by y . The X variable that we are using to predict the Y variable is known as the **explanatory variable** or independent variable. A value of the X variable will be denoted by x .

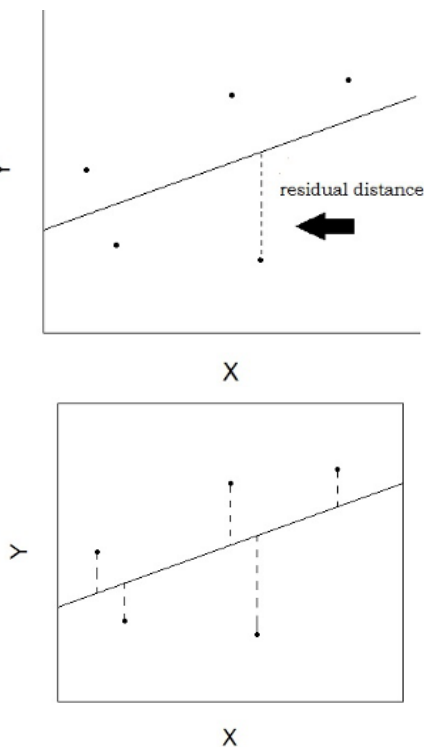
Example: One can determine how old a tree is by counting its rings, but that requires cutting the tree down or taking a core sample. Instead, it would be nice to estimate the tree's age simply by using its diameter. A forester measured 27 trees of the same species that had been cut down and counted the rings to determine the actual ages of the trees.



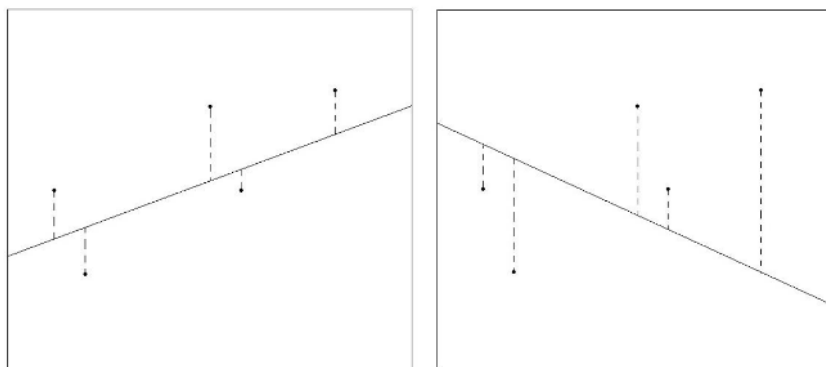
How can we determine which line is best? We must look at the residuals of the line.

A **residual** is the vertical distance between an actual observed value of Y and the predicted value of Y at the observed value of X . It tells us how far off our prediction is from the actual observed value. Formally, a residual is defined to be the difference between the true Y value and the predicted Y value:

$$d_i = y_i - \hat{y}_i$$



The following images show the same 5 points and their residuals for two different lines.



These residuals can be negative (if the true value is under our line) or positive (if the true value is over our line). To get an idea of how good our line is, we should make each residual positive. We do this by squaring them. The line of best fit is the line that minimizes the sum of the squared residuals, $\sum_{i=1}^n d_i^2$. This line is known as the regression line and is also known as the least squares line.

Returning to the tree example, the line of best fit is shown in the plot:

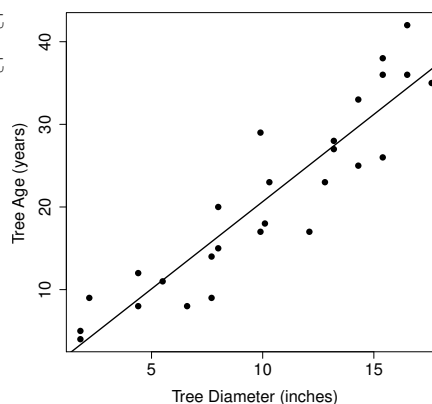
How do we find this equation? We will mostly let Excel calculate these lines for us, but we can do it by hand as well by using these equations:

$$\begin{aligned} m &= r \frac{s_y}{s_x} \\ b &= \bar{y} - m\bar{x} \end{aligned}$$

In this problem, the X variable is and the Y variable is

We also have $r = 0.9298$, $\bar{x} = 10.185$
 $\bar{y} = 21.407$, $s_y = 11.386$

So, the equation for this line is



, $s_x = 4.832$,
.

We can write this equation in the context of the problem. Just replace the y and x with what the variables are:

Example: Use the regression line to predict the age of a tree that has a diameter of 12 inches.

Interpreting the slope and intercept

- The slope of a regression line is the predicted change in the Y variable for each additional one-unit increase in the X variable.
 - If the slope is equal to zero, then there is no predicted change in Y when X increases, so the variables would be considered to be uncorrelated.
 - We can test if the true slope is equal to zero, but that is actually equivalent to testing whether the correlation is equal to zero.
- The intercept of a regression line is the predicted y value when the x value is equal to zero.
 - The intercept often doesn't have a meaningful interpretation in these regression problems because an x value of 0 is often impossible or outside the range of observed values.

SUMMARY OUTPUT

<i>Regression Statistics</i>	
Multiple R	0.712581238
R Square	0.507772021
Adjusted R Square	0.343696028
Standard Error	0.724598632
Observations	5

ANOVA					
	<i>df</i>	<i>SS</i>	<i>MS</i>	<i>F</i>	<i>Significance F</i>
Regression	1	1.624870466	1.624870466	3.094736842	0.176780549
Residual	3	1.575129534	0.525043178		
Total	4	3.2			

	<i>Coefficients</i>	<i>Standard Error</i>	<i>t Stat</i>	<i>P-value</i>	<i>Lower 95%</i>	<i>Upper 95%</i>	<i>Lower 95.0%</i>	<i>Upper 95.0%</i>
Intercept	-0.518134715	1.246886643	-0.415542758	0.705678201	-4.486284506	3.450015076	-4.486284506	3.450015076
ice cream	0.000145078	8.24686E-05	1.759186415	0.176780549	-0.000117374	0.00040753	-0.000117374	0.00040753

Activities

Activity 38-1: Return to Home Prices

Recall the data in the **homeprices.xlsx** document on Canvas that has information about the type, size, and listing price of homes for sale (in thousands) in Cedar City.

- Create a scatter plot with square footage on the x axis and listing price on the y axis. Then right click on a point and select "Add Trendline". This plots the data points along with the line of best fit. Include this plot or a sketch of it here.
- Run a regression analysis using square footage as the x variable and listing price as the y variable. Report the correlation coefficient.
- Write the line of best fit in the context of the problem. Don't forget the hat!
- Use the line of best fit to predict the listing price of a house with a size of 1600 square feet.
- Notice that house number 8 (a duplex) has exactly 1600 square feet and is listed at \$380,000. Using your answer from part d, compute the residual for this home.

- f. Comment on what information this residual tells us about that house. Specifically, is that house listed at a price that is higher or lower than we would expect from its square footage? How much more or less?
- g. Interpret the y intercept of the line of best fit in the context of the problem. Does that interpretation make sense?
- h. Interpret the slope of the line of best fit in the context of the problem.
- i. For the following hypothesis test, would we reject H_0 or fail to reject H_0 ? Explain how you know using both a p -value and a confidence interval given in the regression output.
- H_0 : True Slope = 0
- H_a : True Slope $\neq$ 0

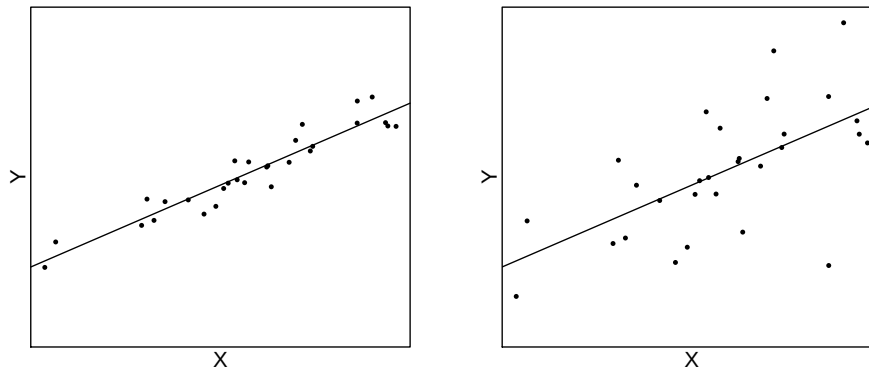
Regression Inference

Lecture

Like most of the data we deal with, regression equations are created using samples and so the predictions we get are nearly always wrong. We would like intervals where we believe the true y value lies for any x value that we plug into the regression equation.

The coefficient of determination

Before we do that, however, we will discuss a measure of how well our model fits the data. Below are two plots that have the exact same line of best fit, but in the first case the line fits the data better because the data is more strongly correlated.



We can be more confident in predictions made in the first plot, even though the predicted value for Y at any X value will be the same in both cases.

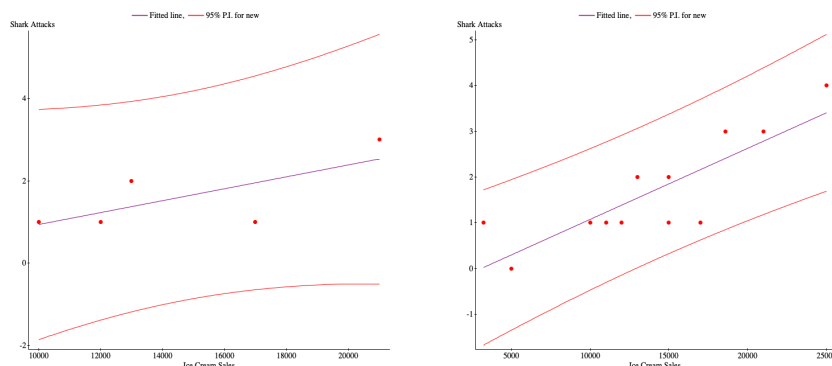
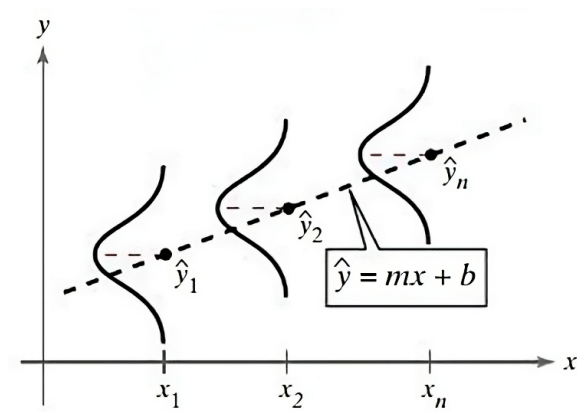
We quantify how well X explains Y with the **coefficient of determination**, which we call r^2 or “r squared”. For simple linear regression, r^2 is literally r (the correlation coefficient), squared. r^2 is interpreted as the proportion of variation in the Y variable that is explained by the X variable.

Intuitively, r^2 tells you how close the data are to the line. Note that since r must be between -1 and 1, r^2 must be between 0 and 1. If r^2 is close to 1, then the data is very close to the regression line. If r^2 is close to 0 then the data is not very close to the regression line. Also note that r^2 will be smaller than r , unless r is exactly 0 or 1.

Prediction intervals

Now that we know how good our predictions are likely to be based on the coefficient of determination, r^2 , we can look at creating intervals where true y 's are likely to land. We can create a confidence interval for the true value of y for any value of x . These confidence intervals for where we believe the true value is are called **prediction intervals**.

The key to creating these intervals is to think of each prediction $\hat{y}_i$ as the center of a t interval that is used to get an idea of where each true y_i value is.



To construct these prediction intervals, we need an idea of how much variation there is in any estimate of y for a given x value. This can be quantified using the **standard error of the estimate**, s_e .

$$s_e = \sqrt{\frac{\sum (y_i - \hat{y}_i)^2}{n - 2}}$$

Once we have that, we can calculate the confidence interval for a true value of y , which we will call y_p , at a value x_p using the following formula:

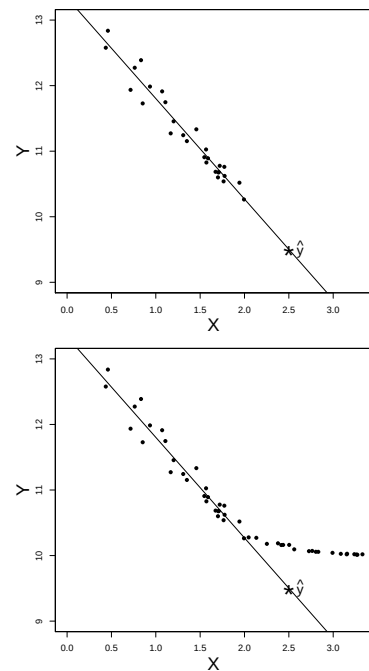
$$\hat{y}_p \pm t_c \cdot s_e \cdot \sqrt{1 + \frac{1}{n} + \frac{(x_p - \bar{x})^2}{\sum (x_i - \bar{x})^2}}$$

Both the point estimate, $\hat{y}_p$ and the margin of error will change depending on which x value we are using for prediction.

This formula is long and obnoxious and Excel does not have built-in functions for this, so we will simply focus on interpreting these.

The 95% prediction interval for the shark attack and ice cream sales example when $x = 15000$: $(-0.87024171, 4.1863039)$.

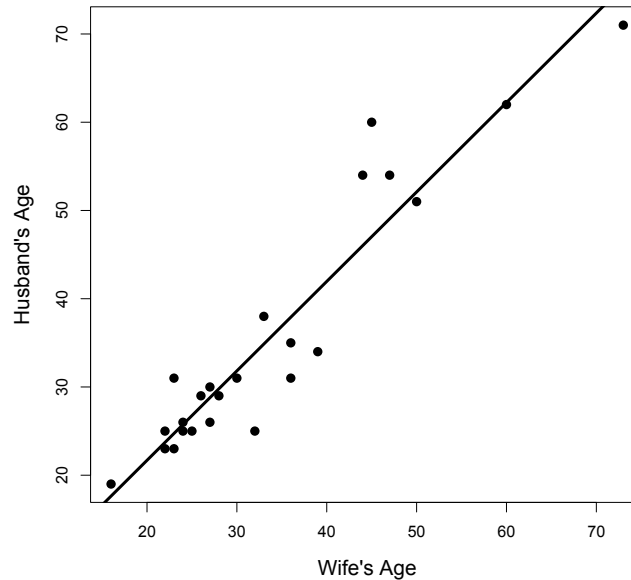
Although we can make predictions and prediction intervals for y based on x values that are outside the range of our data, doing this is called **extrapolation** and caution should be taken when doing so. Why?



Activities

Activity 39-1: Another Return to Marriage Ages

Recall Activity 36-1 that had data regarding the ages at which 24 husbands and wives got married. The scatter plot of those ages is given again below. The correlation is 0.9436 and the regression line is $\widehat{\text{husband age}} = 1.3395 + 1.0146(\text{wife age})$.



- Report the coefficient of determination and interpret this value in the context of the problem.
- Predict the age of the husband when the age of the wife is 30.
- The 95% confidence interval for the slope here is $0.8572 < \text{true slope} < 1.172$. Interpret this interval in the context of the problem.

- d. The 95% prediction interval for the husband's age when the wife is 30 is $21.33 < y_p < 42.23$. Interpret this interval in the context of the problem.
- e. Is that prediction interval from part d practically useful? Explain why or why not?
- f. Find the midpoint of the prediction interval from part d and compare it to answer to part b. Does the relation between those two numbers make sense?
- g. Predict the age of the husband when the age of the wife is 82. Can you be confident in that prediction? Explain why or why not.

Chapter 9 Summary

1. Correlation is a way to quantify how strong the linear relationship is between two quantitative variables.
2. Correlation is always between -1 and 1 . Closer to 0 is weaker.
3. We use regression to predict values of the response variable using values of one or more explanatory variables using the equation $\hat{y} = mx + b$ in simple linear regression or $\hat{y} = b + m_1x_1 + \cdots + m_kx_k$ in multiple linear regression or to quantify the nature and magnitude of relationships between variables by looking at the slopes.
4. The slope is interpreted as follows. Replace all underlined terms with values or context from the problem. When (the X variable) increases by 1 (unit), (the Y variable) is predicted to (increase/decrease) by (slope) (units).
5. The intercept is interpreted as follows. Replace all underlined terms with values or context from the problem. The predicted value of (the Y variable) is (intercept) when (the X variable) is zero (units).
6. R^2 is the coefficient of determination and it is the proportion of variation in the Y variable that is explained by the X variable(s).

Regression Symbols

Symbol	Description
r	Sample correlation coefficient
r^2	Coefficient of determination
ρ	Population coefficient
Y	The response variable
y	A value of the response variable
X	The explanatory variable
x	A value of the explanatory variable
$\hat{y}$	A predicted value of the response variable
$\hat{y}_p$	A particular predicted value of the response variable for prediction intervals
x_p	A particular x value used for a prediction intervals
d_i	A residual value: $y_i - \hat{y}_i$
m	The slope of the regression line
b	The y -intercept of the regression line
s_e	The standard error of the estimate
s_x	The sample standard deviation of X
s_y	The sample standard deviation of Y

Chapter 10: More Chi-Square (χ^2) Tests and the F -Distribution

Topic 40

χ^2 Goodness-of-Fit Test

Lecture

In chapter 7, we were introduced to the idea hypothesis testing for one parameter and in chapter 8, we extended that to include comparing two parameters. We will finish the content in this class by looking at four more hypothesis tests for answering common statistical questions. Each of these tests are based on either the χ^2 distribution or the F distribution, but, just as before, the format of our hypothesis tests remain unchanged:

Hypothesis testing outline

1. State the parameter(s) or variable(s) of interest.
2. Set up the null & alternative hypotheses.
3. State the significance level and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p -value (if we are basing the decision on p -values).
5. Make the statistical decision.
6. Interpret the results.

We reject the null hypothesis in the following situations, otherwise we fail to reject H_0 :

- For any test, if the p -value is less than α .
- For a left-tailed test, if the test statistic is less than the critical value.
- For a right-tailed test, if the test statistic is greater than the critical value.
- For a two-tailed test, if the test statistic is not between the critical values.

χ^2 Goodness-of-Fit Test

The first two tests we will discuss have to do with categorical variables. Recall that categorical variables take the form of labels or descriptions. Variables like gender, political affiliation, state of birth, and model of car are all categorical.

The **χ^2 goodness-of-fit** test is used to test whether a frequency distribution of a categorical variable fits an expected distribution. This test is best shown with an example first.

Example: According to Niche.com, the five most popular majors at SUU fall in the categories of Liberal Arts and Humanities, Education, Nursing, Elementary Education, and Biology. Let's limit our attention to only those majors. Of those majors, Niche.com reports that 71.6% of graduates obtain a Liberal Arts and Humanities degree, 12.4% get an Education degree, 5.9% are nursing graduates, 5.2% get a degree in Elementary Education, and Biology majors make up the other 5.0%. Suppose we want to test Niche.com's distribution of these majors. Sample information is given in the table below:

Major	Observed Frequency	Expected %	Expected Frequency
Liberal Arts	171	71.6%	$285(.716) = 204.06$
Education	51	12.4%	
Nursing	26	5.9%	
El. Ed.	16	5.2%	
Biology	21	5.0%	
Total	285	100%	285

The question we must answer is whether the distribution from our sample is different enough from the expected distribution to suggest the expected distribution _____.

We can do this with a hypothesis test based on the χ^2 distribution. In general, we can write the hypotheses as

H_0 : The distribution of our variable is equal to some expected distribution.

H_A : The distribution of our variable differs from the expected distribution.

Of course, we must add context to these hypotheses. In our case,

The **test statistic** is computed in the following way. Let each of the expected frequencies be denoted by $E_i = np_i$ and each of the observed frequencies will be O_i . Then the test statistic

$$\text{is } \chi^2 = \frac{(O_1 - E_1)^2}{E_1} + \cdots + \frac{(O_k - E_k)^2}{E_k} = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i}.$$

This is also written as

$$\chi^2 = \sum \frac{(O - E)^2}{E}; \quad df = k - 1$$

where k is the number of categories in our distribution. This test is only valid if each of the expected frequencies are at least 5 and we have a random sample.

Now we can finish the test:

Topic 41

χ^2 Test for Independence

Lecture

Continuing with categorical variables, we often want to know whether or not two categorical variables are related. By “are they related?”, we mean “If an observation falls into a certain category for one variable, does this make it more likely to fall into a certain category of the other?”

In other words, we want to know if our two variables are independent or dependent? If they are independent, then knowing that a variable falls into a certain category for the row variable does not affect the probability that it will fall into any of the categories for the column variable, and vice versa.

Let’s consider two categorical variables we have seen before that are included in the following contingency table:

		Marijuana Usage Level			
		Never	Rarely	Frequently	Total
Political Views	Liberal	479	173	119	771
	Conservative	214	47	15	276
	Other	172	45	85	302
	Total	865	265	219	1349

Are the variables “political views” and “marijuana usage level” related? Do they seem to depend on one another?

There is some uncertainty in answering this question since this table is taken from a random sample, and any apparent relationship we observe between variables from a random sample could be due to _____. We can answer this question much more definitively by conducting a hypothesis test for independence.

The hypotheses for the χ^2 tests for independence will always be the same:

H_0 : The variables are independent.

H_A : The variables are dependent.

Remember that we always begin by assuming that the null hypothesis is true. So, we need to know what our contingency table would look like if the variables were _____.

Remember how we determined independence in probability. Two events are independent if:
 $P(A \cap B) = P(A)P(B)$.

Consider the political/marijuana table. If event A is "Liberal" and event B is "Never", then

$$P(A) = \qquad \qquad \qquad P(B) =$$

So, if they were independent, then $P(A \cap B) =$

But we know that $P(A \cap B) = \frac{\# \text{ A and B}}{\text{sample size}}$. That means if these variables were independent, we would expect:

Writing this more generally, we have:

$$\text{Expected frequency} = \frac{(\text{Sum of row 1})(\text{Sum of column 1})}{\text{sample size}}$$

So, if the variables were perfectly independent, we would expect 494.38 observations in the liberal row and the never column. We have 479. Is that difference big enough to consider the variables as dependent?

There are lots of other levels to consider: Rarely and Conservative, Frequently and Other, Never and Other, etc. We need to compare the expected counts to the actual counts for every cell in our table to really get a sense of how much how actual table is deviating from the expected one if they were independent.

In general, the expected count is

$$\text{Expected frequency } (E_{r,c}) = \frac{(\text{Sum of row } r)(\text{Sum of column } c)}{\text{sample size}}$$

Once we have all of our expected counts, we can calculate our test statistic for the test for independence.

$$\chi^2 = \sum \frac{(O - E)^2}{E}; \quad df = (r - 1)(c - 1)$$

where E are the expected counts, O are the observed counts, r are the number of rows and c are the number of columns.

This test is only valid if our sample size is large enough. That is, each of the expected counts must be at least 5. Also, we always treat these independence tests as right-tailed. That is, there will only be one critical value and the p -value is the probability of getting a number larger than our test statistic.

Example: Conduct a hypothesis test to determine whether we have sufficient evidence to suggest that political affiliation and marijuana usage level are dependent.

		Marijuana Usage Level			
		Never	Rarely	Frequently	Total
Political Views	Liberal	479	173	119	771
	Conservative	214	47	15	276
	Other	172	45	85	302
	Total	865	265	219	1349

	Never	Rarely	Freq	Total		Never	Rarely	Freq	Total
Lib.		$\frac{771 \cdot 265}{1349}$	$\frac{771 \cdot 219}{1349}$	771	Lib.		151.46	125.17	771
Cons.		$\frac{276 \cdot 265}{1349}$	$\frac{276 \cdot 219}{1349}$	276	Cons.	176.98	54.22	44.81	276
Other		$\frac{302 \cdot 265}{1349}$	$\frac{302 \cdot 219}{1349}$	302	Other	193.65	59.33	49.03	302
Total	865	265	219	1349	Total	865	265	219	1349

$$\begin{aligned} \chi^2 = & \frac{(173-151.46)^2}{151.46} + \frac{(119-125.17)^2}{125.17} + \frac{(214-176.98)^2}{176.98} + \frac{(47-54.22)^2}{54.22} + \frac{(15-44.81)^2}{44.81} \\ & + \frac{(172-193.65)^2}{193.65} + \frac{(45-59.33)^2}{59.33} + \end{aligned}$$

$$\begin{aligned}
\chi^2 &= && +3.063 + 0.304 \\
&&& +6.430 + 0.961 + 19.831 \\
&&& +2.420 + 3.461 + 26.398 \\
&= &&
\end{aligned}$$

It is important to remember that “dependent” in this sense just means that knowing something about one variable gives you information about the probability of the other. It does not imply that one causes the other! It could be that political affiliation affects marijuana use, or that marijuana use affects political affiliation, or that other, confounding variables affect both of these things.

You can see which cells contributed the most to the χ^2 test statistic by looking at which ones have the largest value of $(O - E)^2/E$. In this case, the “conservative and frequently” and “other and frequently” contributed the biggest values (19.831 and 26.389, respectively). This means that these cells had the largest difference between observed and expected counts. Here, the observed count for “conservative and frequently” was much lower than what we would expect under independence (15 vs. 44.81) and the observed count for “other and frequently” was much higher than what we would expect under independence (85 vs. 49.03).

Activities

Activity 41-1: z Test vs. χ^2 Test

In 1993, Lasko et al. conducted an experiment to investigate a possible relationship between heart disease and baldness. Researchers asked a sample of 663 male heart patients to classify whether they were balding or not. They also asked a control group (not suffering from heart disease) of 772 males the same question. The results are given in the following table:

	Little to No Balding	Some to Extreme Balding	Total
Heart Disease	251 ()	412 ()	
Control	331 ()	441 ()	
Total			

- Fill in the total row and column in the table above. You can ignore those parentheses for now.
- What proportion of men in the study had some to extreme balding?
- What proportion of those with heart disease had some to extreme balding?
- What proportion of those without heart disease (the control) had some to extreme balding?
- Based on your answers above, does it seem like balding and whether a man has heart disease are related? Explain.
- To investigate this further, let's do a test for independence. Before we get started, let's find the expected counts for each cell and put those values in the parentheses.

- g. Conduct a full test, showing all 6 steps, for whether baldness and heart disease are related.
- h. Another way to determine if these two variable are related is by finding the proportions of balding for those with heart disease and those without and to test whether there is a difference. You already found these proportions in parts c and d. Use those to do a 2-sample z test for the difference in proportion. Show all 6 steps.
- i. Comment on how the p -value in part g and is related to the p -value in part h.

When we have two categories for each of our variable in a test for independence, the test will be equivalent to a 2-sample z test for the difference in proportion. When either variable has 3 or more categories, then a test for independence should be used.

Comparing Two Variances and the F Distribution

Lecture

In Chapter 8, we conducted hypothesis tests for the difference in proportion and the difference in means, but we did not do one for the difference in variance. We turn to testing two variances now. It is typical, however, to not find the difference in variances. Rather, we find the ratio of variances.

When testing for the ratio in variances, the alternative hypothesis are usually one of the following:

$$H_A : \sigma_1^2 / \sigma_2^2 > 1$$

$$H_A : \sigma_1^2 / \sigma_2^2 < 1$$

$$H_A : \sigma_1^2 / \sigma_2^2 \neq 1$$

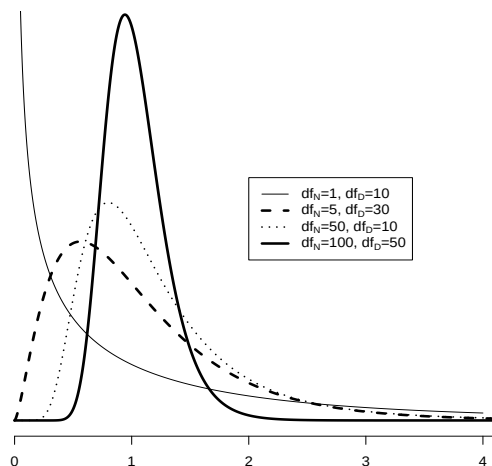
The reason we focus on ratios and not differences is because of the kind of distribution the ratio of variances follows. Remember that when testing a single variance, we needed to use the χ^2 distribution. The difference in χ^2 values does not follow any nice distribution, but the ratio of χ^2 values does. The ratio of χ^2 variables follows an F distribution.

The F Distribution

Like a t or χ^2 distribution, an F distribution is a family of distributions that are characterized by their degrees of freedom.

This family of distributions has the following properties:

1. Like χ^2 , the F distributions are not symmetric, they are skewed right.
2. Since an F variable is the ratio of two χ^2 variables and each χ^2 variable has its own degrees of freedom, an F distribution will have degrees of freedom that correspond to the numerator, df_N , and degrees of freedom that correspond to the denominator, df_D .
3. All F values are at least 0.
4. As both degrees of freedom increase, the F distribution gets more and more symmetric.
5. The mean of all F distributions is approximately equal to 1.
6. If the numerator degrees of freedom equals 1, then the square root of the F distribution is identical to the t distribution.



The test statistic for testing the ratio of variances or standard deviations is

$$F = \frac{s_1^2}{s_2^2}; \quad df_N = n_1 - 1; \quad df_D = n_2 - 1$$

In order for these tests to be valid, we need random, independent samples from normal populations.

Note: Finding critical values for the F distribution is given in the Excel guide and on the summary page at the end of this chapter's notes.

Activities

Activity 42-1: F Critical Values for Hypothesis Tests

Find the critical value(s) for each of the following hypothesis tests. The Excel guide on Canvas will be helpful here.

a. $\alpha = 0.02$; $n_1 = 50$; $n_2 = 64$; $H_0: \sigma_1^2 \geq \sigma_2^2$; $H_a: \sigma_1^2 < \sigma_2^2$.

b. $\alpha = 0.05$; $n_1 = 200$; $n_2 = 10,057$; $H_0: \sigma_1/\sigma_2 = 1$; $H_a: \sigma_1/\sigma_2 \neq 1$.

c. $\alpha = 0.10$; $n_1 = 4$; $n_2 = 17$; $H_0: \sigma_1 \leq \sigma_2$; $H_a: \sigma_1 > \sigma_2$.

Activity 42-2: East High vs. West High

The East High Wildcats and West High Knights are bitter basketball rivals. A question has been posed about which team is more consistent in terms of the number of points they score in basketball games. Samples of 16 East High games and 21 West High games were taken and it was found that the Wildcats score 87 points per game on average with a standard deviation of 6.2 points and the Knights score 82 points per game on average with a standard deviation of 9.3 points. Based on this information, can we conclude that one team is more consistent than the other? Use a significance level of 0.10.

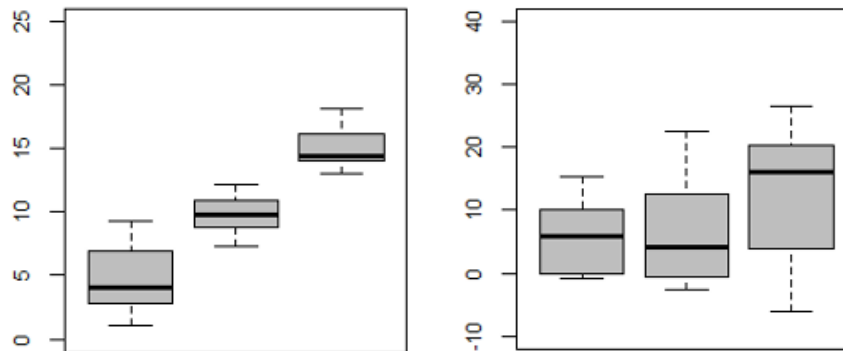
Extra Space for Notes

Analysis of Variance (ANOVA) Test

Lecture

In the goodness-of-fit test, we discussed how to deal with more than two proportions. Now, we will discuss how to deal with more than two means. The procedure for comparing multiple population means is known as the **analysis of variance** or ANOVA. Of course, we know that if we are testing for a difference in two independent means, we can use a two-sample t-test. With ANOVA, we test three or more groups to see if any of groups' means differ significantly from the others.

It may seem strange to call the test that compares *means* an analysis of *variance*, but the variance is a crucial part of the process. Consider the following boxplots. In both of these cases, the sample means for the three boxplots are about 5, 10, and 15, respectively. In the first case, it seems clear that the true means must also differ. In the second case, this isn't so clear.



The thing that makes these two cases so different is that in the first, the boxplots are not very spread out, while in the second they are. More variation reflects greater _____ regarding the values of the true unknown means.

With ANOVA, we compare the average “between group” variance to the average “within group” variance.

Between group variance refers to how much the means of the groups vary from one another.

Within group variance refers to how spread out the data is in each group.

The basic idea is this: if the average amount of variation _____ the groups is substantially larger than the average amount of variation _____ the groups, then the true group means likely differ. If not, then we cannot rule out that they are equal.

As you might guess, we will make this decision using a hypothesis test. For ANOVA, we have:

H_0 : The group means are all equal or $\mu_1 = \mu_2 = \cdots = \mu_k$

H_A : At least one group mean differs from the others.

ANOVA notation

- k : the # of groups we are dealing with.
- $\mu_1, \dots, \mu_k$: the true means for groups 1 through k .
- $\bar{x}_1, \dots, \bar{x}_k$: the sample means for groups 1 through k .
- $s_1, \dots, s_k$: the sample standard deviations for groups 1 through k .
- $n_1, \dots, n_k$: the sample sizes for groups 1 through k .
- N : the overall sample size for all groups combined ($N = n_1 + n_2 + \cdots + n_k$).
- $\bar{\bar{x}}$: the grand mean of all data from all groups ($\frac{\sum x}{N} = \frac{n_1\bar{x}_1 + \cdots + n_k\bar{x}_k}{N}$).

Note that if the sample sizes for each group are equal, then the grand mean is just the average of all of sample means: $\bar{\bar{x}} = \frac{\sum \bar{x}_i}{k}$.

Sum of Squares

The way we go about testing this is by dealing with the sum of squares. "Sums of squares" refers to sums of squared deviations from some mean. ANOVA compares "between" to "within" group variation, and these types of variation are quantified using sums of squares. The three important sums of squares we will look at are:

SS_T : _____ sum of squares.

SS_B : Sum of squares _____ the samples.

SS_W : Sum of squares _____ the samples.

Total Sum of Squares

Total sum of squares is calculated by summing up the squared deviations of every point in the data from the grand mean:

$$SS_T = \sum_{i=1}^N (x_i - \bar{x})^2$$

This quantifies the amount of overall variation in the data, for all of the groups put together. The degrees of freedom associated with SS_T are $df = N - 1$.

Sum of Squares Between Samples

The sum of squares between samples is calculated by summing the squared deviations of each sample mean from the grand mean:

$$SS_B = n_1(\bar{x}_1 - \bar{x})^2 + \cdots + n_k(\bar{x}_k - \bar{x})^2 = \sum_{i=1}^k n_i(\bar{x}_i - \bar{x})^2$$

This quantifies how much the sample means vary. It is what we are referring to when we say “between group variation”. The degrees of freedom associated with SS_B are $df_N = k - 1$.

Sum of Squares Within Samples

The sum of squares within samples is calculated by adding up the squared deviations of each data point around its group mean:

$$SS_W = (n_1 - 1)s_1^2 + \cdots + (n_k - 1)s_k^2 = \sum_{i=1}^k (n_i - 1)s_i^2$$

This quantifies how much the data varies around its group sample means. It is what we are referring to when we say “within group variation”. The degrees of freedom associated with SS_W are $df_D = N - k$.

Mean Squares

Finally, to get an idea of how much variation we have on average, we will divide SS_B and SS_W by their degrees of freedom to obtain **mean squares**:

$$\begin{aligned} \text{Mean Square Between} &= MS_B = \frac{SS_B}{df_N} = \frac{\sum_{i=1}^k n_i(\bar{x}_i - \bar{x})^2}{k - 1} \\ \text{Mean Square Within} &= MS_W = \frac{SS_W}{df_D} = \frac{\sum_{i=1}^k (n_i - 1)s_i^2}{N - k} \end{aligned}$$

Test Statistic

It turns out that when we divide these mean squares, we obtain a value that follows an F distribution. So, we can then compute our test statistic by calculating:

$$F = \frac{MS_B}{MS_W}; \quad df_N = k - 1; \quad df_D = N - k$$

If the F test statistic is large, that means there is much more variation _____ samples than there is _____ them. If that is the case, we should _____ null hypothesis that all of the means are equal.

There are some necessary conditions to use ANOVA:

1. The population of each group is normal.
2. Each population has the same variance (or standard deviation).
3. Each sample is random and independent.

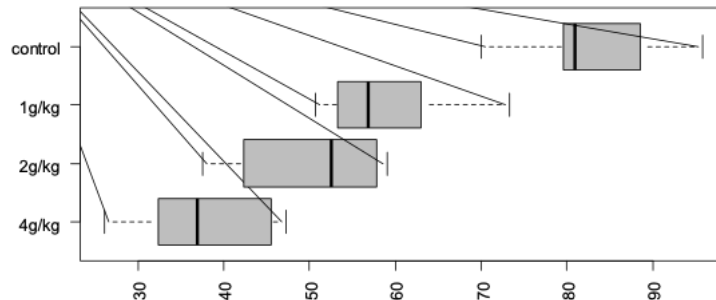
The ANOVA Table

All of the information regarding this ANOVA procedure can be summarized in an ANOVA table:

Source of Variation	Sum of Squares	df	Mean Square	F Statistic
Between	SS_B	$k - 1$	$MS_B = \frac{SS_B}{k-1}$	$F = \frac{MS_B}{MS_W}$
Within	SS_W	$N - k$	$MS_W = \frac{SS_W}{N-k}$	
Total	SS_T	$N - 1$		

Example: A researcher who studies sleep is interested in the effects of ethanol on sleep time. She gets a sample of 20 rats and gives each an injection having a particular concentration of ethanol per body weight. There are 4 treatment groups, with 5 rats per treatment. She records REM sleep time for each rat over a 24-period. The results are given in the table and plotted below:

Treatment	Mean	SD
1) 0 (control)	83.0	9.7
2) 1 g/kg	59.4	9.0
3) 2 g/kg	49.8	9.66
4) 4 g/kg	37.6	8.8



Source of Variation	Sum of Squares	df	Mean Square	<i>F</i> Statistic
Between				
Within				
Total				

Note that this overall ANOVA test does not identify which means differ from which other means or how many differ. It only tells us that at least one is significantly different from at least one other. There are other statistical methods that can be used to deduce which means differ, but we will not consider them here.

Chapter 10 Summary

- We can use a χ^2 to evaluate whether one categorical variable fits an expected distribution using the χ^2 goodness of fit test.
- We can use a χ^2 to evaluate whether two categorical variables are independent using the χ^2 test for independence.
- An F distribution is one more (and the last) distribution we will see in this class and can be used to test different hypotheses.
- The ratio of two χ^2 distributions follows an F distribution, so any F distribution has a numerator degrees of freedom and a denominator degrees of freedom.
- We can use an F , which is the ratio of two χ^2 values to evaluate how two variances are related to one another.
- We can use an F to evaluate whether several (more than 2) group means differ from one another by using the ANOVA procedure.

Test type:	Left Tailed	Right Tailed	Two Tailed
z_0	invNorm(α , 0, 1) =NORM.INV(α , 0, 1)	invNorm($1 - \alpha$, 0, 1) =NORM.INV($1 - \alpha$, 0, 1)	$\pm$ invNorm($1 - \alpha/2$, 0, 1) $\pm$ =NORM.INV($1 - \alpha/2$, 0, 1)
t_0	invT(α , df) =T.INV(α , df)	invT($1 - \alpha$, df) =T.INV($1 - \alpha$, df)	$\pm$ invT($1 - \alpha/2$, df) $\pm$ =T.INV($1 - \alpha$, df)
χ_0^2	=CHISQ.INV(α , df)	=CHISQ.INV($1 - \alpha$, df)	=CHISQ.INV($\alpha/2$, df) =CHISQ.INV($1 - \alpha/2$, df)
F_0	=F.INV(α , df_N , df_D)	=F.INV($1 - \alpha$, df_N , df_D)	=F.INV($\alpha/2$, df_N , df_D) =F.INV($1 - \alpha/2$, df_N , df_D)

To find **p-values**:

Test type:	Left Tailed	Right Tailed	Two Tailed
z test stat	normalcdf(-1000, z , 0, 1) =NORM.DIST(z , 0, 1, TRUE)	normalcdf(z , 1000, 0, 1) =1-NORM.DIST(z , 0, 1, TRUE)	If $z < 0$: double left tailed If $z > 0$: double right tailed
t test stat	tcdf(-1000, t , df) =T.DIST(t , df, TRUE)	tcdf(t , 1000, df) =1-T.DIST(t , df, TRUE)	If $t < 0$: double left tailed If $t > 0$: double right tailed
χ^2 test stat	χ^2 cdf(-1000, χ^2 , df) =CHISQ.DIST(χ^2 , df, TRUE)	χ^2 cdf(χ^2 , 1000, df) =1-CHISQ.DIST(χ^2 , df, TRUE)	If $\chi^2 < df$: double left tailed If $\chi^2 > df$: double right tailed
F test stat	Fcdf(-1000, F , df_N , df_D) =F.DIST(F , df_N , df_D , TRUE)	Fcdf(F , 1000, df_N , df_D) =1-F.DIST(F , df_N , df_D , TRUE)	If $F < 1$: double left tailed If $F > 1$: double right tailed

Congratulations! You made it through all of the content in this class!

Summary Pages

Chapter 1 Summary

- **Statistics** is the science of data.
- **Data** is information.
- **Observational units** or cases are the actual subjects from whom we are collecting information.
- A **variable** is a characteristic of a subject that can be assigned a number or category. It must be collectible on each observational unit.
- Variables are either **quantitative** or **categorical**.
 - Quantitative variables are numeric. These can be **continuous** if the variable can take on decimal values or **discrete** if it can only be a whole number.
 - Categorical (or qualitative) variables do not have to be numbers. These can be **binary** if it only has two categories or **non-binary** if it has 2+ categories.
- The pattern of variation of a variable is known as its **distribution**.
- A **proportion** simply measures a part in comparative relation to the whole. It is like a percentage, but written as a decimal.
- Statistics can be divided into **descriptive statistics** and **inferential statistics**.
- **Observational studies** do not have researchers involved in assigning groups.
 - Descriptive statistics involves simply describing a dataset.
 - Inferential statistics involves generalizing to a larger group.
- The **population** is the large group we want to know something about. Populations are described by **parameters**.
- The **sample** is the smaller group we actually collect information from. Samples are described by **statistics**. Samples should be selected **randomly** to avoid being **biased**.
- **Bias** occurs when the sample is not representative of the population.
- The **response variable** is the one we want to predict.
- The **explanatory variable** is the one we use to predict the response variable.
- **Observational studies**: researchers are not involved in assigning groups.
- **Controlled experiments**: researchers are directly involved in assigning groups.
- A **confounding variable** is one that helps explain the association between variables but is not accounted for in the study.
- **Blinded study**: participants don't know which group they are in. **Double-blinded study**: neither participants nor researcher knows which group the participants are in.

Chapter 2 Summary

1. Data graphics are used to present information from a distribution (or frequency distribution) for all sorts of variables – both categorical and quantitative.
2. The most common graphs for categorical variables are bar charts and pie charts (but pie charts are terrible!).
3. The most common graphs for quantitative variables are dotplots, histograms and boxplots.
4. Proportions (or percentages) are useful when describing categorical variables.
5. When describing a distribution for a quantitative variable, you need to mention **center, spread, shape (including number of peaks), unusual features** (like outliers) and write in context. The order you mention these characteristics does not matter.
6. The most common measures of center are the mean and the median.
7. The most common measures of spread are the standard deviation and the IQR.
8. The most common shapes are skewed left, skewed right, or approximately symmetric.
9. We use the median and IQR when the distribution is skewed and we use the mean and standard deviation when it is symmetric.

Bonus – Symbol Guide

The following table lists symbols and what they are usually denoting. This will be updated as we continue in this class.

Symbol	Description
n	Sample size
N	Population size
x or x_i	Observation
$\bar{x}$	Sample mean
μ	Population mean
s	Sample standard deviation
σ	Population standard deviation
M	Median
Q_1	Lower quartile
Q_3	Upper quartile
IQR	Interquartile range
df	Degrees of freedom
z	Standard score

Chapter 3 Summary

Summary of Probability Identities

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ in general.
2. $P(A \cup B) = P(A) + P(B)$ if A and B are mutually exclusive.
3. $P(A \cap B) = P(A) \cdot P(B|A)$ in general.
4. $P(A \cap B) = P(A) \cdot P(B)$ if A and B are independent.
5. $P(A \cap B) = 0$ if A and B are mutually exclusive.
6. $P(A|B) = \frac{P(A \cap B)}{P(B)}$ in general.
7. $P(A|B) = P(A)$ if A and B are independent.
8. $P(A^c) = 1 - P(A)$ in general.

Summary of Counting Principles

Principle	Description	Formula
Counting principle	If one event can occur in m ways and a second event can occur in n ways, then the number of ways the two events can occur in sequence is $m \cdot n$. This can be extended to more than two events.	$m \cdot n$
Arrangements	The number of arrangements of n distinct objects.	$n!$
Permutations	The number of permutations of n distinct objects taken r at a time, where $r \leq n$.	${}_nP_r = \frac{n!}{(n-r)!}$
Combinations	The number of combinations of r objects selected from a group of n objects without regard to order, where $r \leq n$.	${}_nC_r = \frac{n!}{r!(n-r)!}$

Chapter 4 Summary

Distribution	Summary	Formulas
Binomial Distribution	A binomial experiment satisfies these conditions: 1. The experiment has a fixed number of n of independent trials. 2. There are only two possible outcomes for each trial. Each outcome can be classified as a success or a failure. 3. The probability of success p is the same for each trial. 4. We are counting the number of successes. The parameters of a binomial distribution are n and p.	n = the number of trials. x = the number of successes in n trials. p = the probability of success in a single trial. q = the probability of failure in a single trial = $1 - p$. The probability of exactly x successes in n trials is $P(x) = {}_nC_x p^x q^{n-x} = \frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n.$ $E(X) = \mu = np$ $\text{Var}(X) = \sigma^2 = np(1-p), \quad \text{SD}(X) = \sigma = \sqrt{np(1-p)}$ $P(X = x) = \text{binompdf}(n, p, x) = \text{BINOM.DIST}(x, n, p, \text{FALSE})$ $P(X \leq x) = \text{binomcdf}(n, p, x) = \text{BINOM.DIST}(x, n, p, \text{TRUE})$
Geometric Distribution	A geometric distribution is a discrete probability distribution of a random variable X that satisfies these conditions. 1. A trial is repeated until a success occurs. 2. The repeated trials are independent of each other. 3. The probability of success p is the same for each trial. 4. The random variable x represents the number of the trial in which the first success occurs. The parameter of a geometric distribution is p.	x = the number of trials in which the first success occurs p = probability of success in a single trial. The probability that the first success occurs on trial number x is $P(x) = (1-p)^{x-1} \cdot p, \quad x = 1, 2, 3, \dots$ $E(X) = 1/p$ $\text{Var}(X) = (1-p)/p^2, \quad \text{SD}(X) = \sqrt{1-p}/p$ $P(X = x) = \text{geomtpdf}(p, x)$ $P(X \leq x) = \text{geomtcdf}(p, x)$
Poisson Distribution	The Poisson distribution is a discrete probability distribution of a random variable X that satisfies these conditions. 1. The experiment consists of counting the number of times x an event occurs over a specified interval of time, area, or volume. 2. The probability of the event occurring is the same for each interval. 3. The number of occurrences in one interval is independent of the number of occurrences in other intervals. The parameter of the Poisson distribution is μ.	x = the number of occurrences in the given interval. μ = the mean number of occurrences in a given interval. The probability of exactly x occurrences in an interval is $P(x) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$ $E(X) = \mu$ $\text{Var}(X) = \mu, \quad \text{SD}(X) = \sqrt{\mu}$ $P(X = x) = \text{poissonpdf}(\mu, x) = \text{POISSON.DIST}(x, \mu, \text{FALSE})$ $P(X \leq x) = \text{poissoncdf}(\mu, x) = \text{POISSON.DIST}(x, \mu, \text{TRUE})$

For a random variable X with associated probability mass function $P(x)$,

- For $P(x)$ to be valid, $0 \leq P(x) \leq 1$ and $\sum P(x) = 1$.
- $E(X) = \mu = \sum xP(x)$. • $\text{SD}(X) = \sqrt{\text{Var}(X)} = \sqrt{\sum x^2 P(x) - \mu^2}$.
- $\text{Var}(X) = E((x - \mu)^2) = \sigma^2 = \sum (x - \mu)^2 P(x) = \sum x^2 P(x) - \mu^2$.

For discrete random variables (one that can only take on whole number values),

$P(X < x) = P(X \leq x - 1)$	Example: $P(X < 4) = P(X \leq 3)$
$P(X > x) = 1 - P(X \leq x)$	Example: $P(X > 4) = 1 - P(X \leq 4)$
$P(X \geq x) = 1 - P(X \leq x - 1)$	Example: $P(X \geq 4) = 1 - P(X \leq 3)$

Chapter 5 Summary

1. Continuous distributions have pdfs instead of pmfs that describe how the probabilities are distributed.
2. All pdf values must be at least zero and the total area under a pdf must be 1.
3. For all continuous distributions, $P(X < x) = P(X \leq x)$, $P(X > x) = P(X \geq x)$, and $P(a < X < b) = P(a \leq X < b) = P(a < X \leq b) = P(a \leq X \leq b)$.
4. A normal distribution is bell-shaped and centered at its mean.
5. To find probabilities for a variable X that is normally distributed, use a z -score and table with $z = \frac{x-\mu}{\sigma}$ or a calculator with `normalcdf(lower, upper,  $\mu$ ,  $\sigma$ )`.
6. To find percentiles for a variable X that is normally distributed, use a table and inverse z -score formula $x = \mu + \sigma z$ or a calculator with `invnorm(area left,  $\mu$ ,  $\sigma$ )`.
7. To find probabilities for a sample mean $\bar{X}$ when n is large enough, use a z -score and table with $z = \frac{\bar{x}-\mu}{\sigma/\sqrt{n}}$ or a calculator with `normalcdf(lower, upper,  $\mu$ ,  $\sigma/\sqrt{n}$ )`.
8. To find percentiles for a sample mean $\bar{X}$ when n is large enough, use a table and inverse z -score formula $\bar{x} = \mu + z \cdot \sigma/\sqrt{n}$ or a calculator with `invnorm(area left,  $\mu$ ,  $\sigma/\sqrt{n}$ )`.
9. To find probabilities for a sample proportion $\hat{p}$ when n is large enough, use a z -score and table with $z = \frac{\hat{p}-p}{\sqrt{p(1-p)/n}}$ or a calculator with `normalcdf(low, up,  $p$ ,  $\sqrt{p(1-p)/n}$ )`.
10. To find percentiles for a sample proportion $\hat{p}$ when n is large enough, use a table and inverse z -score formula $\hat{p} = p + z\sqrt{p(1-p)/n}$ or a calculator with `invnorm(area left,  $p$ ,  $\sqrt{p(1-p)/n}$ )`.
11. In Excel, you can find the following functions:
 - $P(X < x) = \text{NORM.DIST}(x, \text{mean}, \text{standard deviation}, \text{TRUE})$
 - $P(X > x) = 1 - \text{NORM.DIST}(x, \text{mean}, \text{standard deviation}, \text{TRUE})$
 - $P(a < X < b) = \text{NORM.DIST}(b, \mu, \sigma, \text{TRUE}) - \text{NORM.DIST}(a, \mu, \sigma, \text{TRUE})$
 - You can find the cutoff value by using `=NORM.INV(area left, mean, SD)`
12. Central Limit Theorem Statements:
 - (1) $\bar{x}$ is approximately normally distributed with mean μ and standard deviation $\sigma/\sqrt{n}$.
 - (2) $\hat{p}$ is approximately normally distributed with mean p and standard deviation $\sqrt{\frac{p(1-p)}{n}}$.

Chapter 6 Summary

Confidence intervals both give us an idea of an interval of plausible values for our unknown population parameter and help us quantify how uncertain we are in our estimates.

Constructing a confidence interval for a population proportion, p :

1. Identify the statistics $\hat{p}$ and n or calculate the sample proportion as $\hat{p} = \frac{x}{n}$ where x is the number of “successes.”
2. Verify that the sampling distribution of $\hat{p}$ can be approximated by a normal distribution by checking that $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$.
3. Find the critical value z_c that corresponds to the given level of confidence.
4. Find the left and right endpoints by calculating $\hat{p} \pm z_c \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$.
5. Write the confidence interval and provide an interpretation.

Note: Steps 3 and 4 can be done with the 1-PropZInt command in the calculator.

Constructing a confidence interval for a population mean, μ :

1. Identify the statistics $\bar{x}$, s , and n or calculate them.
2. Verify that the sample is random and that $n \geq 30$ or the data come from a normal distribution.
3. Find the critical value t_c that corresponds to the confidence level with $df = n - 1$.
4. Find the left and right endpoints by calculating $\bar{x} \pm t_c \cdot s / \sqrt{n}$.
5. Write the confidence interval and provide an interpretation.

Note: Steps 3 and 4 can be done with the TInterval command in the calculator.

Constructing a confidence interval for a population variance and standard deviation, σ^2 and σ :

1. Identify the statistic s or s^2 or calculate them.
2. Verify that the sample is random and that the data come from a normal distribution.
3. Find the critical values χ_L^2 and χ_R^2 that correspond to the given level of confidence with $df = n - 1$ using Excel.
4. Find the left endpoint by calculating $\frac{(n-1)s^2}{\chi_R^2}$ for the variance or $\sqrt{\frac{(n-1)s^2}{\chi_R^2}}$ for the SD.
5. Find the right endpoint by calculating $\frac{(n-1)s^2}{\chi_L^2}$ for the variance or $\sqrt{\frac{(n-1)s^2}{\chi_L^2}}$ for the SD.
6. Write the confidence interval and provide an interpretation.

Chapter 7 Summary

Hypothesis testing is another statistical inference procedure. In each hypothesis test, we assume the null hypothesis is true and then gather evidence that it is not based on a sample. Each hypothesis test follows the following 6 steps:

1. State the parameter(s) of interest.
2. Set up the null & alternative hypotheses.
3. State the significance level and the corresponding critical value (if we are basing the decision on critical values).
4. Compute the test statistic and p -value (if we are basing the decision on p -values).
5. Make the statistical decision.
6. Interpret the results.

The test statistic's form depends on what population parameter we are testing. Refer to the Inference Summary Sheet for information about the test statistic for each parameter.

Every hypothesis test will be a left-tailed, right-tailed, or two-tailed test. Pay special attention to what the claim is and remember that the statement of equality always goes in the null hypothesis.

To find **critical values**, use the t-table or use technology:

Test type:	Left Tailed	Right Tailed	Two Tailed
z_0	invNorm(α , 0, 1) =NORM.INV(α , 0, 1)	invNorm($1 - \alpha$, 0, 1) =NORM.INV($1 - \alpha$, 0, 1)	$\pm$ invNorm($1 - \alpha/2$, 0, 1) $\pm$ =NORM.INV($1 - \alpha/2$, 0, 1)
t_0	invT(α , df) =T.INV(α , df)	invT($1 - \alpha$, df) =T.INV($1 - \alpha$, df)	$\pm$ invT($1 - \alpha/2$, df) $\pm$ =T.INV($1 - \alpha$, df)
χ_0^2	=CHISQ.INV(α , df)	=CHISQ.INV($1 - \alpha$, df)	=CHISQ.INV($\alpha/2$, df) =CHISQ.INV($1 - \alpha/2$, df)

To find **p-values**:

Test type:	Left Tailed	Right Tailed	Two Tailed
z test stat	normalcdf(-1000, z , 0, 1) =NORM.DIST(z , 0, 1, TRUE)	normalcdf(z , 1000, 0, 1) =1-NORM.DIST(z , 0, 1, TRUE)	If $z < 0$: double left tailed If $z > 0$: double right tailed
t test stat	tcdf(-1000, t , df) =T.DIST(t , df, TRUE)	tcdf(t , 1000, df) =1-T.DIST(t , df, TRUE)	If $t < 0$: double left tailed If $t > 0$: double right tailed
χ^2 test stat	χ^2 cdf(-1000, χ^2 , df) =CHISQ.DIST(χ^2 , df, TRUE)	χ^2 cdf(χ^2 , 1000, df) =1-CHISQ.DIST(χ^2 , df, TRUE)	If $\chi^2 < df$: double left tailed If $\chi^2 > df$: double right tailed

Chapter 8 Summary

The formulas for hypothesis testing and confidence intervals will be different depending on what kind of parameter is being investigated, but the overall process is always the same. Refer to the Inference Summary Sheet for information about the test statistic for each parameter. Here is an updated symbol table.

Symbol	Description	Symbol	Description
n, n_1, n_2	Sample sizes	s, s_1, s_2	Sample standard deviations
N, N_1, N_2	Population sizes	$\sigma, \sigma_1, \sigma_2$	Population standard deviations
df	Degrees of freedom	s^2, s_1^2, s_2^2	Sample variances
E	Margin of error	$\sigma^2, \sigma_1^2, \sigma_2^2$	Population variances
x, x_1, x_2	Number of "successes"	χ^2	Test statistic for variance/standard deviation
$\hat{p}, \hat{p}_1, \hat{p}_2$	Sample proportion	χ_L^2	Left (smaller) chi-square critical value
p, p_1, p_2	Population proportions	χ_R^2	Right (larger) chi-square critical value
$\bar{p}$	Proportion of all successes from both samples	$\bar{d}$	Sample mean of the differences
z	Test statistic for proportion(s)	μ_d	Population mean of the differences
z_c	z critical value for confidence intervals	s_d	Standard deviation of the differences
z_0	z critical value for hypothesis tests	H_0	Null hypothesis
$\bar{x}, \bar{x}_1, \bar{x}_2$	Sample means	H_A	Alternative hypothesis
μ, μ_1, μ_2	Population means	α	Significance level
t	Test statistic for mean(s)	μ_0	Hypothesized mean
t_c	t critical value for confidence intervals	p_0	Hypothesized proportion
t_0	t critical value for hypothesis tests	σ_0	Hypothesized standard deviation
$q, \hat{q}, q_0, \bar{q}$	$1 - p, 1 - \hat{p}, 1 - q_0, 1 - \bar{p}$	σ_0^2	Hypothesized variance

Chapter 9 Summary

1. Correlation is a way to quantify how strong the linear relationship is between two quantitative variables.
2. Correlation is always between -1 and 1 . Closer to 0 is weaker.
3. We use regression to predict values of the response variable using values of one or more explanatory variables using the equation $\hat{y} = mx + b$ in simple linear regression or $\hat{y} = b + m_1x_1 + \cdots + m_kx_k$ in multiple linear regression or to quantify the nature and magnitude of relationships between variables by looking at the slopes.
4. The slope is interpreted as follows. Replace all underlined terms with values or context from the problem. When (the X variable) increases by 1 (unit), (the Y variable) is predicted to (increase/decrease) by (slope) (units).
5. The intercept is interpreted as follows. Replace all underlined terms with values or context from the problem. The predicted value of (the Y variable) is (intercept) when (the X variable) is zero (units).
6. R^2 is the coefficient of determination and it is the proportion of variation in the Y variable that is explained by the X variable(s).

Regression Symbols

Symbol	Description
r	Sample correlation coefficient
r^2	Coefficient of determination
ρ	Population coefficient
Y	The response variable
y	A value of the response variable
X	The explanatory variable
x	A value of the explanatory variable
$\hat{y}$	A predicted value of the response variable
$\hat{y}_p$	A particular predicted value of the response variable for prediction intervals
x_p	A particular x value used for a prediction intervals
d_i	A residual value: $y_i - \hat{y}_i$
m	The slope of the regression line
b	The y -intercept of the regression line
s_e	The standard error of the estimate
s_x	The sample standard deviation of X
s_y	The sample standard deviation of Y

Chapter 10 Summary

- We can use a χ^2 to evaluate whether one categorical variable fits an expected distribution using the χ^2 goodness of fit test.
- We can use a χ^2 to evaluate whether two categorical variables are independent using the χ^2 test for independence.
- An F distribution is one more (and the last) distribution we will see in this class and can be used to test different hypotheses.
- The ratio of two χ^2 distributions follows an F distribution, so any F distribution has a numerator degrees of freedom and a denominator degrees of freedom.
- We can use an F , which is the ratio of two χ^2 values to evaluate how two variances are related to one another.
- We can use an F to evaluate whether several (more than 2) group means differ from one another by using the ANOVA procedure.

Test type:	Left Tailed	Right Tailed	Two Tailed
z_0	invNorm(α , 0, 1) =NORM.INV(α , 0, 1)	invNorm($1 - \alpha$, 0, 1) =NORM.INV($1 - \alpha$, 0, 1)	$\pm$ invNorm($1 - \alpha/2$, 0, 1) $\pm$ =NORM.INV($1 - \alpha/2$, 0, 1)
t_0	invT(α , df) =T.INV(α , df)	invT($1 - \alpha$, df) =T.INV($1 - \alpha$, df)	$\pm$ invT($1 - \alpha/2$, df) $\pm$ =T.INV($1 - \alpha$, df)
χ_0^2	=CHISQ.INV(α , df)	=CHISQ.INV($1 - \alpha$, df)	=CHISQ.INV($\alpha/2$, df) =CHISQ.INV($1 - \alpha/2$, df)
F_0	=F.INV(α , df_N , df_D)	=F.INV($1 - \alpha$, df_N , df_D)	=F.INV($\alpha/2$, df_N , df_D) =F.INV($1 - \alpha/2$, df_N , df_D)

To find **p-values**:

Test type:	Left Tailed	Right Tailed	Two Tailed
z test stat	normalcdf(-1000, z , 0, 1) =NORM.DIST(z , 0, 1, TRUE)	normalcdf(z , 1000, 0, 1) =1-NORM.DIST(z , 0, 1, TRUE)	If $z < 0$: double left tailed If $z > 0$: double right tailed
t test stat	tcdf(-1000, t , df) =T.DIST(t , df, TRUE)	tcdf(t , 1000, df) =1-T.DIST(t , df, TRUE)	If $t < 0$: double left tailed If $t > 0$: double right tailed
χ^2 test stat	χ^2 cdf(-1000, χ^2 , df) =CHISQ.DIST(χ^2 , df, TRUE)	χ^2 cdf(χ^2 , 1000, df) =1-CHISQ.DIST(χ^2 , df, TRUE)	If $\chi^2 < df$: double left tailed If $\chi^2 > df$: double right tailed
F test stat	Fcdf(-1000, F , df_N , df_D) =F.DIST(F , df_N , df_D , TRUE)	Fcdf(F , 1000, df_N , df_D) =1-F.DIST(F , df_N , df_D , TRUE)	If $F < 1$: double left tailed If $F > 1$: double right tailed

Standard Normal Probabilities

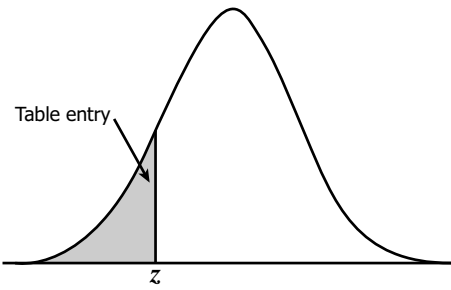


Table entry for z is the area under the standard normal curve to the left of z .

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
-3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
-3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
-3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
-3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
-2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
-2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
-2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
-1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
-1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
-1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
-1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
-1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
-1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
-1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
-1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
-1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
-0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
-0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
-0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
-0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
-0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
-0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
-0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
-0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
-0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
-0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Standard Normal Probabilities

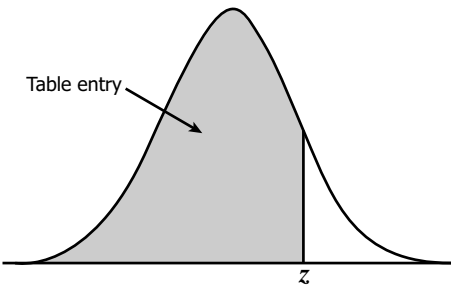


Table entry for z is the area under the standard normal curve to the left of z .

[illegible]

Inference Summary Sheet

Type of Inference	Parameter(s)	TI Calculator Function	CI Form or Test Statistic	Necessary Conditions
Confidence Interval	Proportion p (single)	1-PropZInt	$\hat{p} \pm z_c \sqrt{\frac{\hat{p}\hat{q}}{n}}$	1. Random sample. 2. $n\hat{p} \geq 5$ & $n\hat{q} \geq 5$.
Confidence Interval	Proportions $p_1 - p_2$ (independent)	2-PropZInt	$\hat{p}_1 - \hat{p}_2 \pm z_c \sqrt{\frac{\hat{p}_1\hat{q}_1}{n_1} + \frac{\hat{p}_2\hat{q}_2}{n_2}}$	1. Independent samples. 2. Random samples. 3. $n\hat{p}_1 \geq 5$ & $n\hat{q}_1 \geq 5$. 4. $n\hat{p}_2 \geq 5$ & $n\hat{q}_2 \geq 5$.
Confidence Interval	Mean μ (single)	TInterval	$\bar{x} \pm t_c \cdot s/\sqrt{n}$ $df = n - 1$	1. Random sample. 2. Normal data or $n \geq 30$.
Confidence Interval	Means $\mu_1 - \mu_2$ (independent)	2-SampTInt	$\bar{x}_1 - \bar{x}_2 \pm t_c \cdot \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$ $df = \min(n_1 - 1, n_2 - 1)$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Confidence Interval	Means μ_d (dependent)	TInterval	$\bar{d} \pm t_c \cdot s_d/\sqrt{n}$ $df = n - 1$	1. Dependent samples. 2. Random samples. 3. Normal data or $n \geq 30$.
Confidence Interval	Variance or SD σ^2 or σ (single)	None	Left: $\frac{(n-1)s^2}{\chi_R^2}$, Right: $\frac{(n-1)s^2}{\chi_L^2}$ (Take square roots for SD.) $df = n - 1$	1. Random sample. 2. Data are normal.
Hypothesis Test	Proportion p (single)	1-PropZTest	$z = \frac{\hat{p} - p_0}{\sqrt{p_0q_0/n}}$	1. Random sample. 2. $np_0 \geq 5$ & $nq_0 \geq 5$.
Hypothesis Test	Proportion $p_1 - p_2$ (independent)	2-PropZTest	$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\bar{p}\bar{q}}{n_1} + \frac{\bar{p}\bar{q}}{n_2}}}$ $\bar{p} = \frac{\text{successes}_1 + \text{successes}_2}{n_1 + n_2}$	1. Independent samples. 2. Random samples. 3. $n_1\bar{p} \geq 5$ & $n_1\bar{q} \geq 5$. 4. $n_2\bar{p} \geq 5$ & $n_2\bar{q} \geq 5$.
Hypothesis Test	Mean μ (single)	TTest	$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$ $df = n - 1$	1. Random sample. 2. Normal data or $n \geq 30$.
Hypothesis Test	Mean $\mu_1 - \mu_2$ (independent)	2-SampTTest	$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$ $df = \min(n_1 - 1, n_2 - 1)$	1. Independent samples. 2. Random samples. 3. Normal data or $n_1 \geq 30$ and $n_2 \geq 30$.
Hypothesis Test	Means μ_d (dependent)	TTest	$t = \frac{\bar{d}}{s_d/\sqrt{n}}$ $df = n - 1$	1. Dependent samples. 2. Random samples. 3. Normal data or $n \geq 30$.
Hypothesis Test	Variance or SD σ^2 or σ	None	$\chi^2 = \frac{(n-1)s^2}{\sigma_0^2}$ $df = n - 1$	1. Random sample. 2. Data are normal.

Common Critical Z-Values

Confidence Level	90%	95%	98%	99%
Critical Value (z_c)	1.645	1.96	2.326	2.576

Central Limit Theorem Statements:

$\hat{p}$ is approximately normal with mean p and standard deviation $\sqrt{pq/n}$.

$\bar{x}$ is approximately normal with mean μ and standard deviation $\sigma/\sqrt{n}$.

Chapters 9 and 10 Inference Procedures

Type of Inference	Parameter(s)	TI Calculator Function	CI Form or Test Statistic	Necessary Conditions
Hypothesis Test	Correlation ρ	LinRegTTest	$t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$ $df = n - 2$	1. Random sample. 2. X and Y are bivariate normal.
Hypothesis Test	Distribution of proportions (goodness of fit)	χ^2 GOF-Test	$\chi^2 = \sum \frac{(O - E)^2}{E}$ $df = k - 1$	1. Random samples. 2. $E_i = np_i \geq 5; i = 1, \dots, k$.
Hypothesis Test	Independence of variables	χ^2 -Test	$\chi^2 = \sum \frac{(O - E)^2}{E}$ $df = (r - 1)(c - 1)$	1. Random sample. 3. Expected frequencies each at least 5.
Hypothesis Test	Variances or SDs σ_1^2/σ_2^2 or σ_1/σ_2 (independent)	2-SampFTest	$F = \frac{s_1^2}{s_2^2}$ $df_N = n_1 - 1$ $df_D = n_2 - 1$	1. Independent samples. 2. Random samples. 3. Both populations normal.
Hypothesis Test	Means $\mu_1, \dots, \mu_k$ (ANOVA)	ANOVA	$F = \frac{MS_B}{MS_W}$ $df_N = k - 1$ $df_D = N - k$	1. Independent samples. 2. Random samples. 3. Normal populations. 4. All populations have same variance.

Note that the goodness-of-fit test, the test for independence, and the ANOVA test are always right-tailed tests.

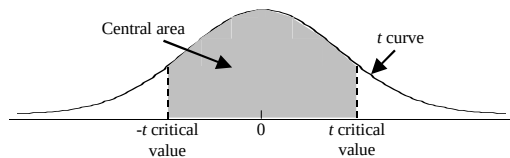
ANOVA formulas:

$$\begin{aligned}\bar{x} &= \frac{\sum_{i=1}^N x_i}{N} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + \dots + n_k\bar{x}_k}{N} = \frac{\sum_{i=1}^k n_i\bar{x}_i}{N} \\ SS_B &= n_1(\bar{x}_1 - \bar{x})^2 + \dots + n_k(\bar{x}_k - \bar{x})^2 = \sum_{i=1}^k n_i(\bar{x}_i - \bar{x})^2 \\ SS_W &= (n_1 - 1)s_1^2 + \dots + (n_k - 1)s_k^2 = \sum_{i=1}^k (n_i - 1)s_i^2 \\ MS_B &= \frac{SS_B}{k - 1} \\ MS_W &= \frac{SS_W}{N - k} \\ F &= \frac{MS_B}{MS_W}; \quad df_N = k - 1; \quad df_D = N - k\end{aligned}$$

The ANOVA Table

Source of Variation	Sum of Squares	df	Mean Square	F Statistic
Between	SS_B	$k - 1$	$MS_B = \frac{SS_B}{k-1}$	$F = \frac{MS_B}{MS_W}$
Within	SS_W	$N - k$	$MS_W = \frac{SS_W}{N-k}$	
Total	SS_T	$N - 1$		

t critical values



Confidence area captured:		0.90	0.95	0.98	0.99
Confidence level:		90%	95%	98%	99%
Degrees of Freedom	1	6.314	12.706	31.821	63.657
	2	2.920	4.303	6.965	9.925
	3	2.353	3.182	4.541	5.841
	4	2.132	2.776	3.747	4.604
	5	2.015	2.571	3.365	4.032
	6	1.943	2.447	3.143	3.707
	7	1.895	2.365	2.998	3.499
	8	1.860	2.306	2.896	3.355
	9	1.833	2.262	2.821	3.250
	10	1.812	2.228	2.764	3.169
	11	1.796	2.201	2.718	3.106
	12	1.782	2.179	2.681	3.055
	13	1.771	2.160	2.650	3.012
	14	1.761	2.145	2.624	2.977
	15	1.753	2.131	2.602	2.947
	16	1.746	2.120	2.583	2.921
	17	1.740	2.110	2.567	2.898
	18	1.734	2.101	2.552	2.878
	19	1.729	2.093	2.539	2.861
	20	1.725	2.086	2.528	2.845
	21	1.721	2.080	2.518	2.831
	22	1.717	2.074	2.508	2.819
	23	1.714	2.069	2.500	2.807
	24	1.711	2.064	2.492	2.797
	25	1.708	2.060	2.485	2.787
	26	1.706	2.056	2.479	2.779
	27	1.703	2.052	2.473	2.771
	28	1.701	2.048	2.467	2.763
	29	1.699	2.045	2.462	2.756
	30	1.697	2.042	2.457	2.750
	40	1.684	2.021	2.423	2.704
	60	1.671	2.000	2.390	2.660
	70	1.667	1.994	2.381	2.648
	80	1.664	1.990	2.374	2.639
	90	1.662	1.987	2.368	2.632
	100	1.660	1.984	2.364	2.626
	1000	1.646	1.962	2.330	2.581
z critical values	∞	1.645	1.960	2.326	2.576
α for 2-tailed tests		0.10	0.05	0.02	0.01
α for 1-tailed tests		0.05	0.025	0.01	0.005